

DESIGN MANUAL

ON

UNIVERSAL

ACCESSIBILITY

2026



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PREFACE

1. This Design Manual (this Manual) is revamped from the “Design Manual: Barrier Free Access 2008 (2025 Edition)” (DMBFA).
2. The DMBFA, which sets out the barrier-free design requirements for persons with a disability (PwDs), the elderly and people with other forms of physical infirmities or limitation such as pregnant women and families with young children, was first published in 2008. Several amendments were made by the Buildings Department (BD) through the issuance of Practice Note for Authorized Persons, Registered Structural Engineers and Registered Geotechnical Engineers APP-41 from 2012 to 2025 taking into account feedback from the building sector, the rehabilitation sector and relevant stakeholder organisations arising from their practical uses and experiences.
3. Subsequent to the 2017 Policy Address which announced that the Government had tasked the Rehabilitation Advisory Committee to formulate a new PwDs and Rehabilitation Programme Plan (RPP), three stages of public engagement were carried out from March 2018 to January 2020. The Government formally released the new RPP in July 2020, accepting in principle the strategic directions and recommendations therein. For accessibility of community environment and services, the Government acting through the Labour and Welfare Bureau carried out a Consultancy Study for Enhancement of the Accessibility of Physical Environment in Hong Kong which recommended revising or enhancing the DMBFA, and where practicable, promoting the universal design concepts.
4. In this connection, the BD commissioned a consultancy study in 2024 to undertake a holistic review of the DMBFA, with a view to enhancing the design standards to align with the latest needs and international trends, and ensuring that concepts of universal design are incorporated in a way that is feasible and pragmatic in the local context. The DMBFA has therefore been revamped and expanded to incorporate universal design concepts with the objective of creating a more inclusive and accessible built environment. This culminates in the formulation of this Manual, which aims to promote an inclusive built environment that is accessible and functional for a broader spectrum of the community.
5. Save for the provisions already covered by regulations 72 and 73, the Third and Fourth Schedules of the Building (Planning) Regulations (B(P)R), the requirements under this Manual are, at present, advisory in nature. For those obligatory provisions stipulated in this Manual that are not yet covered by regulations 72 and 73, the Third and Fourth Schedules of the B(P)R, the BD will propose inclusion of such provisions in the B(P)R through legislative amendments. A summary of the obligatory provisions proposed to be included in the B(P)R is provided in Appendix E for reference.

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CHAPTER 1 FOREWORD

Disability Discrimination Ordinance (Cap. 487)

- 1.1 The Disability Discrimination Ordinance was enacted in August 1995. It prohibits, among other things, discrimination against PwDs by failing to provide means of access to any premises that the public or a section of the public is entitled or allowed to enter or use, or by refusing to provide appropriate facilities. However, there is no discrimination in relation to the provision of access to premises if the premises are so designed or constructed as to be inaccessible to PwDs and any alteration to the premises to provide such access would impose unjustifiable hardship on the persons who would have to provide it. A person who believes he or she has been discriminated against in relation to access to premises or the provision of facilities may lodge a complaint with the Equal Opportunities Commission or may institute legal proceedings in the court. Reference may be made to this Manual as one sees fit in considering whether it is reasonable to require the provision of such access or facilities.
- 1.2 For a new building or for the alterations and additions to an existing building, section 84 of the Disability Discrimination Ordinance stipulates that:

“s84. Building approvals

- (1) Notwithstanding any provision in any other Ordinance (whether enacted before or after the commencement of this Ordinance) but subject to subsection (3), a public authority which has the power to approve building works shall not, in respect of those works, approve building plans, whether for a new building or for the alterations or additions to an existing building unless the person seeking approval satisfies the public authority that such access as is reasonable in the circumstances to the building or premises will be provided for persons with a disability.*
- (2) In considering whether reasonable access will be provided under subsection (1), the public authority may take into account:*
 - (a) whether it is practicable to provide such access within the curtilage of the building, bearing in mind the physical location and immediate environs of the building; and*
 - (b) whether providing such access would impose unjustifiable hardship on the person seeking approval or on any other person.*

(Cont'd)

(3) *Subsection (1) has no application to:*

- (a) *buildings of 13 m or less in height above ground level which are used, or intended to be used, for occupation by a single family; or*
- (b) *temporary buildings or contractor's sheds referred to in Part VII of the Building (Planning) Regulations (Cap. 123 sub. leg.).*

(4) *In this section, "public authority" includes:*

- (a) *the Director of Lands;*
- (b) *the Building Authority;*
- (c) *the Housing Authority;*
- (d) *the Director of Architectural Services."*

1.3 This Manual applies to the design and construction of new buildings or alterations and additions to existing buildings. Relevant Government authorities and departments will also make reference to the design requirements therein in the design and construction of their buildings.

CHAPTER 2 PRELIMINARY

2.1 Interpretation

For the purpose of this Manual, the following terms are defined as:

“Access” means any means to enable persons, regardless of their age, size, ability or disability to approach, enter and leave the building and to use the facilities therein without assistance or undue difficulties.

“Access route” is a path provided from a prominent point on the lot boundary, which is accessible to a public street or pedestrian way, directly to an entrance which is commonly used by the public, or to a point adjacent to an entrance which is commonly used by the public, and to an accessible lift.

“Accessible” describes a site, building, facility or portion thereof that is barrier-free, can be approached, entered, used and exited by persons, regardless of their ages, sizes, abilities or disabilities, in compliance with this Manual.

“Accessible lift” means a lift fully complied with the obligatory design requirements in [Chapter 7](#).

“Accessible route” is a continuous unobstructed path which is easily identifiable for persons, regardless of their ages, sizes, abilities or disabilities to approach, enter and leave the building and to use the facilities therein without assistance or undue difficulties.

“Assistive listening system” means a system which enables sound signs that are amplified in both volume and signal to be transmitted to persons with hearing impairment without interference from excessive background noise or excessive reverberation.

“Common areas” are those areas open to and available for the common use and enjoyment of all occupiers of the building, including both indoor and outdoor areas within the development.

“Door” includes one leaf of a pair of double doors.

“Ground floor” means any ground floor, lower ground floor, upper ground floor provided with an access route.

“Large scale specified premises” are specified premises which have an occupant capacity of more than 10 000 people.

“Luminance contrast” is the amount of light reflected from the surface of an object compared to the amount of light reflected from the surface of its surrounding background, determined by measuring the light reflectance value.

“Persons with a disability” (PwDs) means persons who on account of injury, disease, or congenital deformity, are impaired in vision, hearing or locomotion. Such persons shall include persons with ambulant disabilities, wheelchair users, persons with visual impairment, the blind, persons with hearing impairment and the deaf.

“Persons with ambulant disabilities” are persons with ambulant impairment who may require the aid of devices such as prostheses, orthoses, sticks or crutches for walking.

“Public services escalator or passenger conveyor” means an escalator or passenger conveyor that is part of a public traffic system including entrance and exit points (for example for connecting a traffic station and a building), and is for operating regularly for not less than 140 hours/week with a load reaching 100% of the brake load during periods lasting for at least 0.5 hour during any time interval of 3 hours. Reference is made to the Code of Practice for Energy Efficiency of Building Services Installation issued by Electrical and Mechanical Services Department.

“Required staircase” means an access staircase in a firefighting and rescue stairway or a staircase required for means of escape in case of fire.

“Slip-resistance” is the frictional force that opposes the movement of an object across a surface, usually with reference to the sole or heel of a shoe on a pedestrian surface. The requirements for slip-resistant surfaces in different locations are specified in [Appendix C](#) of this Manual.

“Specified premises” are premises with relatively high traffic flow and high number of occupancies, including:

- (a) Shopping malls and other retail premises with gross floor area of more than 30 000 m²; and
- (b) Cinemas, theatres, concert halls, sport facilities, stadiums, museums and theme parks with an occupant capacity of more than 3 000 people.

“Tactile guide path” means a system of standardised pattern applied to or built onto walking surfaces through the combined use of tactile directional tiles/blocks, positional tiles/blocks and tactile hazard warning tiles/blocks with slip-resistant surface in accordance with [Appendix C](#) of this Manual to assist in wayfinding and orientation for persons with visual impairment.

“Tactile warning strip” means a standardised pattern applied to or built onto walking surfaces using tactile hazard warning tiles/blocks with slip-resistant surface in accordance with [Appendix C](#) of this Manual to alert persons with visual impairment of certain physical or potential hazards within the built environment.

“Wet areas”, in relation to a residential unit means internal areas that is subject to periodic or constant wetting, including bathroom, toilet and kitchen.

“Wheelchair users” are those persons who depend on wheelchairs for mobility.

“Unjustifiable Hardship” has the same meaning as ‘unjustifiable hardship’ referred to in section 4 of the *Disability Discrimination Ordinance Cap. 487*, which reads:

‘For the purposes of this Ordinance in determining what constitutes unjustifiable hardship, all relevant circumstances of the particular case are to be taken into account including:

- (a) the reasonableness of any accommodation to be made available to a person with a disability;*
- (b) the nature of the benefit or detriment likely to accrue or be suffered by any persons concerned;*
- (c) the effect of the disability of the person concerned; and*
- (d) the financial circumstances of and the estimated amount of expenditure (including recurrent expenditure) required to be made by the person claiming unjustifiable hardship.’*

CHAPTER 3 EXTENT OF APPLICATION

3.1 Extent of Application

3.1.1 The requirements set out in this Manual are classified into:

(1) Mandatory Section

- Obligatory Design Requirements

All Obligatory Design Requirements shall be complied with. Where alternative designs are proposed in lieu of strict compliance with the Obligatory Design Requirements, such alternative designs must be able to facilitate the access to, and use of, the buildings and its facilities by persons, regardless of their ages, sizes, abilities or disabilities to an extent not inferior to that provided by the Obligatory Design Requirements.

(2) Advisory Section

- Advisory Design Requirements

Advisory Design Requirements are included for the reference of professionals, developers and building owners who intend to provide access and/or special facilities that are enhanced from the Obligatory Design Requirements for the use by all intended users.

3.1.2 Subject to paragraph 3.2, a new building or any alterations and additions to an existing building shall be designed and constructed in accordance with the Mandatory Section set out in this Manual provided that:

- (1) the means of access and facilities stipulated in the Mandatory Section shall be provided to the categories of buildings specified in the left column of Table 1 and to the extent specified in the right column thereof, excluding the parts of buildings stated in paragraph 3.2.2; and
- (2) the extent of application of additional assistive provisions to various uses of buildings shall be as specified in Table 2.

TABLE 1

CATEGORY OF BUILDINGS & EXTENT OF THE APPLICATION	
Category of Buildings	Extent of application
Domestic buildings	All common areas and facilities of buildings of more than four (4) storeys. Main entrance, common areas and facilities of the ground floor*, and access route to buildings which do not exceed four (4) storeys. <i>Notwithstanding the above, the requirements stated in <u>paragraph 5.6</u> apply to all main entrance doors and wet areas of residential units.</i>
Non-domestic buildings	All parts of such buildings including outdoor areas*.
Composite buildings	Non-domestic parts of such buildings including outdoor areas*. All common areas and facilities of the domestic parts of such buildings*, if the domestic parts exceed four (4) storeys. Main entrance, common areas and facilities of the ground floor* and access route to buildings, if the domestic parts do not exceed four (4) storeys. <i>Notwithstanding the above, the requirements stated in <u>paragraph 5.6</u> apply to all main entrance doors and wet areas of residential units.</i>

* Except for landscape features such as planters, water features, rocks, cliffs, boulders, caves, vegetations, etc.

TABLE 2

“Y” denotes “Applicable”

“-“ denotes “Not Applicable”

EXTENT OF APPLICATION OF ADDITIONAL ASSISTIVE PROVISIONS TO VARIOUS USES OF BUILDINGS						
Uses of buildings	Required Additional Assistive Provisions listed below:					
	Braille & tactile floor plan [para. 10.10(1)]	Tactile guide path [para. 4.3 and 10.3]	Visual Display Board [para. 10.10(2)]	Accessible Public Information / Service Counter [para. 10.6]	Visual Fire Alarm System [para. 10.8]	Assistive Listening System [para. 10.7]
1. Domestic use	-	-	-	-	-	-
2. Common areas of Office	-	-	-	-	Y	-
3. Department store and shopping complex	Y	Y	-	Y	Y	-
4. Hotel, guesthouse, hostel and bank	Y	-	-	Y	Y	-
5. Place for worship	Y	-	-	Y	Y	-
6. Cinema, theatre, concert hall, stadium, museum, theme park and purpose- built family amusement centre	Y	Y	Y	Y	Y	Y
7. School, college, university and public library	Y	Y	-	Y	Y	-
8. Factory, workshop and place for industrial use	-	-	-	-	Y	-

TABLE 2 (Cont'd)

EXTENT OF APPLICATION OF ADDITIONAL ASSISTIVE PROVISIONS TO VARIOUS USES OF BUILDINGS						
Uses of buildings	Required Additional Assistive Provisions listed below:					
	Braille & tactile floor plan [para. 10.10(1)]	Tactile guide path [para. 4.3 and 10.3]	Visual Display Board [para. 10.10(2)]	Accessible Public Information / Service Counter [para. 10.6]	Visual Fire Alarm System [para. 10.8]	Assistive Listening System [para. 10.7]
9. Sports complex and public swimming pool complex	Y	Y	-	Y	Y	-
10. Restaurant and food court	Y	-	-	-	Y	-
11. Indoor market and supermarket	Y	-	-	-	Y	-
12. Hospital, purpose-built clinic	Y	Y	Y	Y	Y	Y
13. Residential home for the elderly and welfare centre	Y	Y	-	Y	Y	-
14. Club house	Y	-	-	Y	Y	-
15. Transport station, interchange, passenger terminal	Y	Y	Y	Y	Y	Y
16. Carpark	Y	-	-	-	Y	-

3.2 Exemptions

- 3.2.1 The Obligatory Design Requirements shall not apply to the following buildings:
- (1) Buildings of 13 m or less in height above ground level which are used, or intended to be used, for occupation by a single family; or
 - (2) Temporary buildings or contractor's sheds referred to in Part VII of the Building (Planning) Regulations (Cap.123 sub. leg.).
- 3.2.2 The Obligatory Design Requirements shall not apply to the areas or parts of a building where there is a relatively high risk to PwDs in these areas/parts including:
- (1) Commercial kitchen, cold room and cinema projector room.
 - (2) Area only used for building services and maintenance (testing, inspection, verification, repair and overhaul) including:
 - (a) plant, cooling tower and power plant;
 - (b) equipment and lift motor room, and electrical transformer room and switch room, battery room, machinery room, plant room and pump room;
 - (c) boiler room;
 - (d) non-tenantable spaces accessed only by ladder, catwalk or crawl space;
 - (e) access route for maintenance, pit, lift shaft and ventilation shaft;
 - (f) sub-station, telecommunication equipment room, metering area;
 - (g) machinery areas of conveyance and transportation system such as automated rail system, tram and cable car; and
 - (h) unmanned utilities facilities such as equipment/utility towers and tunnels, or the like.
 - (3) Area used for storage of raw material or produce or for bulk storage where:
 - (a) the stored material is hazardous; or
 - (b) the public is not permitted to enter, such as waste containment area, chemical store, or the like.
 - (4) Mezzanine floor used only for storage, plant and equipment installation or the like.
 - (5) Raised platform used primarily for purpose of security or safety management, including, but not limited to, guard tower or fixed lifeguard stand.
 - (6) Swimming pool (the water containing pool only).

- (7) Areas to which PwDs may not reasonably be expected to have access in view of the relatively high risk operations in the buildings such as:
 - (a) rail and racing tracks;
 - (b) areas of heavy/high risk industrial process such as train depot, ship yard, vehicle and aircraft repair and maintenance, cross decking and automatic storage and retrieval system areas of warehouses/logistic centre; and
 - (c) controlled areas of keeping/treatment of large/lived animals such as stables, sanctuary, zoo enclosure and restricted area of aquarium, or the like.
- (8) Any path providing access only to an exempted area.

CHAPTER 4 ARRIVING AT THE BUILDING

4.1 General

This Chapter aims to provide design requirements to ensure proper access for all people, with or without disabilities, to approach and enter a building independently to reach and use its facilities without undue difficulty.

MANDATORY SECTION

Obligatory Design Requirements

4.2 Provision of Access Route

- (1) Access shall be provided from a prominent point or points on the lot boundary, which is accessible to a public street or pedestrian way, directly to at least one entrance which is commonly used by the public or to a point directly adjacent to one entrance which is commonly used by the public and to an accessible lift, and such access route shall cover both internal and external areas, unless it is impractical to do so because of difficult terrain or unusual characteristics of the site.
- (2) In case where the main entrance is not an accessible entrance or it is impractical to provide a commonly used entrance at prominent point on the lot boundary due to difficult terrain or unusual characteristics of the site (e.g. in the case of a sloping site or presence of steep access road/driveway), this would be acceptable provided that PwDs could reach the building by means of vehicle and adequate directional signs shall be posted at conspicuous location of the main entrance to show clearly the location of and the route to an accessible entrance.

4.3 Tactile Guide Path

- (1) For categories of buildings as required in Table 2, such access shall be provided with a tactile guide path from a point of access at the lot boundary to the main entrance of the building and from the main entrance to lift zone, the nearest accessible toilet, public information/service counter, braille and tactile floor plan and staircase as specified in paragraph 10.3.

4.4 Car Parking Areas

Ratio of Accessible Car Parking Spaces

- (1) Accessible car parking spaces shall be provided with proper access, proper designation and directional signage in the carparks. The required number of accessible car parking spaces shall be provided as follows:

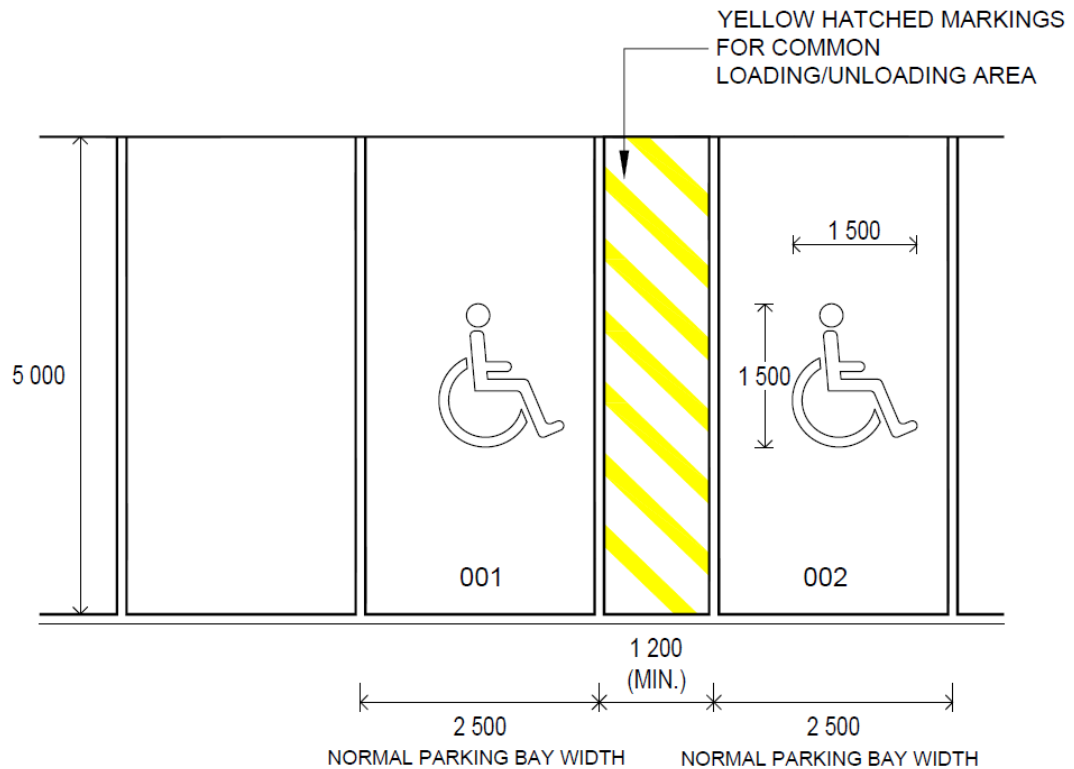
Total No. of Car Parking Spaces in Lot	Required No. of Accessible Car Parking Spaces
1 - 50	1
51 - 150	2
151 - 250	3
251 - 350	4
351 - 450	5
Above 450	6

- (2) For large scale specified premises as defined in Chapter 2, the required number of accessible car parking spaces shall be provided as follows:

Total No. of Car Parking Spaces in Large Scale Specified Premises	Required No. of Accessible Car Parking Spaces
1 - 25	1
26 - 50	2
51 - 100	3
for each additional increment of 100 or part thereof	One additional space

Requirements for Accessible Car Parking Spaces

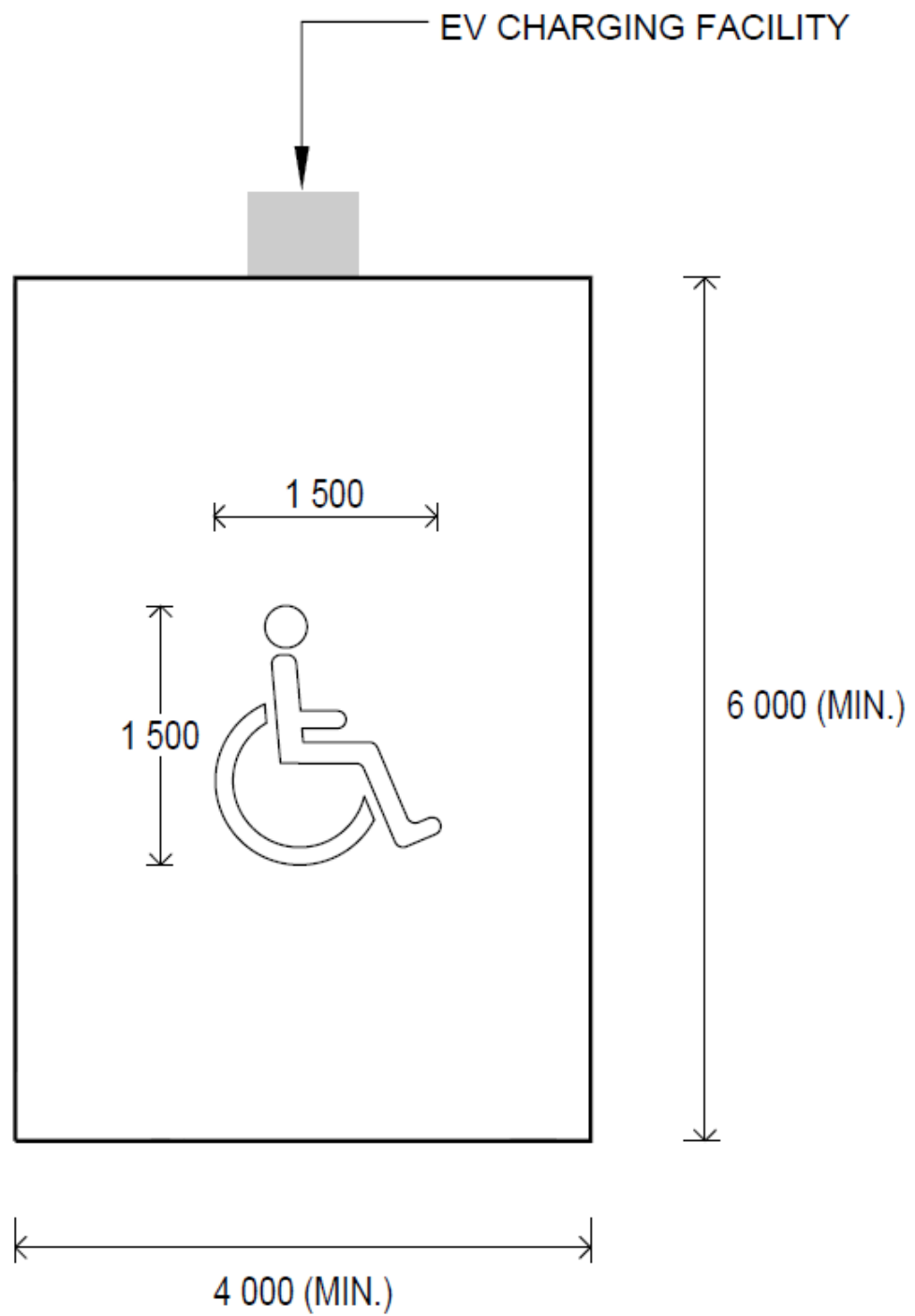
- (3) An accessible car parking spaces shall be located in proximity and with an accessible route to the lobby with an accessible lift or entrance.
- (4) An accessible car parking space shall have a minimum width of 3 500 mm and a minimum depth of 5 000 mm.
- (5) Where a common loading/unloading area is provided between two accessible car parking spaces, such car parking spaces shall be not less than 2 500 mm in width.
- (6) The common loading/unloading area shall be of at least 1 200 mm wide and be marked with yellow hatched markings as shown in **Figure 1**.



*ALL DIMENSIONS ARE IN mm

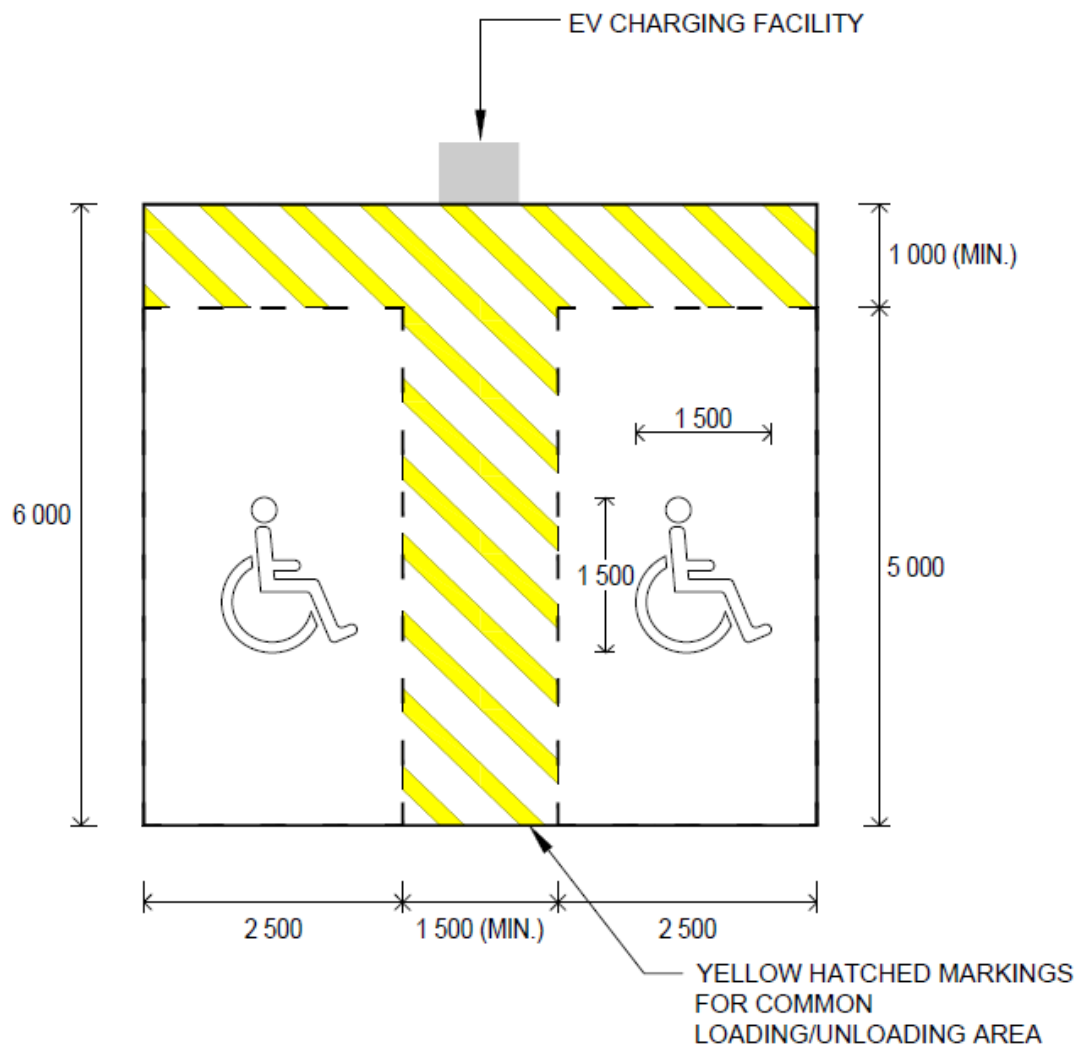
Figure 1 - Common Loading/ Unloading Area of Accessible Car Parking Spaces

- (7) In case the accessible car parking spaces are provided with electric vehicle (EV) charging facilities, each parking space shall be not less than 6 000 mm in length and 4 000 mm in width as shown in **Figure 2**.
- (8) Where a common loading/unloading area is provided between two accessible car parking spaces with EV charging facilities, each parking spaces shall be not less than 2 500 mm in width.
- (9) The common loading/unloading area shall be at least 1500 mm wide and marked with yellow hatched markings as shown in **Figure 3**.



*ALL DIMENSIONS ARE IN mm

Figure 2 - Dimensions of Accessible Car Parking Space with EV Charging Facility

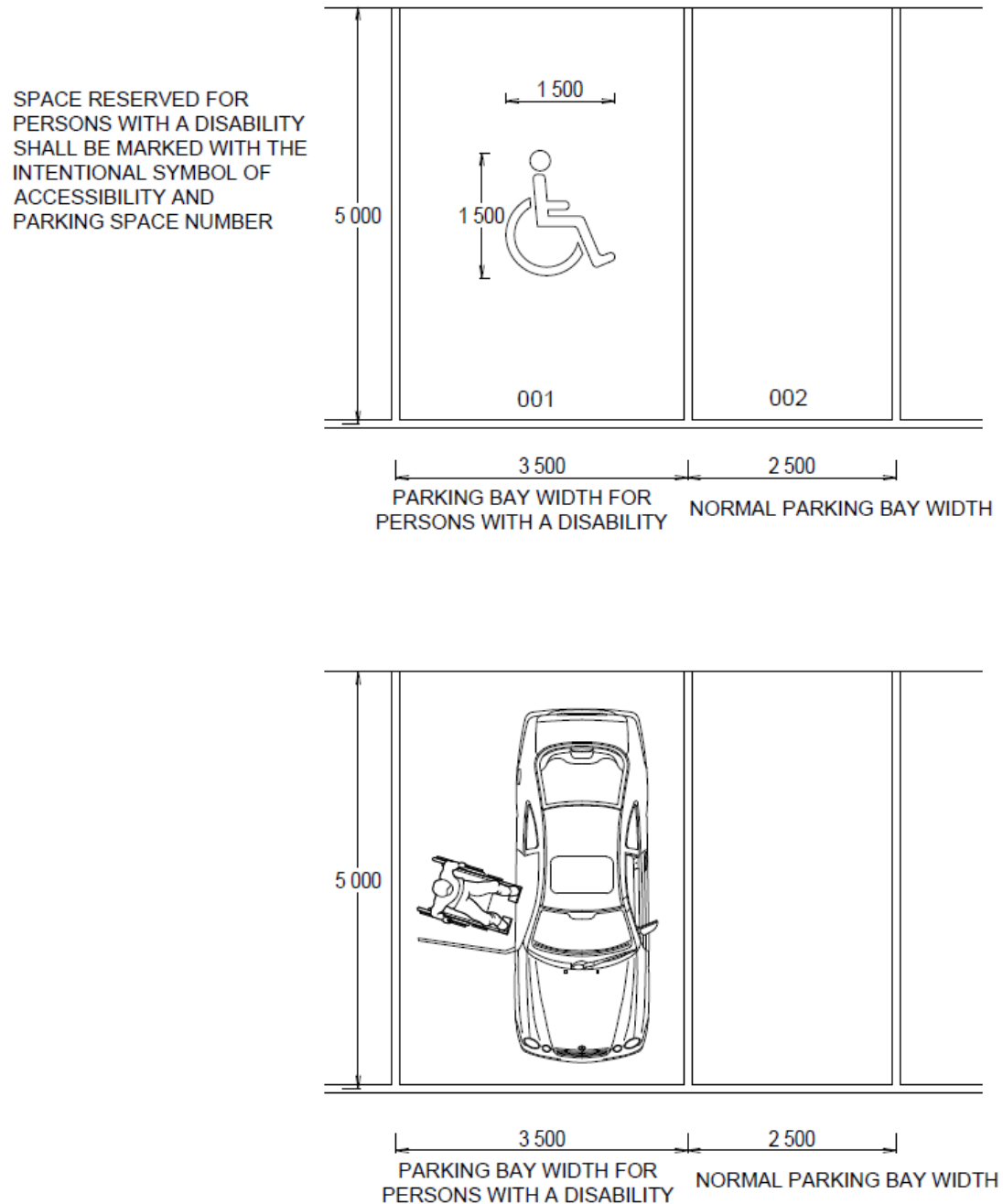


*ALL DIMENSIONS ARE IN mm

Figure 3 - Common Loading/ Unloading Area of Accessible Car Parking Spaces with EV Charging Facilities

Marking Requirements for Accessible Car Parking Spaces

- (10) The accessible car parking space shall be clearly marked with the international symbol of accessibility and the parking space number on the floor as shown in **Figure 4**.



*ALL DIMENSIONS ARE IN mm

Figure 4 - Dimensions and Identification of Accessible Car Parking Space

ADVISORY SECTION

Advisory Design Requirements

4.5 Picking Up and Setting Down Areas

- (1) Picking up and setting down area adjoining communal main entrance of a building including the associated driveway should be provided with shelters.
- (2) Covered walkway/passageway should be provided for the main initial access at the lot boundary to the main entrance of the building; and for connecting the picking up and setting down area adjoining communal main entrance and other parts of the building.
- (3) Sufficient and clear directional signage indicating the access to other parts of the building should be provided at the picking up and setting down area.

4.6 Accessible Car Parking Spaces

- (1) At the entrance of public parking facilities, signage should be displayed in prominent positions to indicate the designated numbers of the accessible car parking spaces (see **Figure 5**).
- (2) Indication/directional signage along driveway showing the way leading to the accessible car parking spaces reserved should be provided.

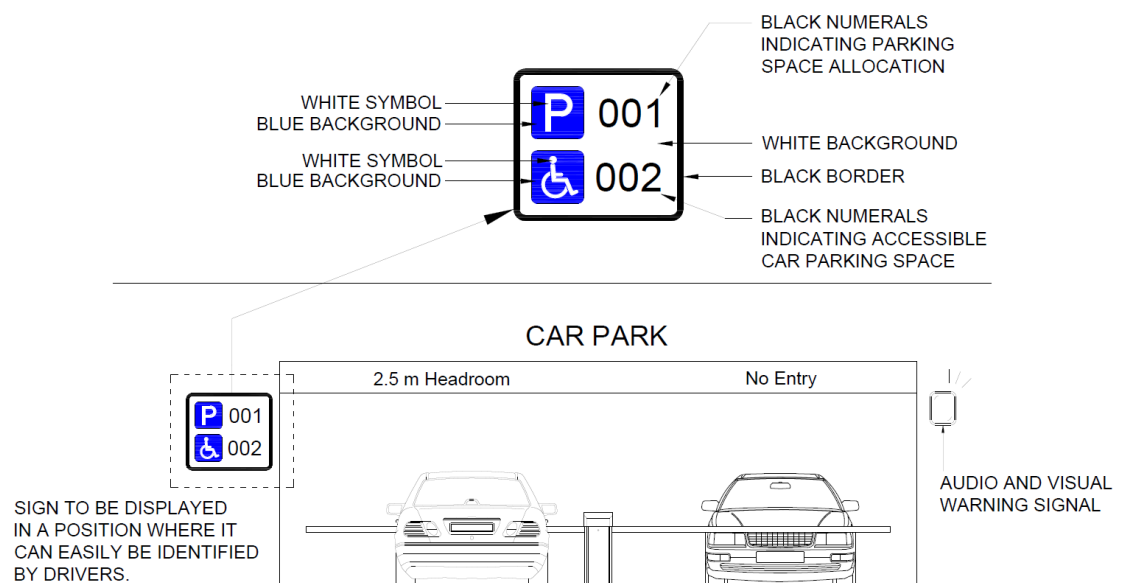


Figure 5 - Parking Space Indication Signage

- (3) Provision of accessible car parking spaces for buildings other than large scale specified premises should follow paragraph 4.4(2) as far as practicable.
- (4) Audio and visual warning signal should be provided at vehicle entrances of car park to alert pedestrians to the ingress and egress of vehicles.

CHAPTER 5 HORIZONTAL CIRCULATION AROUND AND WITHIN A BUILDING

5.1 General

This Chapter aims to provide design requirements to ensure that all building developments shall be designed, constructed, maintained and managed so that it is accessible. All aspects of horizontal circulation, including corridors, lobbies and paths shall be designed to facilitate ease of movement and to avoid congestion. The elements mentioned in this Chapter shall also be applicable to the outdoor common areas of the buildings.

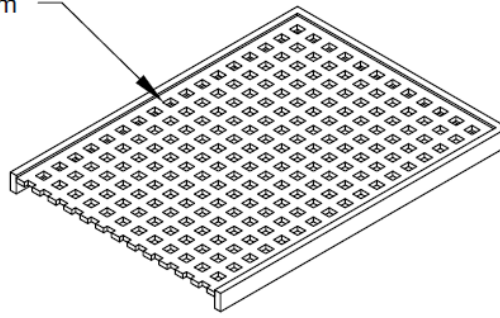
MANDATORY SECTION

Obligatory Design Requirements

5.2 Floor Surfaces of Access

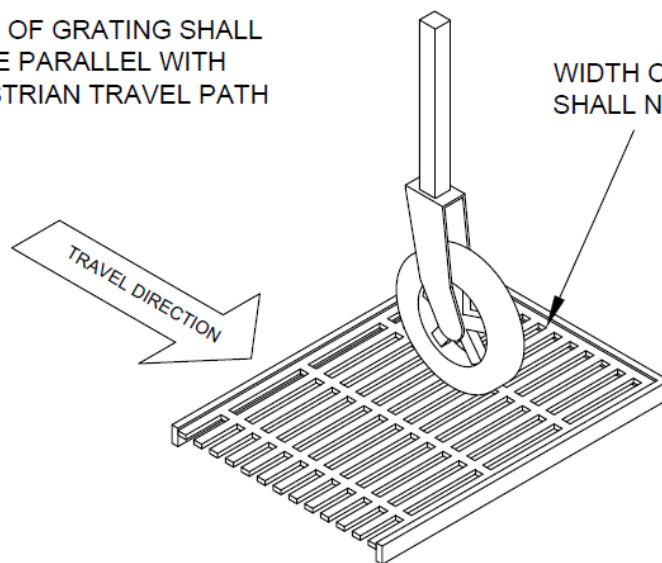
- (1) The surface of an accessible route including access route shall be firm.
- (2) All accessible routes in both internal and external common areas of a building shall have slip-resistant floor surfaces in accordance with Appendix C of this Manual.
- (3) On footpath, cover to a channel shall be flush with the surface of the footpath. Any hole in such cover or between such covers shall have a dimension of not more than 20 mm.
- (4) Slots of grating shall have a width of not more than 13 mm and shall not be parallel with pedestrian travel path (see **Figure 6**).

HOLES SHALL NOT EXCEED
20 mm X 20 mm



SLOTS OF GRATING SHALL
NOT BE PARALLEL WITH
PEDESTRIAN TRAVEL PATH

WIDTH OF SLOTS OF GRATING
SHALL NOT EXCEED 13 mm WIDE



*ALL DIMENSIONS ARE IN mm

Figure 6 - Grating Size and Orientation

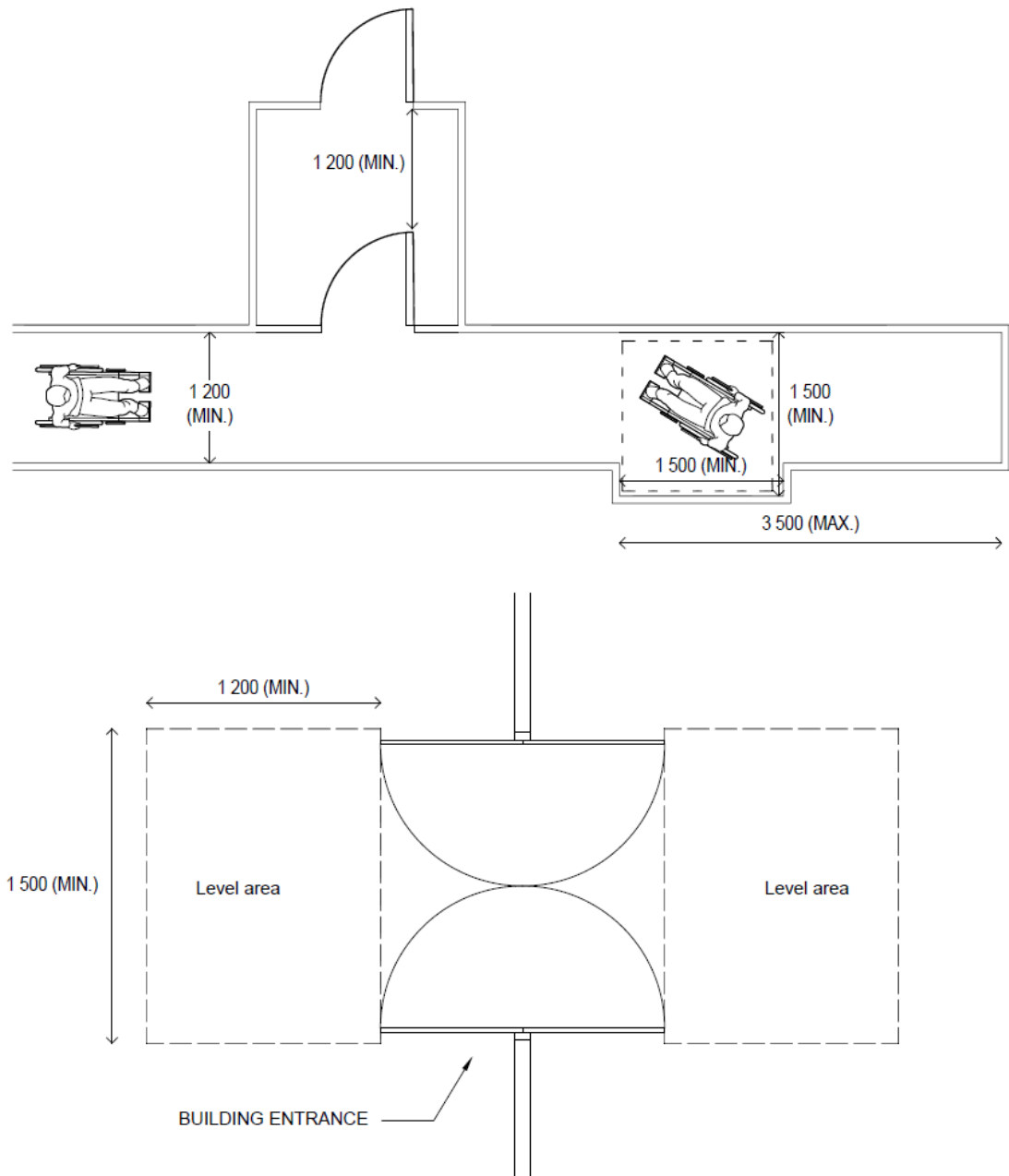
5.3 Width of Access

- (1) The clear width of an accessible route shall be not less than 1 200 mm.
- (2) Subject to other provisions under this Manual, such access shall be free from protrusion hazards, steps, kerbs other than dropped kerbs, steep ramps, handrails, doors or doorways which will impede the passage of a wheelchair, or other form of barrier which will prevent access by PwDs.

5.4 Space and Headroom of Access

Manoeuvring Space of Access

- (1) Space shall be allowed for manoeuvring wheelchairs in access such as corridor, lobby, path and the like as follows (see **Figure 7**):
 - (a) area shall have a clear width of not less than 1 200 mm;
 - (b) a space not less than 1 500 mm x 1 500 mm shall be provided within 3 500 mm of every dead end;
 - (c) any lobby in a corridor shall have a length of not less than 1 200 mm, excluding space for door swings;
 - (d) a level area, extending not less than 1 200 mm beyond the swings of the doors and not less than 1 500 mm in width shall be provided on both sides of every entrance of a building; and
 - (e) this paragraph shall not apply to lobby which lead to staircase only.



*ALL DIMENSIONS ARE IN mm

Figure 7 - Manoeuvring Space of Access

For the purpose of this paragraph, “dead end” is a corridor, lobby or path where the means of exit for PwDs is in one direction only.

Protrusion

- (2) No appliances, fixtures and fittings shall project beyond 90 mm from the surface of any wall in corridors, paths and lobbies below a level of 2 000 mm above the finished floor level unless they are unavoidable, in which case they shall also be extended downwards to the finished floor level or guided by tactile flooring materials.

Controlled Passage

- (3) For cashier counter, security device installed at shop entrance or turnstile controlled passage accessible to the public, each shall have at least one path of minimum 850 mm in width for the use by wheelchair users and clearly marked with international symbol of accessibility, unless an alternative passage adjacent to the controlled passage is provided (see **Figure 8**).

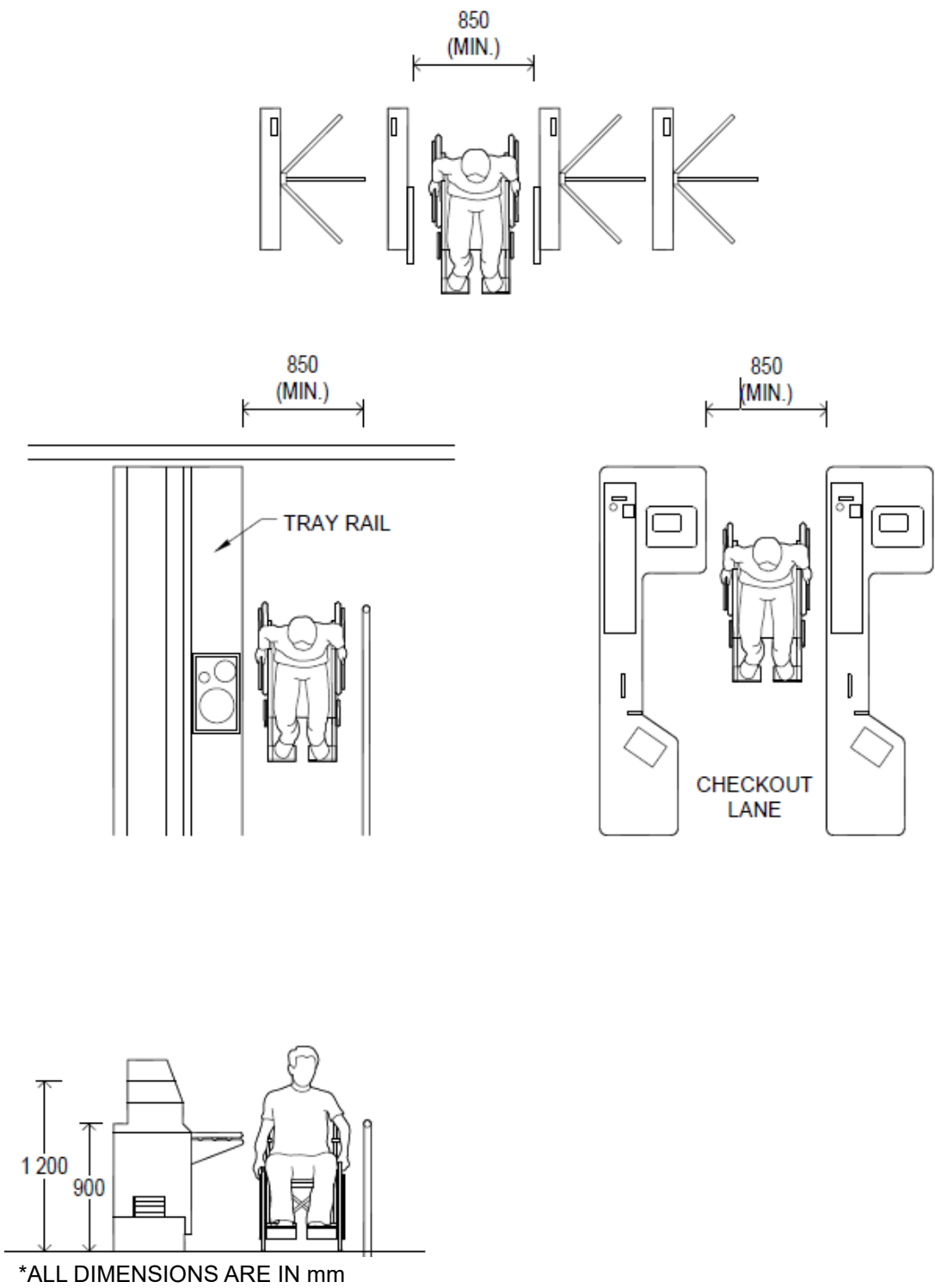
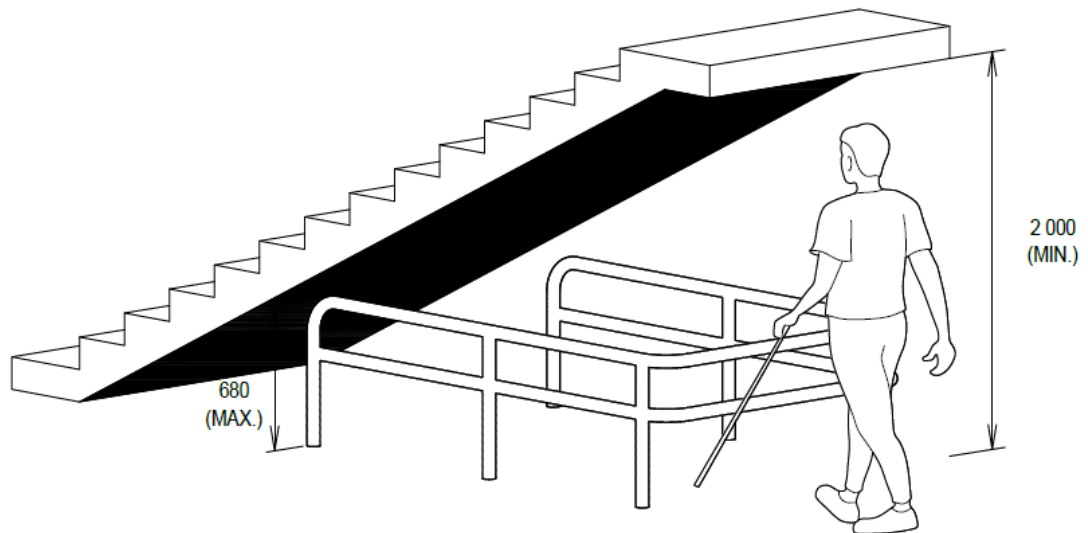


Figure 8 - Controlled Passage

Headroom

- (4) For any area with headroom less than 2 000 mm from the finished floor level, a warning guardrail or other barrier shall be provided for detection, having its leading edge at or below 680 mm above the finished floor level (see **Figure 9**).



*ALL DIMENSIONS ARE IN mm

Figure 9 - Overhead Hazard

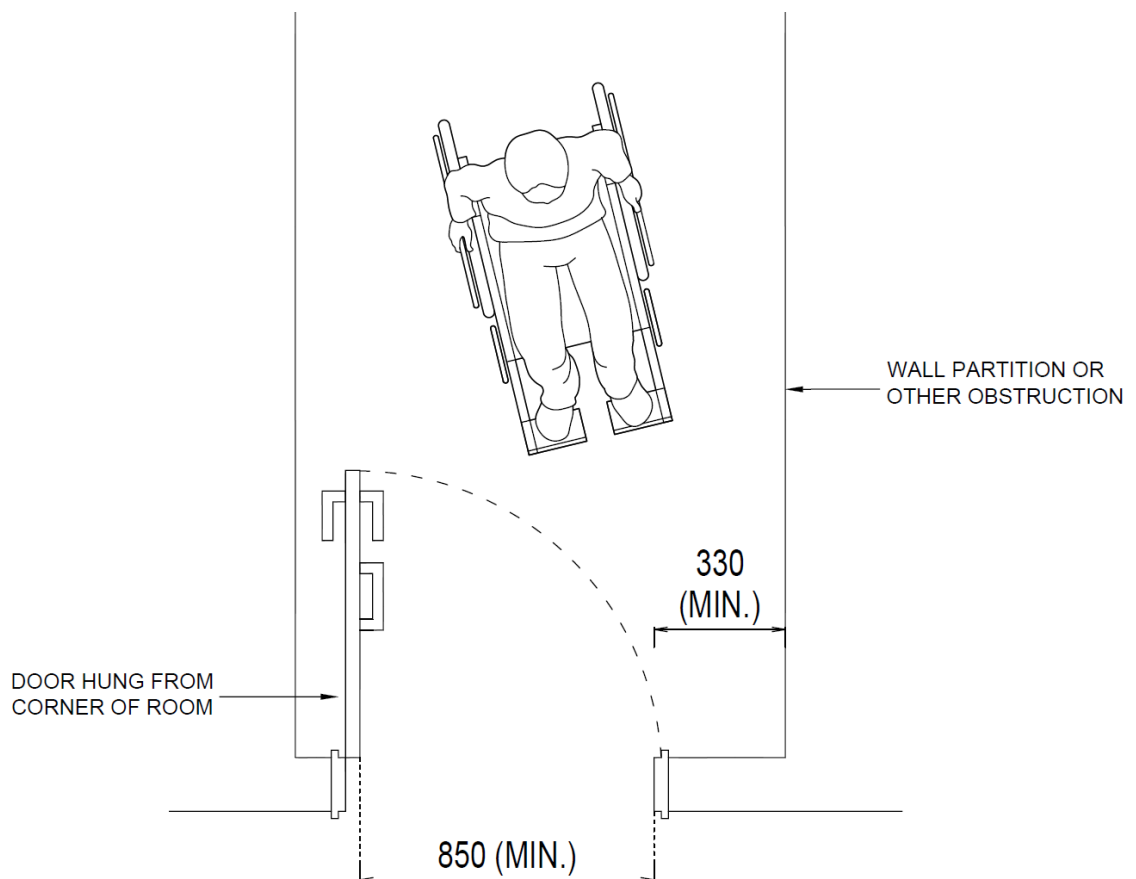
5.5 Door

Width of Doors

- (1) All doors on accessible routes, including one leaf of a pair of double doors, shall have a clear width of not less than 850 mm between the open door and opposite jamb or the other leaf (see **Figure 10**).

Unobstructed Area

- (2) The unobstructed area adjacent to the door frame on the leading face of a single door, other than a double-action door or an automatic door, shall not be less than 330 mm in width (see **Figure 10**).



*ALL DIMENSIONS ARE IN mm

Figure 10 - Plan of Door Suitable for Wheelchair

Door Thresholds

- (3) If door threshold (see **Figure 11**) is provided on accessible route, it shall not exceed 15 mm in height, having a bevelled and round edge on each side at gradient not steeper than 1:2 to facilitate passage of wheelchairs.



*ALL DIMENSIONS ARE IN mm

Figure 11 - Example of Door Threshold

Automatic Main Entrance Doors

- (4) An automatic door shall be provided to at least one main entrance of a building which is commonly used by the public, except for an industrial building.

5.6 Residential Unit

- (1) A door of the main entrance of a residential unit, including one leaf of a pair of double doors, shall have a clear width of not less than 850 mm between the open door and the opposite jamb or the other leaf.
- (2) If a door threshold is provided at the main entrance of a residential unit, it shall not exceed 15 mm in height, having a bevelled and round edge on each side at a gradient not steeper than 1:2 to facilitate the passage of wheelchair.
- (3) Wet areas of a residential unit, including any bathroom, kitchen and toilet shall be provided with slip-resistant floor surfaces in accordance with Appendix C of this Manual.

ADVISORY SECTION

Advisory Design Requirements

5.7 Surfaces

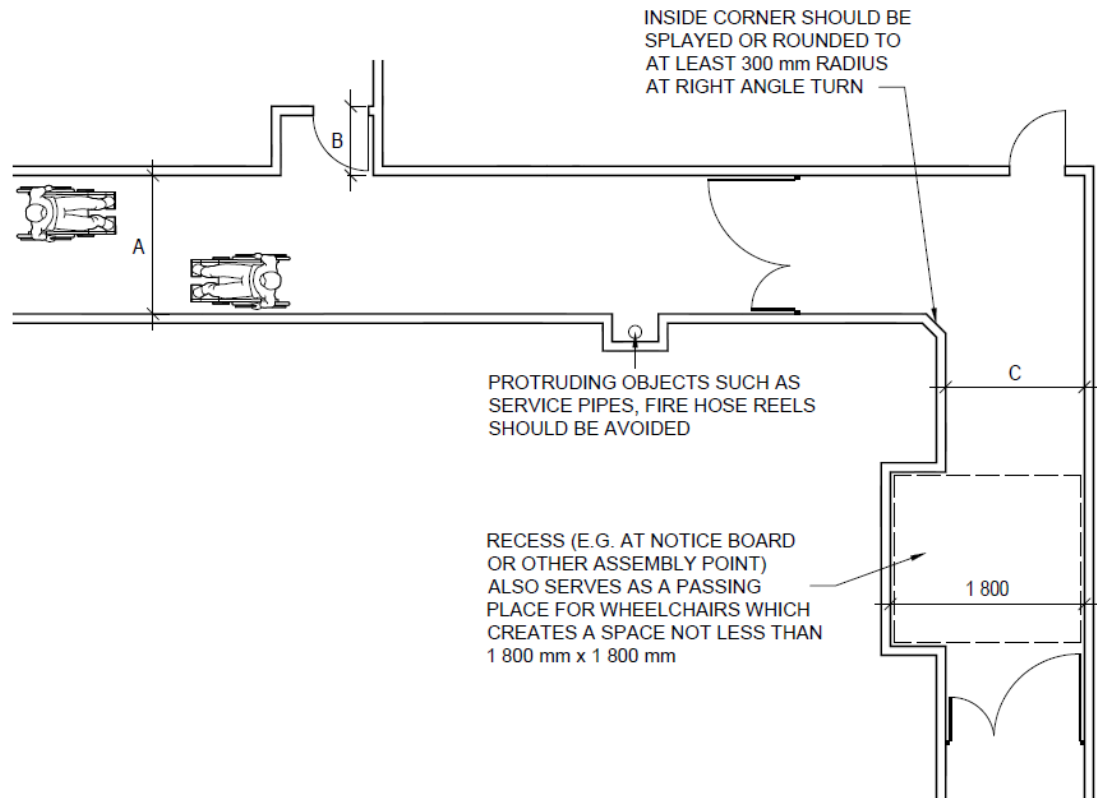
- (1) Surface paved with loose gravel or stone is hazardous and should be avoided.
- (2) Hazards on floor, caused by unnecessary projection or recess, or by unexpected change in level should be avoided.

5.8 Width of Access

- (1) The clear width of an accessible route should not be less than 1 500 mm.

5.9 Width of Access in Specified Premises

- (1) The clear width of accessible routes including access routes within specified premises should be not less than 1 800 mm. At right angle turns, the inside corner should be splayed or rounded to at least 300 mm radius (see **Figure 12**).



- A A CLEAR WIDTH OF 1 800 mm WILL ALLOW TWO WHEELCHAIR USERS TO PASS EACH OTHER.
- B DEPTH OF RECESS SHOULD NOT BE LESS THAN THE WIDTH OF THE DOOR LEAF.
- C A CLEAR WIDTH OF CORRIDOR SHOULD NOT BE LESS THAN 1 500 mm.

*ALL DIMENSIONS ARE IN mm

Figure 12 - Dimension and Space Allowance for Corridor in Building

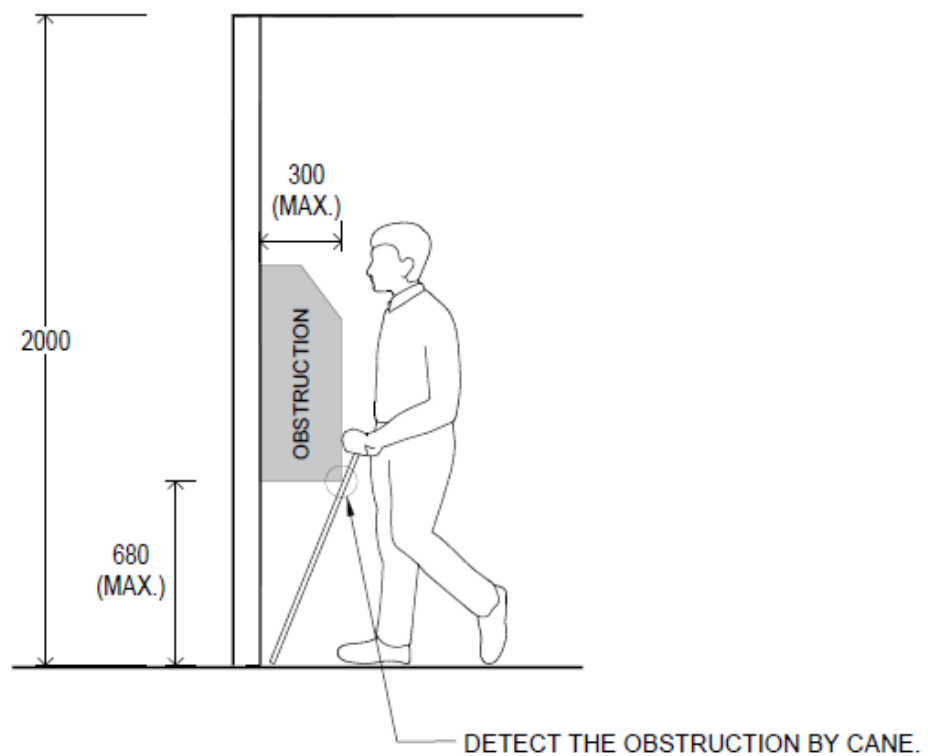
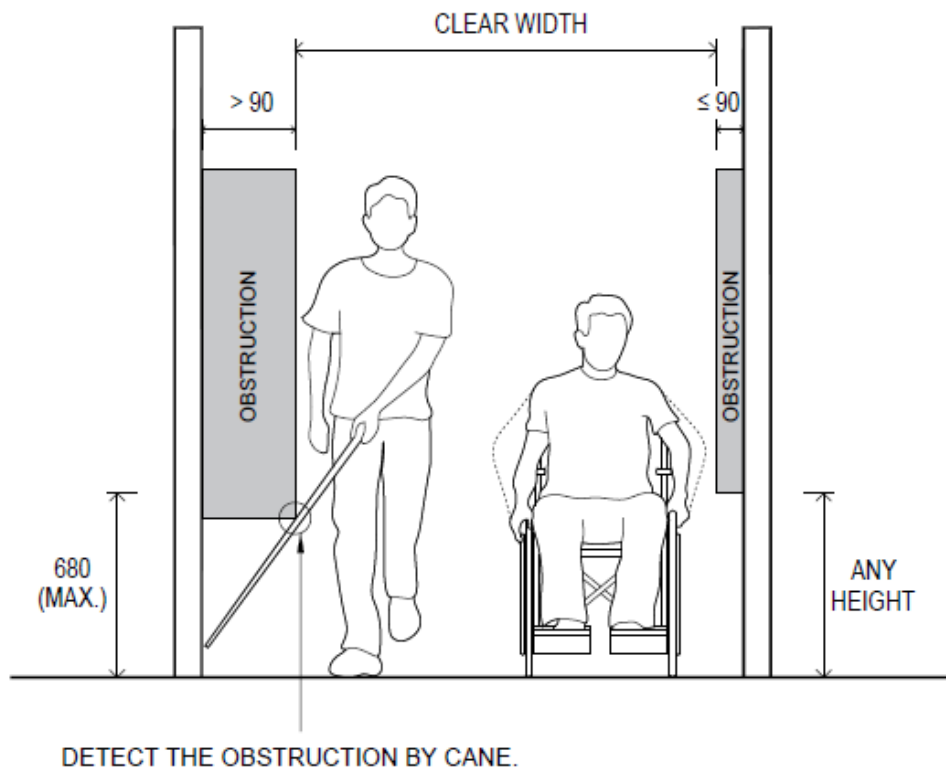
Luminance Contrast

- (2) A minimum luminance contrast of 30% should clearly define between wall, floor and door surfaces.

Protruding Objects

- (3) Protruding object should not reduce the statutory required clear width and height of access and manoeuvring space (see **Figure 13**).
- (4) Protruding object should include but not limited to sign, telephone enclosure, drinking fountain, fire extinguisher, underside of stairway or escalator.

- (5) Where it is reasonable to anticipate visit of persons with visual impairment, protruding object should be avoided at pedestrian areas including walkway, hall, corridor, aisle, lobby, mall and all areas open to the public.



*ALL DIMENSIONS ARE IN mm

Figure 13 - Examples of Design to Cater for Protrusion Hazards

Round Edges

- (6) The counters or devices installed at controlled passages and the warning guard rails provided for low headroom should be designed with round edges.

5.10 Resting Places

- (1) Seats (including fold-down seats) or lean-on railings should be provided in recessed areas at maximum intervals of 50 m in long corridors and public passageways:
 - (a) within all types of buildings; and
 - (b) connecting buildings.

5.11 Door

Width of Doors

- (1) All doors, including one leaf of a double-leaf door, should have a clear width of not less than 850 mm between the open door and opposite jamb or the other leaf.

Unobstructed Area

- (2) An unobstructed area adjacent to the door frame on the leading face of a single door should not be less than 600 mm in width.

Sliding/Folding/Double-action Doors

- (3) Sliding/folding/double-action doors should be provided as far as practicable, in particular for toilets, bathrooms and shower compartments, with door handles or grab rails should be positioned between 950 mm and 1 050 mm above the finished floor level (measured from the top surface of the grip).

Automatic Doors

- (4) Automatic doors including both swinging doors and sliding doors should be provided to all entrances of a building which are commonly used by the public and to all doorways in common areas as far as practicable. Automatic doors should:
 - (a) remain open for a minimum of 5 to 10 seconds;
 - (b) have a guardrail where it opens into a route of travel (see **Figure 14**);
 - (c) have a sign showing it is an automatic door and satisfying with the requirement on luminance contrast under paragraph 10.9(10);

- (d) be operated via a manual large button control to open and be controlled by timers or sensors to close, or be equipped with sensor systems that detect approaching persons or objects for opening and closing, thereby facilitating passage and reducing the risk of accidents during operation; and
- (e) avoid the public being caught by the door.

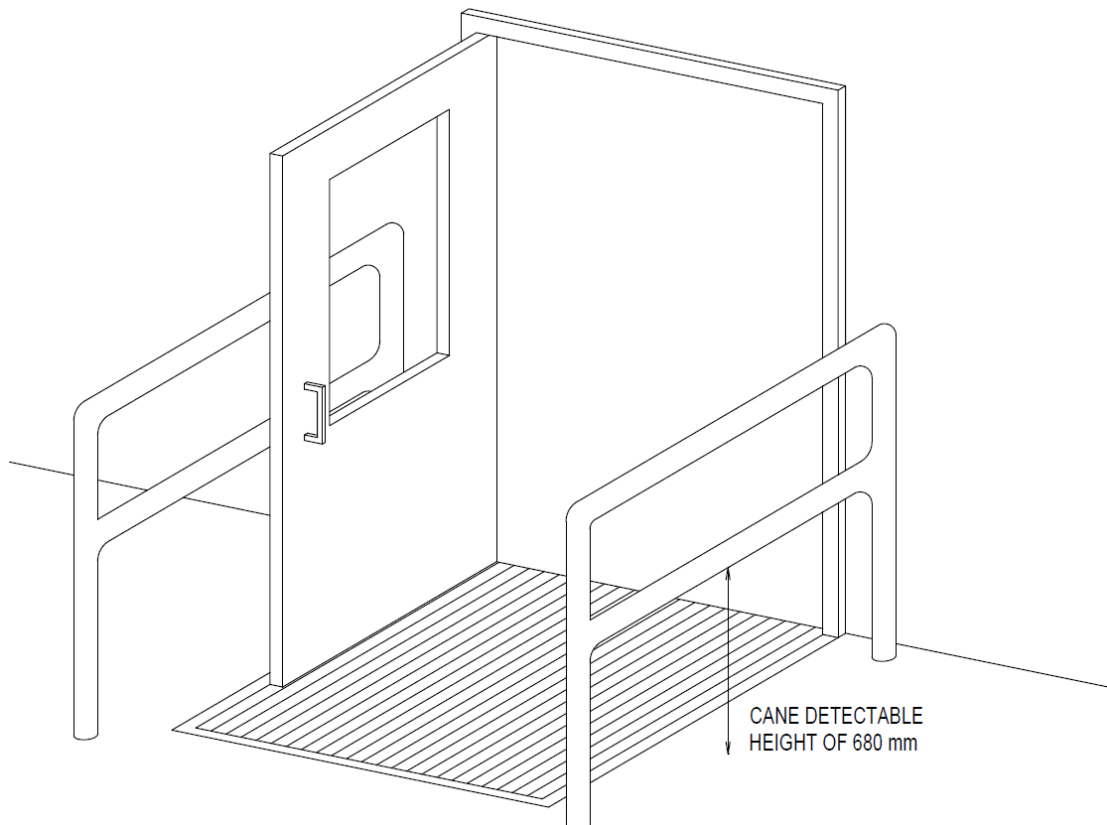


Figure 14 - Guardrails at Out-swinging Automatic Doors

Door Thresholds

- (5) If door threshold (see **Figure 11**) is provided, it should not exceed 15 mm in height, having a bevelled and round edge on each side at gradient not steeper than 1:2 to facilitate passage of wheelchairs.

CHAPTER 6 VERTICAL CIRCULATION OF A BUILDING

6.1 General

This Chapter aims to provide design requirements to ensure that all floor levels can be accessible via lifts, ramps or the like. Vertical circulation routes of buildings shall be designed, constructed, and maintained to ensure that they are easily identifiable, comprehensible and usable by all individuals. Vertical circulation encompasses ramps, dropped kerbs, steps and staircases.

MANDATORY SECTION

Obligatory Design Requirements

6.2 Ramps

Width

- (1) A ramp, which is a sloping walkway leading from one level to another, shall not be less than 1 200 mm in width.

Landing

- (2) A clear space of not less than 1 500 mm x 1 500 mm shall be provided at the head and foot of every ramp, i.e. door swing and alike shall not be allowed to swing onto the landing.

Running Slope and Length

- (3) No ramp shall be steeper than 1 in 12 gradient, except in the following situations of minor rise:

Maximum slope	Maximum length	Maximum rise
1:10 i.e., 10%	1 500 mm	150 mm
1:8 i.e., 12.5%	600 mm	75 mm

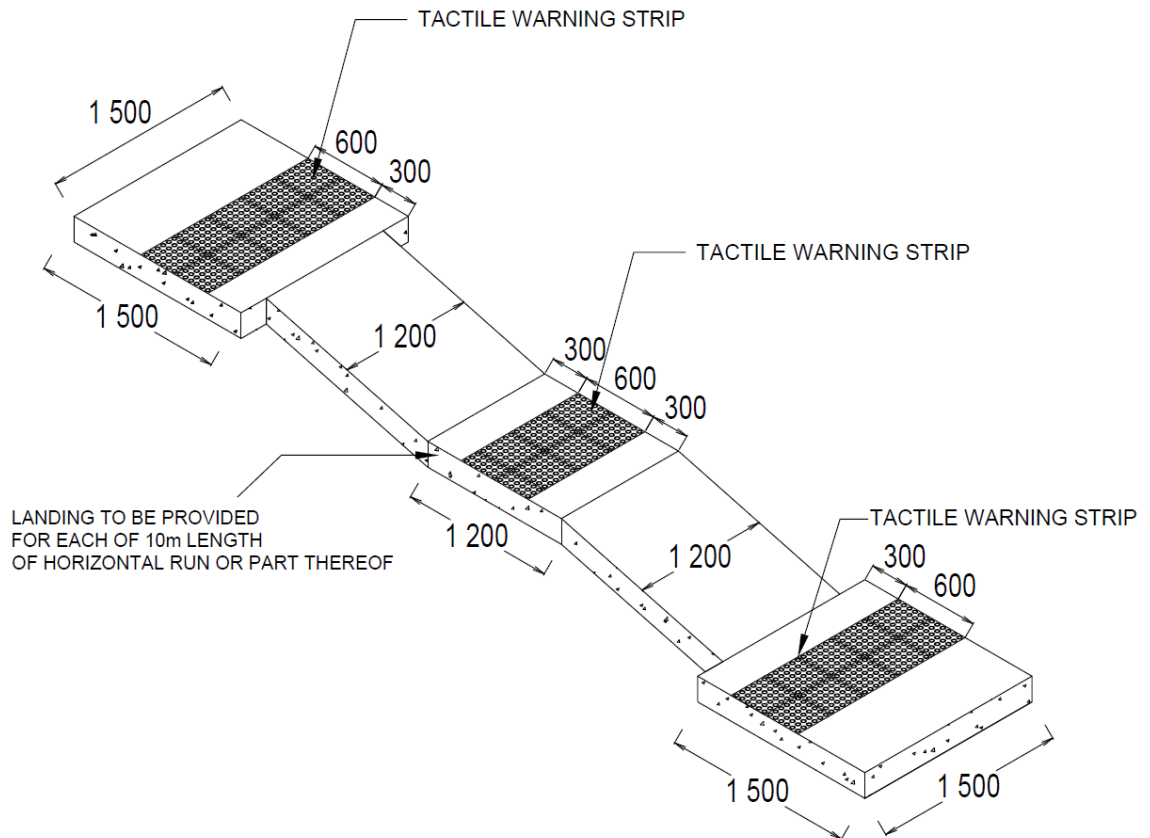
Requirements for Ramps

- (4) Combination of ramps of minor rise as stated in paragraph 6.2(3) above shall not be permitted.
- (5) If the gradient of a ramp is 1 in 20 or steeper, the ramp shall be provided with:
- (a) a landing of not less than 1 200 mm long for each 10 m length of horizontal run or part thereof; (see **Figure 15**)

- (b) tactile warning strips at the head, foot and landings; (see **Figure 15**) and
 - (c) handrails on both sides complying with the requirements in paragraph 10.2.
- (6) The requirements under paragraph 6.2(5) above shall not apply to ramp access to lift or ramp with a length less than 300 mm.

Protection and Surfaces

- (7) Any ramp with a rise greater than 200 mm, leading down towards an area where there may be vehicular traffic, shall have a railing or barrier across the full width of its lower end, and be not less than 1 500 mm from the foot of the ramp.
- (8) Raised traction strip shall be avoided.
- (9) A kerb of at least 100 mm high, or a rail 200 mm above ramp level shall be provided on both sides to prevent wheelchair from slipping over the edge.
- (10) No appliances, fixtures and fittings shall project beyond 90 mm from the surface of any wall below a level of 2 000 mm above the ramp level unless they are unavoidable, in which case they shall also be extended downwards to the ramp level or be guided by tactile flooring materials.
- (11) The floor and wall along ramps shall be in contrasting colours.
- (12) A ramp including its landing shall have a firm and slip-resistant floor surface. The slip-resistant floor surfaces shall be provided in accordance with Appendix C of this Manual.



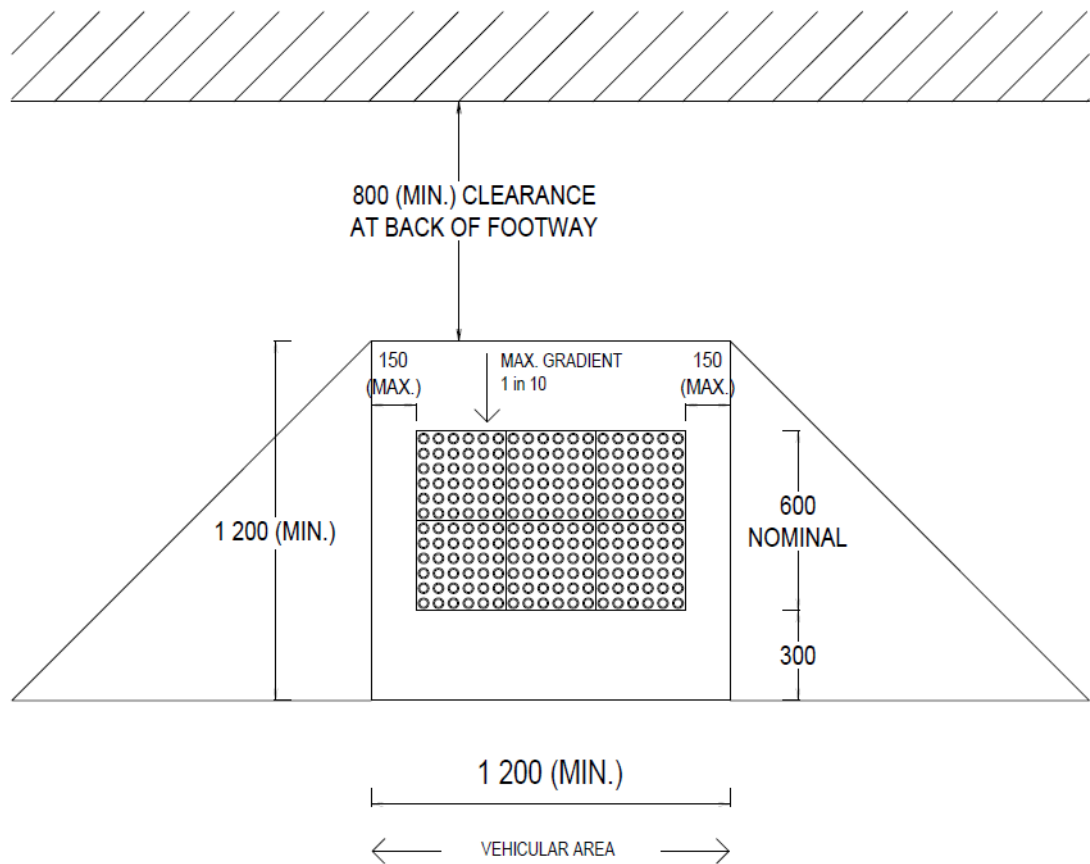
*ALL DIMENSIONS ARE IN mm

Figure 15 - Tactile Warning Strips and Landings for Ramps

6.3 Dropped Kerb

General Requirements

- (1) A dropped kerb is a ramp built on a footpath or pavement to accommodate the change in level towards vehicular areas. Changes in level at kerbs shall be by a dropped kerb as illustrated in **Figure 16**. Dropped kerb shall be provided at pedestrian crossing and at each end of the footpath of a private street or access road. Kerb separating footpath or ramp from vehicular area shall also be a dropped kerb.
- (2) Dropped kerb shall be constructed as follows:
 - (a) not less than 1 200 mm in length and 1 200 mm in width;
 - (b) with a clearance of at least 800 mm long at the back of the footway;
 - (c) ramped at a gradient not steeper than 1:10;
 - (d) with a level difference of not more than 15 mm with the vehicular areas;
 - (e) provided with a tactile warning strip at 300 mm from the vehicular areas and not more than 150 mm from the edge of the side ramp; and
 - (f) provided with a tactile warning strip of the nominal width of 600 mm at the ramp.



*ALL DIMENSIONS ARE IN mm

Figure 16 - Dropped Kerb

Location

- (3) Dropped kerb shall be so located to enable users to have an unobstructed view of traffic approaching from any direction and avoid placing planters and the like in close proximity to a dropped kerb.

Surface

- (4) Raised traction strips shall be avoided.

6.4 Steps and Staircases

Dimension and Orientation

- (1) The required staircases and the main circulation staircase in common areas of a building shall:
 - (a) for required staircases, be constructed with treads not less than 225 mm in width (measured at the centre of the flight) from the face of one riser to the face of the next riser and with risers not more than 175 mm in height;
 - (b) for main circulation staircases, be constructed with treads not less than 300 mm in width (measured at the centre of the flight) from the face of one riser to the face of the next riser and with risers not more than 150 mm in height;
 - (c) have risers built with vertical or receding face not more than 15 mm from the vertical, without a projecting nosing;
 - (d) have not more than 16 steps in any flight without the introduction of a landing;
 - (e) be provided with non-slip nosing in contrasting colour; and
 - (f) be provided with handrails on both sides complying with the requirements in paragraph 10.2.

Colour Contrast

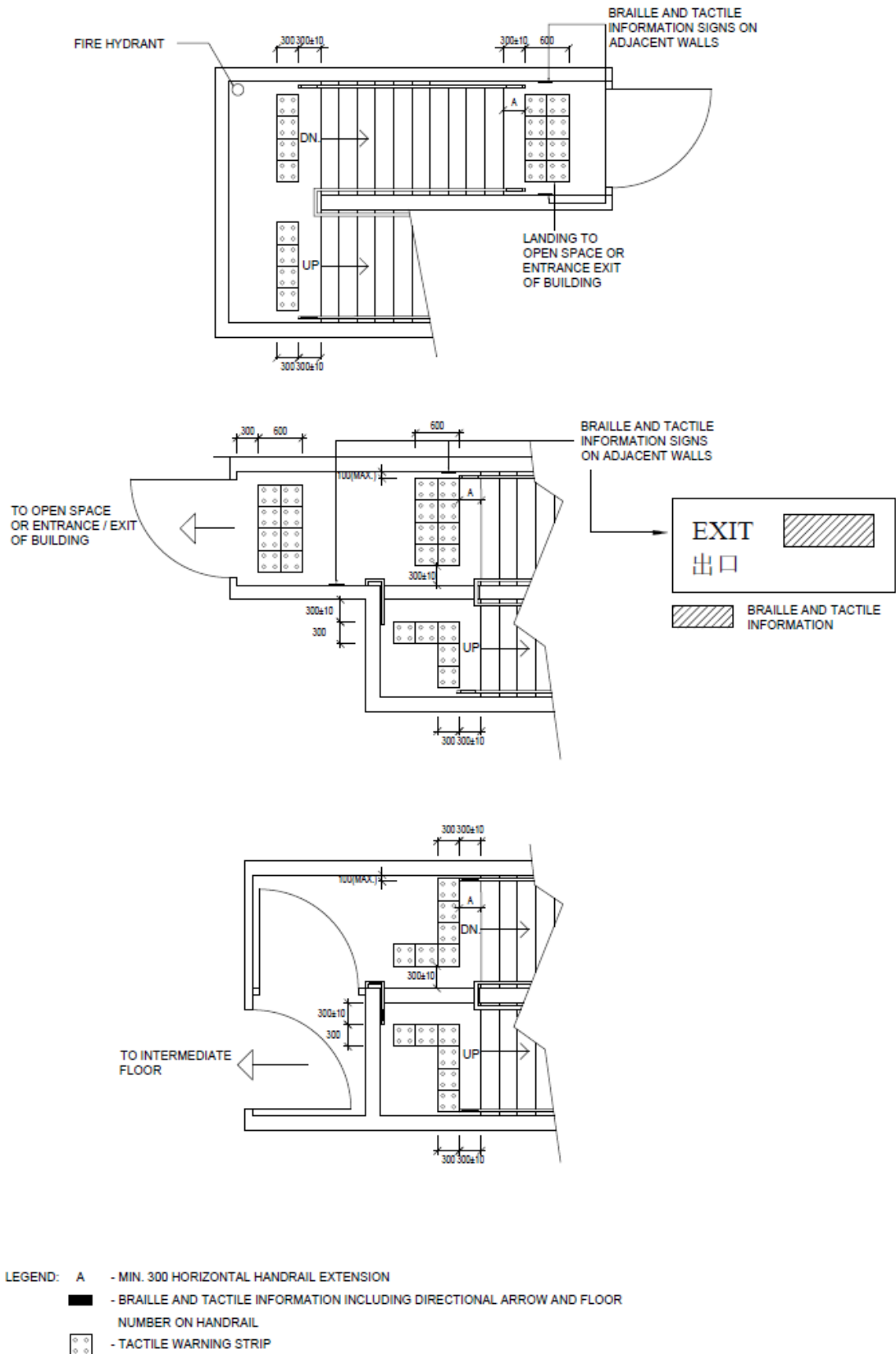
- (2) Treads and walls of a staircase shall be in contrasting colours.

Tactile Warning Strip

- (3) Tactile warning strips shall be provided at landings and at both the bottom and top ends of a staircase, regardless of the number of steps it comprises. For landings leading to a floor or those enclosed by wall, railing or balustrade, tactile warning strips of 300 mm in width shall be provided; for those leading to an open space or the entrance/exit of a building, the tactile warning strips shall be 600 mm in width (see **Figure 17**). In this case, Braille and tactile information signs shall be provided on the adjacent wall to indicate the presence of an opening. For a staircase with intermediate steps between two flights, the provision of tactile warning strips shall follow the arrangement in **Figure 17**.

Avoidance of Projection

- (4) No appliances, fixtures or fittings shall project beyond 90 mm from the surface of any wall in a staircase below a level of 2 000 mm above the treads of the staircase unless they are unavoidable, in which case they shall also be extended downwards to the level of the treads.



*ALL DIMENSIONS ARE IN mm

Figure 17 - Arrangement of Tactile Warning Strips and Handrails at Staircases

ADVISORY SECTION

Advisory Design Requirements

6.5 Ramps

- (1) A ramp should have a running slope 1:12 (8.33%) to 1:20 (5%)

Maximum slope	Maximum length	Maximum rise
1:20 i.e., 5.00%	10 000 mm	500 mm
1:16 i.e., 6.25%	6 400 mm	400 mm
1:14 i.e., 7.14%	4 200 mm	300 mm
1:12 i.e., 8.33%	1 800 mm	150 mm

- (2) Width of a ramp should be preferably at least 1 500 mm.
- (3) Tactile warning strips at the head, foot and landing of a ramp should have a minimum luminance contrast of 70% with the adjoining surfaces.
- (4) The floor and wall along a ramp should have a minimum luminance contrast of 30%.

6.6 Dropped Kerbs

Floor Surfaces

- (1) All dropped kerbs should have slip-resistant surfaces in accordance with Appendix C of this Manual.

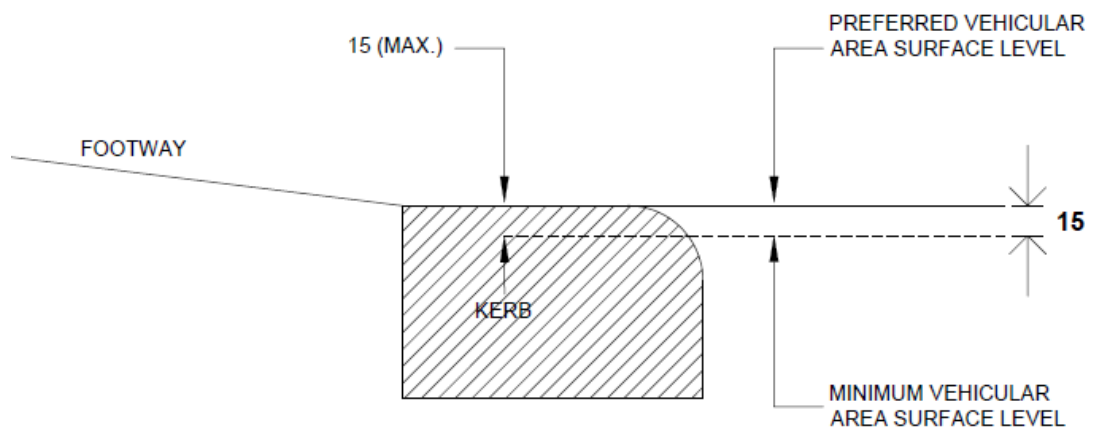
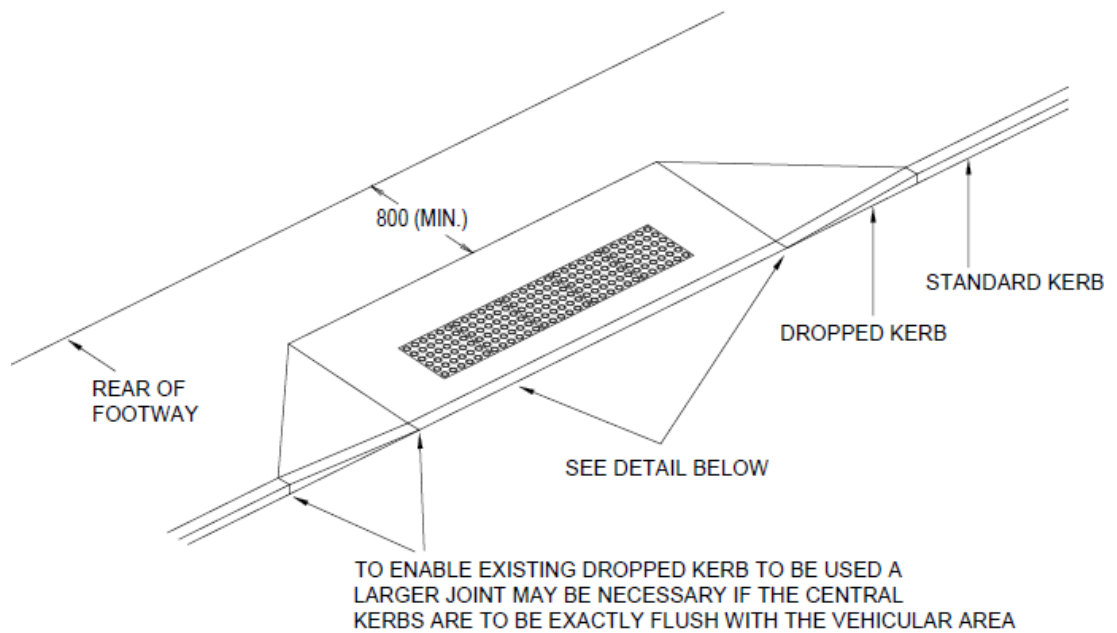
Luminance Contrast

- (2) Tactile warning strip should have a minimum luminance contrast of 70% with the adjoining surfaces.

Conjunction with Pedestrian Crossings

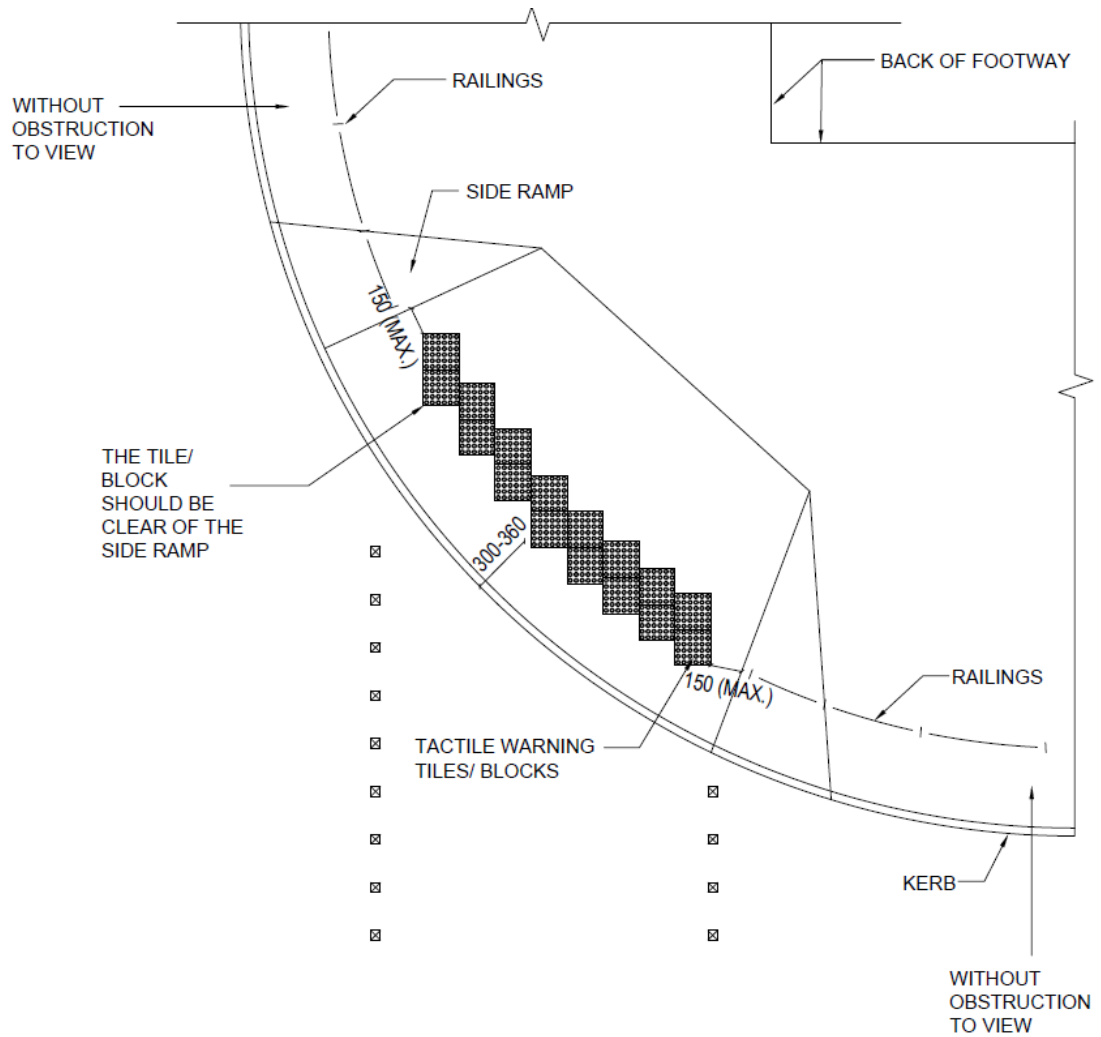
- (3) Dropped kerb should be provided where necessary and in conjunction with pedestrian crossings, which should include visible, audible and tactile crossing devices with traffic lights.

Example of dropped kerb are shown in **Figure 18**, **Figure 19** and **Figure 20**.



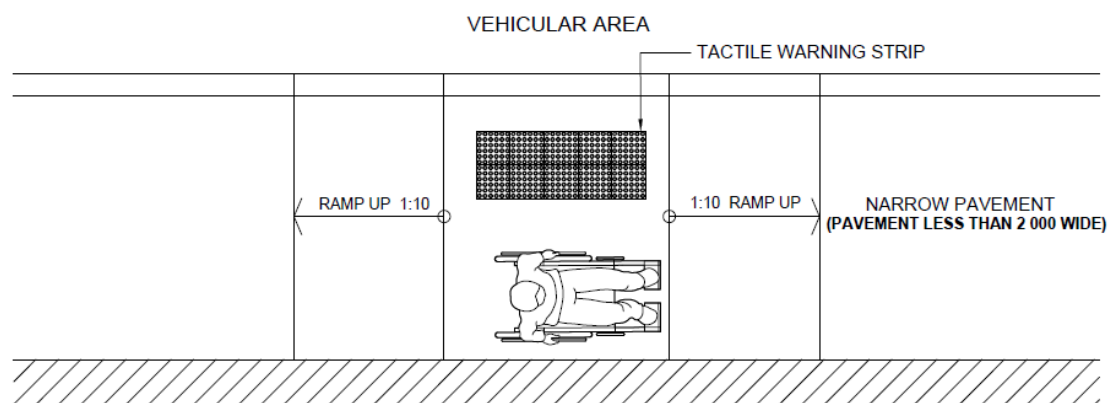
*ALL DIMENSIONS ARE IN mm

Figure 18 - Examples of Design of Dropped Kerb



*ALL DIMENSIONS ARE IN mm

Figure 19 - Dropped Kerb at Street Corner



*ALL DIMENSIONS ARE IN mm

Figure 20 - Dropped Kerb for Access Road and Narrow Pavement

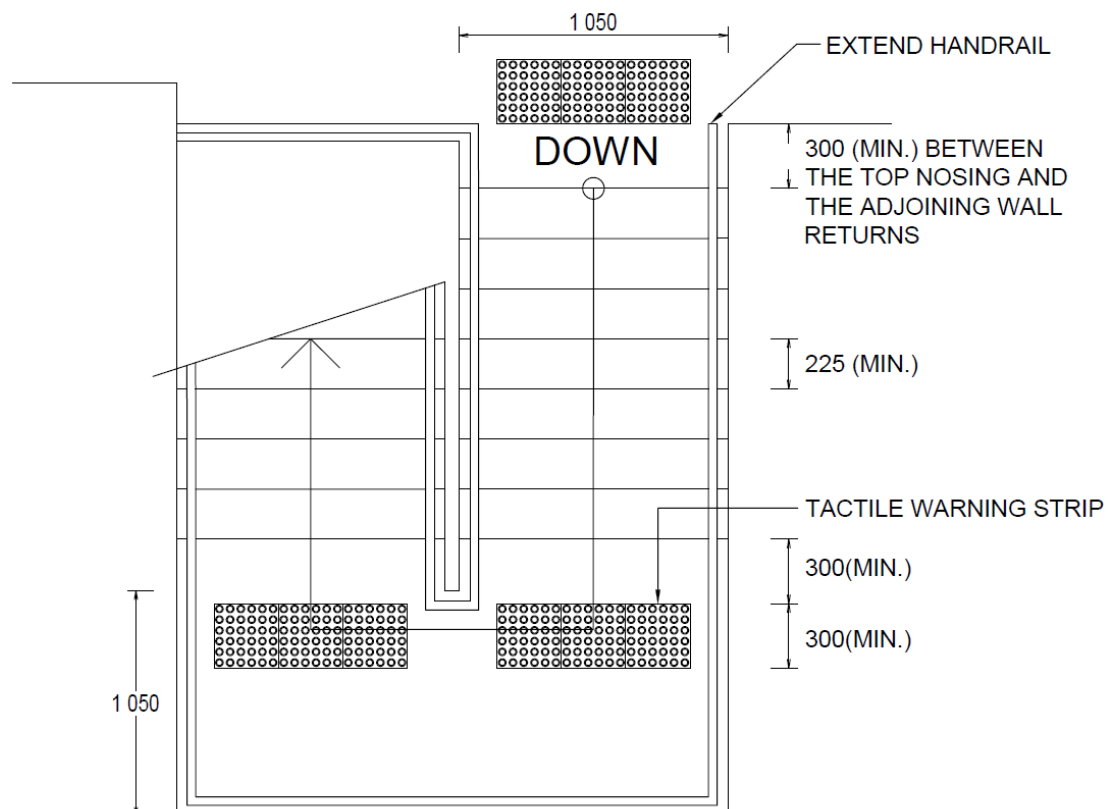
6.7 Steps and Staircases

Dimension and Orientation

- (1) For any external steps and stairs, riser should be reduced to 150 mm high and tread be increased to 300 mm wide for greater ease of use.
- (2) Individual flight should not exceed 1 800 mm in height nor a total of 12 risers.
- (3) The top nosing of any flight should be built not less than 300 mm from the point at which the adjoining wall returns (see **Figure 21**).
- (4) Winder, spiral staircase and splayed step should be avoided.

Luminance Contrast

- (5) Non-slip nosing should have a minimum luminance contrast of 30% with the adjoining surfaces.
- (6) Treads of staircase should have a minimum luminance contrast of 30% with the walls.



*ALL DIMENSIONS ARE IN mm

Figure 21 - Example of Staircase Plan for PwDs

CHAPTER 7 LIFTS AND ESCALATORS

7.1 General

This Chapter aims to provide design requirements to ensure efficient horizontal and vertical circulation by lifts, escalators and passenger conveyors within a building, with a view to safeguarding safe and accessible vertical movement between levels, specifically accommodating individuals with mobility challenges, including those with ambulatory disabilities and wheelchair users.

MANDATORY SECTION

Obligatory Design Requirements

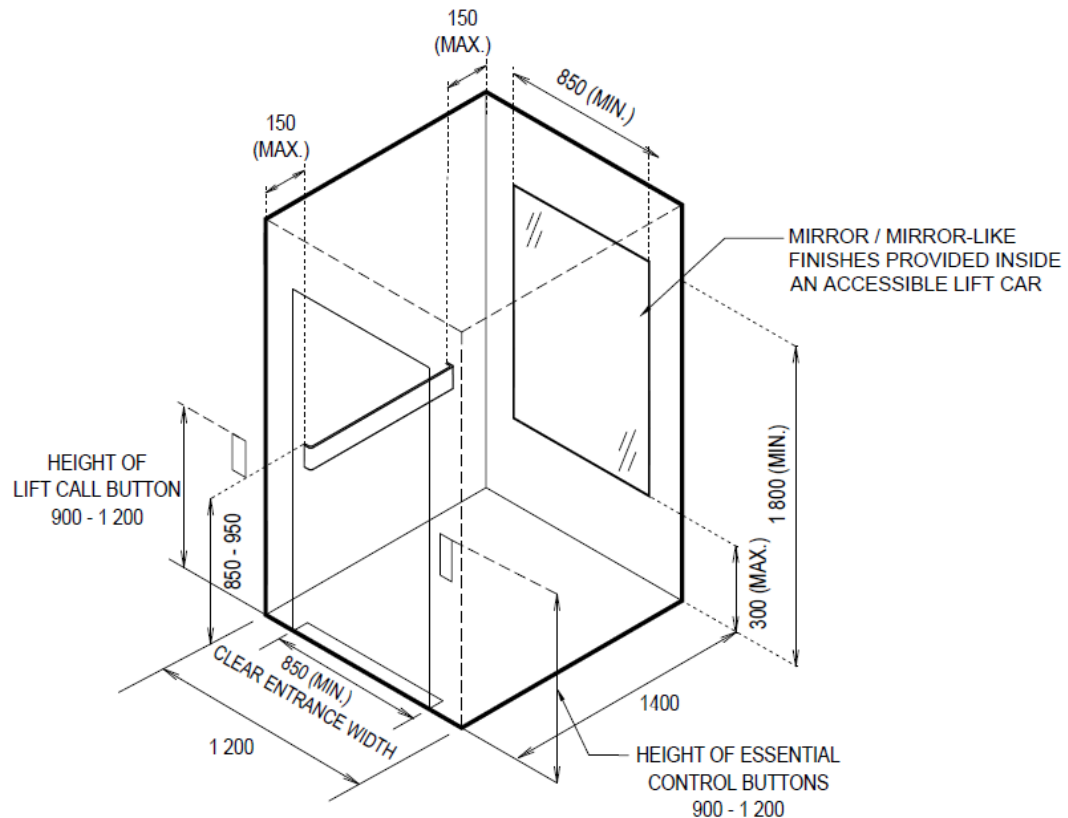
7.2 Lifts

Accessible Lifts

- (1) Every floor of a building shall be accessible by at least one accessible lift which shall fully comply with all the obligatory design requirements as stipulated in this chapter and have direct access to main lift lobby.

All other passenger lifts in the building shall comply with paragraph 7.2(5) to paragraph 7.2(18).

- (2) A lift shall have minimum internal car dimensions of 1 400 mm deep x 1 200 mm wide, with a minimum clear entrance width of 850 mm, and shall have handrails extending to within 150 mm of the corners at the rear and sides of the car. The top of the gripping surface of the handrails shall be at a height of 850 mm to 950 mm, with a space of 30 mm to 50 mm between the handrails and wall (see **Figure 22**).



*ALL DIMENSIONS ARE IN mm

Figure 22 - Accessible Lift

- (3) Where there are more than three lifts in a building, access shall be provided to every floor by at least one lift having minimum internal car dimensions of 1 500 mm x 1 400 mm (either wide or deep) with a minimum clear entrance width of 850 mm.
- (4) A non-breakable mirror or mirror-like surface shall be provided inside an accessible lift car to allow visual feedback for wheelchair users when reversing backwards into the lift lobby. Such provision shall have a clear width of not less than 850 mm. The bottom edge of the mirror should be installed not more than 300 mm above the floor level of the lift car; and the top edges shall be installed not less than 1 800 mm above the floor level of the lift car.

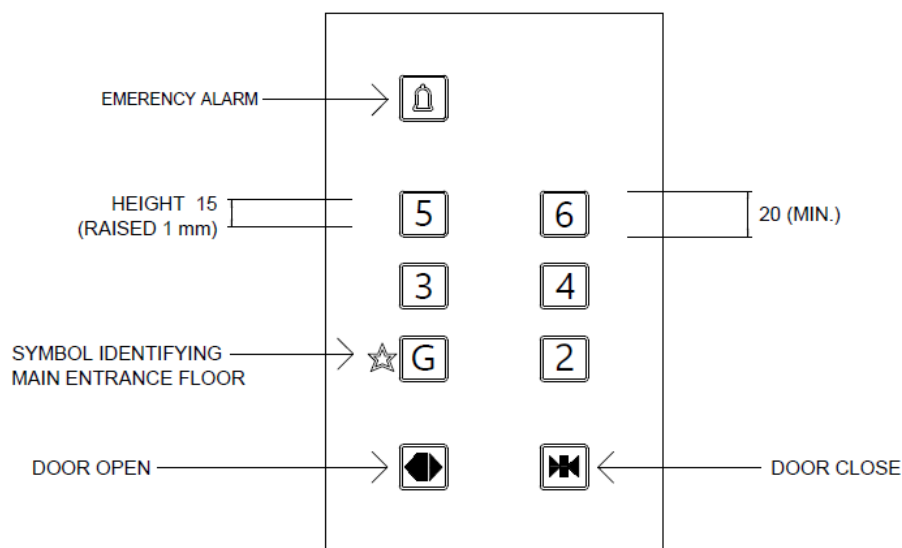
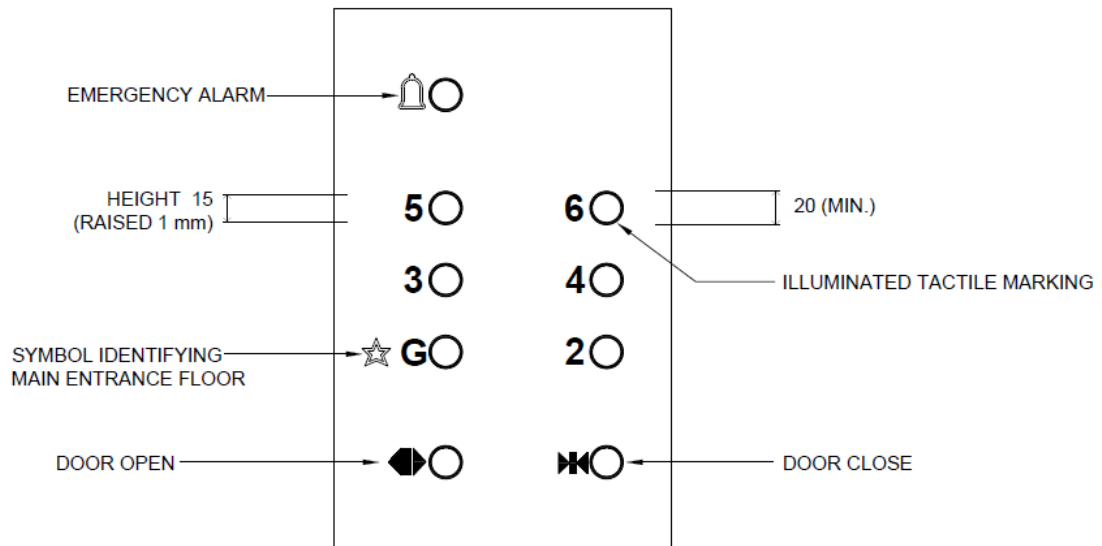
Lift Doors

- (5) Lift car doors and landing doors shall be of the horizontally sliding type, power-operated and automatically controlled.
- (6) An audible signal shall be provided to signify the closing of the doors to alert persons.
- (7) A detection device shall be provided to re-open the lift doors in the event of hitting any obstacle.

- (8) The detection device shall be positioned at a height of between 500 mm and 600 mm above the floor of the lift car.

Lift Control Buttons

- (9) Essential lift control buttons including floor numbering buttons, emergency alarm push button and door opening push button in the lift car shall not be less than 900 mm and not more than 1 200 mm above the floor of the car.
- (10) Lift call buttons at the lift halls shall not be less than 900 mm and not more than 1 200 mm above the floor of the finished floor level of the lift hall.
- (11) Provision of secondary control panel for over-spilled floor numbering buttons is always acceptable.
- (12) All lift control buttons shall have a minimum dimension of 20 mm (see **Figure 23**).
- (13) Braille and tactile markings shall be placed either on or to the left of the control buttons.
- (14) Such Braille and tactile markings shall be in Arabic numerals and/or symbols. Tactile markings shall have a minimum dimension of 15 mm high and be raised 1 mm minimum.
- (15) The tactile marking of the push buttons for the main entrance floor shall be identified with a symbol in a star shape (see **Figure 23**).
- (16) The emergency alarm push button shall be in a tactile bell shape (see **Figure 23**).



*ALL DIMENSIONS ARE IN mm

Figure 23 - Tactile Graphic for Lift Control Buttons

Emergency Call Button in Lifts

- (17) An emergency alarm push button together with a buzzer, an indication light for acknowledgement and an intercom shall be provided in each lift car and be connected to the building management office or the caretaker's office. The building management office or the caretaker's office shall be equipped with a buzzer, indication light(s) (one for each lift) and an intercom connected to the lift car(s).

- (18) The indication light for acknowledgement shall be in the form of a blinking light adjacent to the intercom speaker and a notice “When light blinks, it indicates your emergency call has already been received. Please be patient and wait for the rescue.” in English and Chinese shall be provided next to the blinking light. This system shall be powered by an emergency electricity supply system in the event of power failure.

7.3 Indication and Notification for Accessible Lifts

Indication and Notification for Accessible Lifts

- (1) An illuminated visual indicator and an audible signal shall be provided at the lift entrance to indicate the arrival of the lift car and its direction of travel. The audible signal shall sound once for UP direction and twice for DOWN direction, and shall be activated before the arrival of the lift. The audible signal can be broadcasted from a device in the lift car as an alternative provided it can be heard from the lift lobby.
- (2) Tactile and Braille floor designations shall be provided on the jambs on both sides of each lift entrance, by means of Arabic numerals, minimum 60 mm high, raised 1 mm, and at 1200 mm above the finished floor level.

Indication in Accessible Lift Cars

- (3) Illuminated visual indicators shall be provided to indicate the direction of travel and the position of the car.
- (4) Characters on the position indicator shall have a minimum height of 50 mm.
- (5) An audio indication of the stopping floor shall be provided in English, Cantonese and Putonghua. The volume of audio indication in an accessible lift shall be not less than 35 dB(A) and not more than 80 dB(A), measured at a height of 1 800 mm above the finished floor level at the farthest point in the lift car from the annunciator. The sound of the audio indication should have a frequency between 300 Hz and 3 000 Hz.

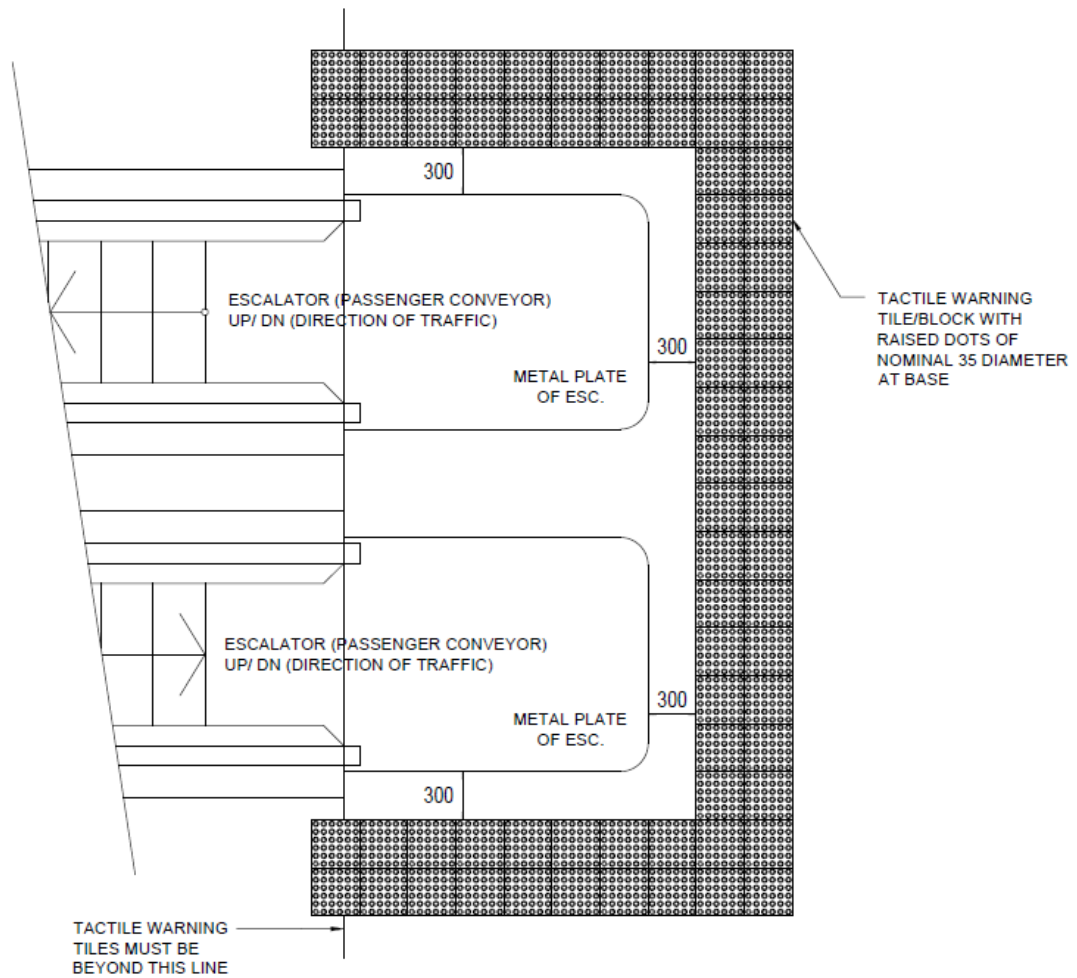
Identification of Accessible Lifts in Lift Lobbies

- (6) If a building contains some lifts that do not comply with this Chapter, then each of those lifts that do comply shall be identified at each landing served, by not less than one international symbol for accessibility.
- (7) If all the lifts in a building comply with this Chapter (including a building with only one lift), then at least one international symbol for accessibility shall be provided at each lift lobby on the entrance levels of the building.

7.4 Escalators and Passenger Conveyors

Tactile Warning Strips

- (1) Tactile warning strips shall be provided at both bottom and top ends of an escalator. The provision of tactile warning strips is illustrated in **Figure 24**.
- (2) Tactile warning strips shall be provided at both ends of a passenger conveyor. The provision of tactile warning strips is illustrated in **Figure 24**.



*ALL DIMENSIONS ARE IN mm

Figure 24 - Tactile Warning Strip to Escalator or Passenger Conveyor

- (3) Escalator step edge shall be clearly defined with colour in sufficient luminance contrast with the steps, e.g. yellow warning strips or alternative solutions.

ADVISORY SECTION

Advisory Design Requirements

7.5

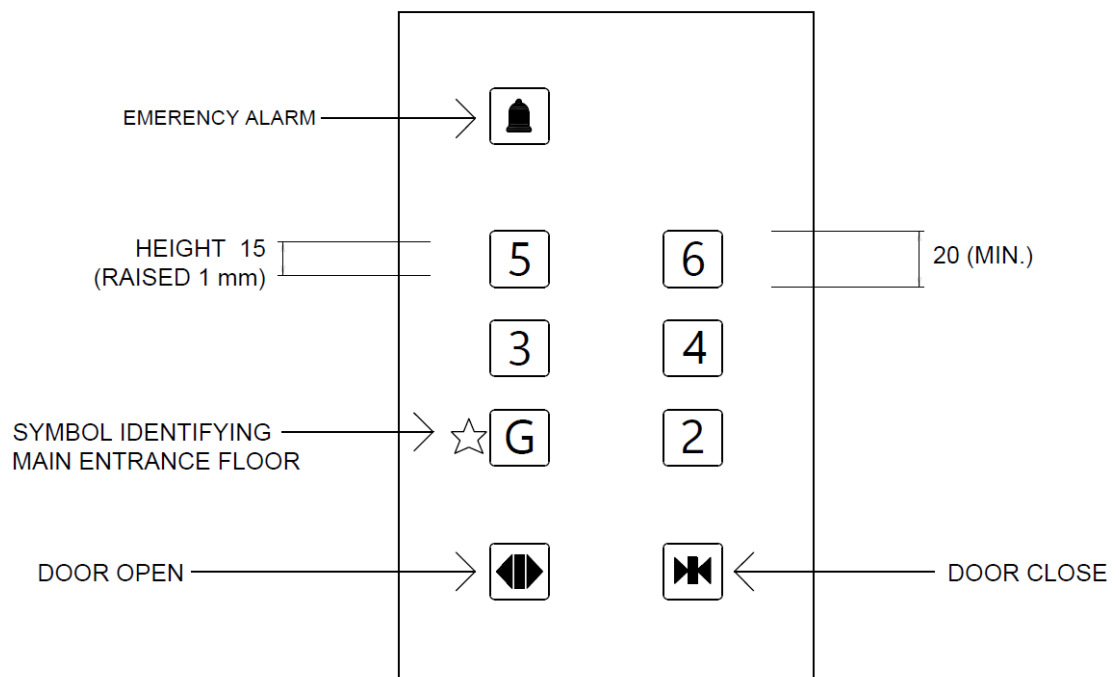
Lifts

Provision of Accessible Lifts

- (1) At least two accessible lifts should be provided in large scale specified premises as defined in Chapter 2.

Lift Control Buttons

- (2) The graphics for tactile markings for open-door and close-door push buttons, emergency alarm button, and main entrance level are shown in **Figure 25** for reference.
- (3) Call button panels should be provided at both sides of door openings.
- (4) In addition to the essential lift control buttons required under paragraph 7.2(9), another set of essential lift control buttons should be installed on one of the side walls of lift car, preferably in the middle and on the right side from the entering direction. For two lift door openings design, the additional set of essential lift control buttons should be on any side wall.
- (5) Lift control buttons in lift cars should have a minimum luminance contrasts of 50% with the background surfaces.
- (6) All lift control buttons including the numbers on floor numbering buttons should be back-lit.
- (7) The function to cancel lift calls by multi-tapping the floor numbering buttons should not be adopted in accessible lifts.
- (8) All lift control buttons should be installed with electrostatic discharge coating or conductive material to minimise the risk of electrostatic discharge shock.



*ALL DIMENSIONS ARE IN mm

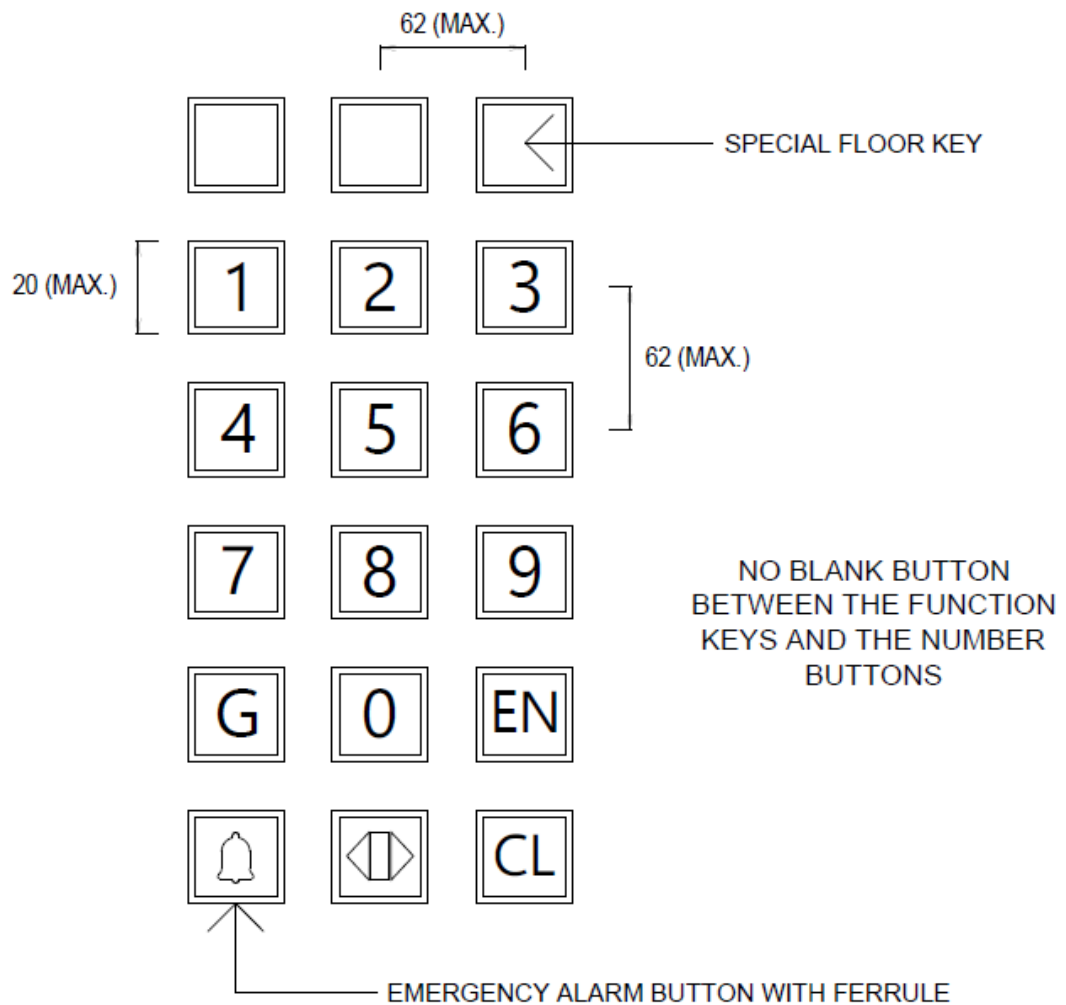
Figure 25 - Tactile Graphic for Lift Control Buttons

Keypad design

(9) In cases where difficulties are encountered to fully comply with the obligatory requirements of installation of lift control buttons in high-rise buildings, keypad control device in conjunction with a conventional lift control panels in lifts for PwDs should be provided. Proposed standardised positions of buttons for keypad control device are shown in **Figure 26** for reference.

(10) The keypad should:

- (a) have control buttons of minimum dimension of 20 mm;
- (b) have Braille and tactile markings following the standard as stipulated in paragraph 7.2;
- (c) be installed between 900 mm and 1 200 mm from finished floor level of the lift car;
- (d) have adequate luminance contrast between the tactile markings on the buttons and the background;
- (e) have voice announcement and visual indication of the floors registered; and
- (f) have instruction for the keypad control panel in the form of Braille or voice announcement beside the panel.



*ALL DIMENSIONS ARE IN mm

Figure 26 - Proposed Standardised Position of Buttons for Keypad Control Device

Illumination Level

- (11) The level of illumination at the car controls, platform, car threshold, landing sill and lift landing should be 150 lux minimum.

Wheelchair Turning Space

- (12) An unobstructed wheelchair turning space of 1 500 mm x 1 500 mm should be provided in front of accessible lift car door.

Lift Doors

- (13) Where planning allows, lift cars may be provided with opposing doors to allow a wheelchair user to leave without having to reverse.

- (14) The minimum time for lift doors to remain fully open at a landing should be 3 seconds.
- (15) An additional detection device positioned at a height of 200 mm above the floor of the lift car should be provided.

Depth of Accessible Lift Car

- (16) The clear depth of an accessible lift car should be not less than 1 500 mm.

Tactile Warning

- (17) Tactile warning tiles should be placed in front of the lift door of the accessible lift complying with this Chapter at each landing for ease of identification by persons with visual impairment.

Control

- (18) A separate call button for the accessible lift complying with this Chapter should be installed on each floor to ensure that the accessible lift will stop at the called floor. Priority of attendance to call for the accessible lift should be given to this button.

Indication in Lift Cars

- (19) The visual indicators and audio indication required under paragraph 7.3(3) to paragraph 7.3(5) should be provided in all other passenger lift cars.

Visual Communication System

- (20) A visual communication system such as the provision of monitor for video call in the lift car should be provided to all passenger lifts to assist persons with hearing impairment in lift car to communicate with the building management office or the caretaker's office in emergency situation.

7.6 Destination Control System for Lifts

- (1) As an alternative to providing lift control buttons inside the lift car as stipulated in paragraph 7.2, the adoption of a Destination Control System (DCS) should comply with the following:

- (a) The keypad control of the DCS to be installed in a conspicuous place in the lift hall should have the built-in design for the use of PwDs.
- (b) The keypad should be of standard telephone keypad type with raised dot for easy orientation at number 5 key. An appropriate Braille notice “lift button for the visually impaired” should be provided.
- (c) The keypad should have an “international wheelchair” button to activate the system specially designed for the PwDs and a push button with a tactile symbol in a “star” shape for identifying the main entrance floor for the use of visually impaired users. The height of the tactile symbol should not be smaller than the height of the number in the keypad. Upon activation of the system by pressing the abovementioned buttons, only accessible lifts should be assigned.
- (d) Tactile Arabic numerals with suitable design (e.g. 15 mm high) and normally engraved by 1 mm instead of Braille should be provided on the keypad.
- (e) The user should be able to know that the DCS has been operated. The call registration should be confirmed by a visible and audible signal. The audible signal should be given on every individual call registration even if the call is already registered. Also, a speech output system (in Cantonese, Putonghua and English) of the keypad for verification of the designated floor should be provided.
- (f) All essential buttons including emergency alarm push button, intercom controls, and the door opening push button should be provided inside the lift car.
- (g) The keypad at the lift hall and all the essential buttons in the lift car should be located between 900 mm and 1 200 mm above the finished floor level of the lift hall or the floor of the lift car in compliance with paragraph 7.2(10).
- (h) The design of the keypad should comply with the international standards for the use of PwDs such as the European Standard EN 81-70:2018, Safety rules for the construction and installations of lifts - Part 70: Accessibility to lifts for persons including persons with disability. Adequate signal and indication for use should be provided including an illuminated visual indicator and an audible signal.
- (i) Braille and tactile plate indicating the user’s instruction should be mounted on the wall adjacent to the keypad reachable by the visually impaired users. The instruction should be in both English and Chinese.
- (j) When the “wheelchair” button is pushed, the gong sound and voice announcement (in Cantonese, Putonghua and English) lift complying with paragraph 7.3(1) should be activated on the arrival of the accessible lift. To allow the PwDs to catch the lift, time allowed for keeping the lift door open and the response time for the keypad should be suitably extended.

- (k) Tactile guide path should be provided to guide the visually impaired users to the keypad in the lift hall and then to the designated accessible lift.
- (2) Pursuant to paragraph 7.6(1) above, for the adoption of DCS without any keypad control in lift hall (e.g. DCS being activated by access cards), which fails to comply with the requirements in paragraph 7.6(1)(a), (b), (c), (d) and (i), the following additional requirements should be complied with:
 - (a) A public information or service counter complying with paragraph 10.6 should be provided near the lift hall.
 - (b) At least one accessible card reader should be provided at the entrance of the lift hall or the turnstile-controlled passage and be located between 900 mm and 1200 mm above finished floor level. The card reader should have a minimum luminance contrast of 30% with the background finishes. Braille and tactile information signs should be provided adjacent to the accessible card reader to indicate clearly its location. Other easily discernible design of the accessible card reader, such as back-lit, should be favourably considered.
 - (c) Notwithstanding Table 2, tactile guide paths in compliance with paragraph 10.3 should be installed from the public information or service counter to the accessible card reader(s) provided according to paragraph 7.6(2)(b), and from such accessible card reader(s) to the designated accessible lift(s). The end point of the tactile guide path should be not more than 300 mm in front of the accessible card reader and the designated accessible lift(s).
 - (d) Notwithstanding the non-provision of keypad control, an “international wheelchair” button should be provided in a conspicuous place in the lift hall. Upon pressing the “international wheelchair” button, audible announcements (in Cantonese, Putonghua and English) should be provided to guide the PwDs on tapping the access card on the accessible card reader, and to inform the registered floor, the number of the assigned lift and the direction to the assigned lift. Only accessible lifts should be assigned.
 - (e) The “international wheelchair” button provided according to paragraph 7.6(2)(d) should be located between 900 mm and 1 200 mm above finished floor level and adjacent to each accessible card reader. The minimum dimension of such button should be not less than 30 mm, with minimum of 50% luminance contrast to the background, and with raised tactile and Braille information.
 - (f) The enquiry hotline of the building management office should be shown on the access card in Braille and tactile markings. Specifications of Braille cells are provided in **Figure 56**.

7.7 Escalators and Passenger Conveyors

- (1) Adequate distance for horizontal movement of steps should be provided at the top and bottom of escalator to ensure that passengers – especially those leaving the escalator – have adequate space and time to board and alight safely.
- (2) The angle of inclination of escalator should not exceed 30°.
- (3) Moving handrail should be in luminance contrast with the background.

Audio Indicator

- (4) Clear audible indicators for direction (e.g. consistent and distinct sounds or signals) should be provided at both ends of escalators and passenger conveyors. These indicators should produce two distinguishable sounds and be installed on the left side of the entry direction.

Alternative Access

- (5) Where there is an accessible lift that provides alternative accessible route for PwDs, it should be near to, and clearly shown by a sign posted at the entry to the escalator.

Speed

- (6) The maximum speed of all escalators or passenger conveyors other than the public services escalators or passenger conveyors as defined in Chapter 2 should be not more than 0.5 m per second.

7.8 Vertical Lifting Platform

Automatic Door

- (1) The automatic door should have a minimum clear entrance width of 900 mm.
- (2) The automatic door should have single door or 2-door design and the kinetic energy of the door should not exceed 10J.

Platform size

- (3) The platform should have minimum size of 1 100 mm (wide) x 1 500 mm (deep).

Safety Barrier

- (4) The safety barriers should be not less than 1 100 mm in height.
- (5) The safety barriers should have provision of a flip-up ramp if a door is not provided, to act as a safety barrier and serves as an accessible ramp for wheelchair users.

Control

- (6) Lifting platform buttons and emergency call button should comply with Chapter 7.2. For automatic swing doors, the control buttons at the landing side should be installed at the wall adjacent to the door opening such that the user will not be interfered by the door movement. For a door swinging into an area with internal corner, the buttons should be located at least 600 mm from the internal corner of a room and the button should have a clear 1 000 mm minimum distance from the swing of the door. The hold-to-run button should be of a minimum dimension of 50 mm.
- (7) The vertical lifting platform should be self-operable by users holding a common key.
- (8) Call button and CCTV should be provided at every landing to a vertical lifting platform. The call button should activate an intercom and video contact with a caretaker's office or public information/service counter provided under paragraph 10.6 for assistance.
- (9) Automatic homing device should be installed so that in the event of a power failure, the platform lift will descend to the entrance level.

Grab Bars

- (10) Grab bars complying with paragraph 10.2(3) should be placed at a height of 900 mm from the finished floor level and be fixed on both sides and at the rear of the platform (door(s) excepted). The vertical distance between the grab bar and the control buttons should not be less than 100 mm.

Maximum Travel and Loading

- (11) The maximum travel should be 7 000 mm.
- (12) The rated load of the lifting platform should not exceed 500 kg.

Sign and Installations

- (13) A sign to indicate the rated load should be provided at a prominent position next to the lift control buttons inside the lift.
- (14) Clear instructions for use and emergency alarm should be provided inside the lifting platform.

Undertaking from Owners

- (15) The owner should undertake quality management and maintenance to facilitate the reliable use of the lifting platform. The platform should be kept in service during the opening hours of the premises and routine servicing/repair should be arranged in closed times.

CHAPTER 8 DESIGN REQUIREMENTS FOR SPECIFIC BUILDING TYPES

8.1 General

This Chapter provides specific design requirements in addition to those set out in other chapters of this Manual for specific building types, including auditorium and related facilities, hotels, hostels and guesthouses.

In respect of auditorium for audience and backstage facilities of theatre, cinema, concert hall, sports stadium, games hall and other entertainment related premises as well as lecture hall and conference hall, the obligatory and advisory design requirements are set out in paragraph 8.2 and paragraph 8.4 respectively.

In respect of guest rooms designated for the use of PwDs in hotels, hostels and guesthouses, the obligatory and advisory design requirements are set out in paragraph 8.3 and paragraph 8.5 respectively.

MANDATORY SECTION

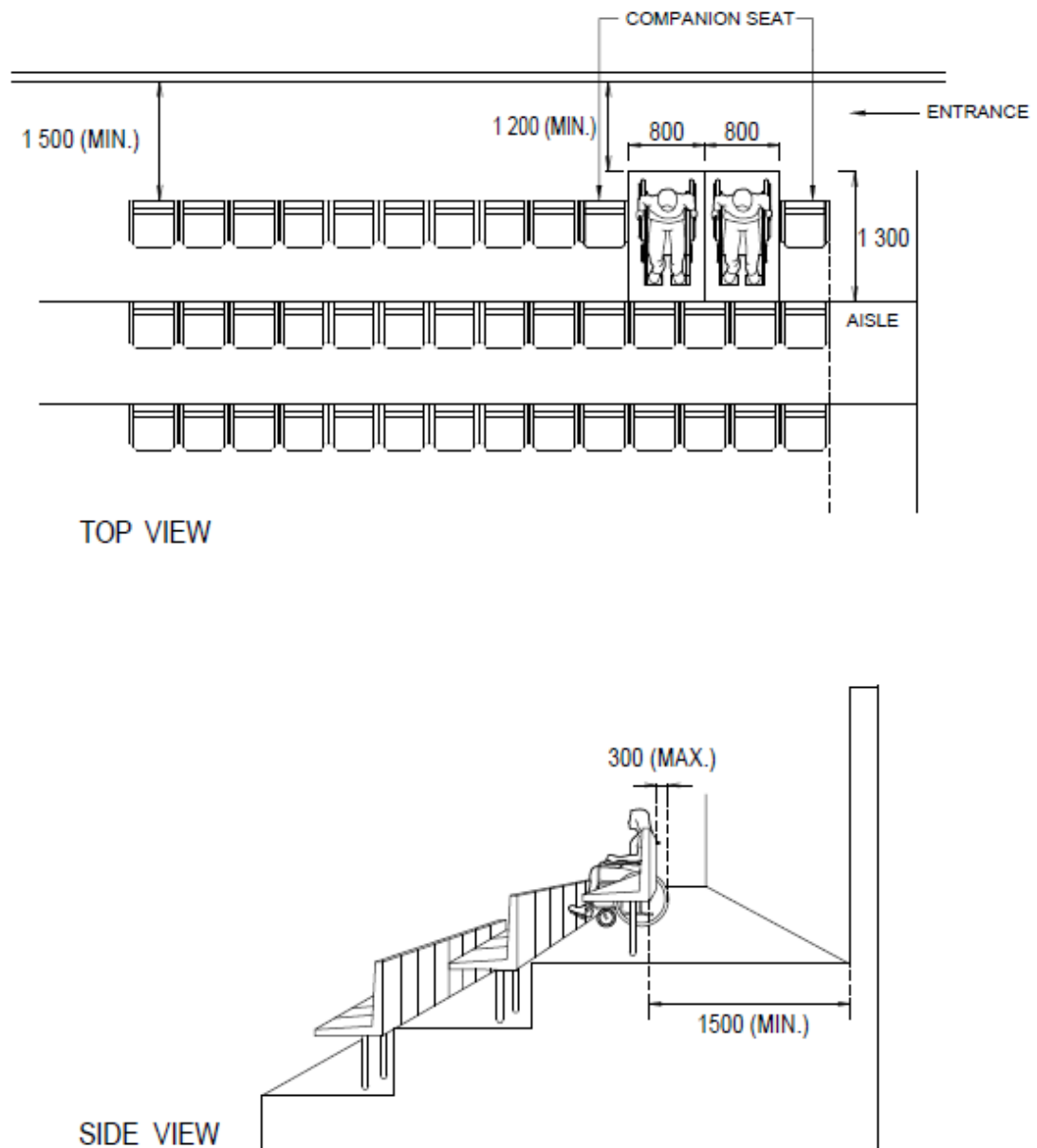
Obligatory Design Requirements

8.2 Auditorium and Related Facilities

Wheelchair Spaces

- (1) A minimum of four wheelchair spaces shall be provided at spectator level in the auditorium with not more than 800 fixed seats. For auditorium with more than 800 fixed seats at spectator level, two wheelchair spaces shall be provided for every 400 fixed seats and any part thereof. (For example, at least six wheelchair spaces shall be provided if there are 900 fixed seats.) The spaces shall be grouped in pairs (not less than two) and also not separated from the seats for other audiences. Each wheelchair space shall have unobstructed line of vision to the stage areas at which the performance or activities of the event takes place, and be of minimum size of 800 mm x 1 300 mm, with the side of 800 mm facing towards the stage podium or screen. The minimum width of the passage leading to a wheelchair space from an accessible entrance of the auditorium shall not be less than 1 500 mm (see **Figure 27**).
- (2) Companion seats shall be provided adjacent to the wheelchair space, which can be located on either side or both sides to allow flexible seating arrangement. The minimum ratio of companion seats to wheelchair spaces shall be at 1:1 with companion seats matching the dimensions of standard audience seating. These companion seats can be designed to be removable.

- (3) Readily removable seats can be installed in wheelchair spaces when the spaces are not occupied by wheelchair users.



*ALL DIMENSIONS ARE IN mm

Figure 27 - Wheelchair Spaces in Auditorium

Access for Wheelchair Users

- (4) The auditorium, the stage, backstage facilities, changing rooms, rehearsal rooms, dressing rooms, rest rooms, toilets and shower rooms shall be accessible to wheelchair users. If there is an access connecting two or more of them, an equivalent accessible route shall be provided for wheelchair users, e.g. by a ramp in compliance with [Chapter 6](#) or an accessible lift in compliance with [Chapter 7](#).

- (5) When one or more dressing room are provided in the premises, a space of not less than 1 500 mm x 1 500 mm shall be provided for the manoeuvring of wheelchairs in at least one dressing room or 5% of the total number of dressing rooms, whichever is greater.

Braille and Tactile Fire Exit Maps and Tactile Warning Strips

- (6) Braille and tactile fire exit maps as shown in **Figure 58** shall be provided at all entrances to the auditorium.
- (7) Tactile warning strips complying with paragraph 6.4(3) shall be provided at both the top and bottom ends of the staircases leading to the stage.
- (8) For large scale specified premises, tactile warning strips shall be installed at the top and bottom landings of aisle staircases.

Seating Plate

- (9) A seat number plate with Arabic and Braille numerals shall be installed at the top or back of every seat at the spectator level in the auditorium. The seats immediately next to the gangways shall also be provided with Braille and tactile information showing the letter/number of the row. Specifications of Braille cells are provided in **Figure 56**.

8.3 Hotels, Hostels and Guesthouses

Minimum Provision for Accessible Guestrooms

- (1) A minimum of two accessible guest rooms with full facilities for PwDs shall be provided in a hotel, hostel and guesthouse. Two such accessible rooms shall be provided for every 100 guest rooms and any part thereof. (For example, a total of four such accessible rooms shall be provided if there are 150 guest rooms.)
- (2) Visual door bells shall be installed at a prominent and easily accessible location in accessible guest rooms.
- (3) The bathroom and shower facilities serving an accessible guest room shall comply with paragraph 9.3.

ADVISORY SECTION

Advisory Design Requirements

8.4 Auditorium and Related Facilities

Wheelchair Spaces

- (1) Removable seats should be provided at accessible spectator levels in an auditorium for the use by large group of wheelchair users for special functions/occasions. Adequate numbers of removable seats to accommodate sixteen wheelchair spaces should be provided in the auditorium with not more than 800 fixed seats. For the auditorium with more than 800 fixed seats, adequate numbers of removable seats to accommodate eight wheelchair spaces should be provided for every additional 400 fixed seats and any part thereof.
- (2) Eight wheelchair spaces should be provided at spectator levels in an auditorium with not more than 800 fixed seats. For auditorium with more than 800 fixed seats, four wheelchair spaces should be provided for every additional 400 fixed seats and any part thereof.

Access for Wheelchair Users

- (3) Access should be provided to the low and high levels of the auditorium.
- (4) Wheelchair spaces should be provided at the central areas of the spectator level accessible with aisle in the auditorium.

Provisions for Hearing Impaired and Visually Impaired

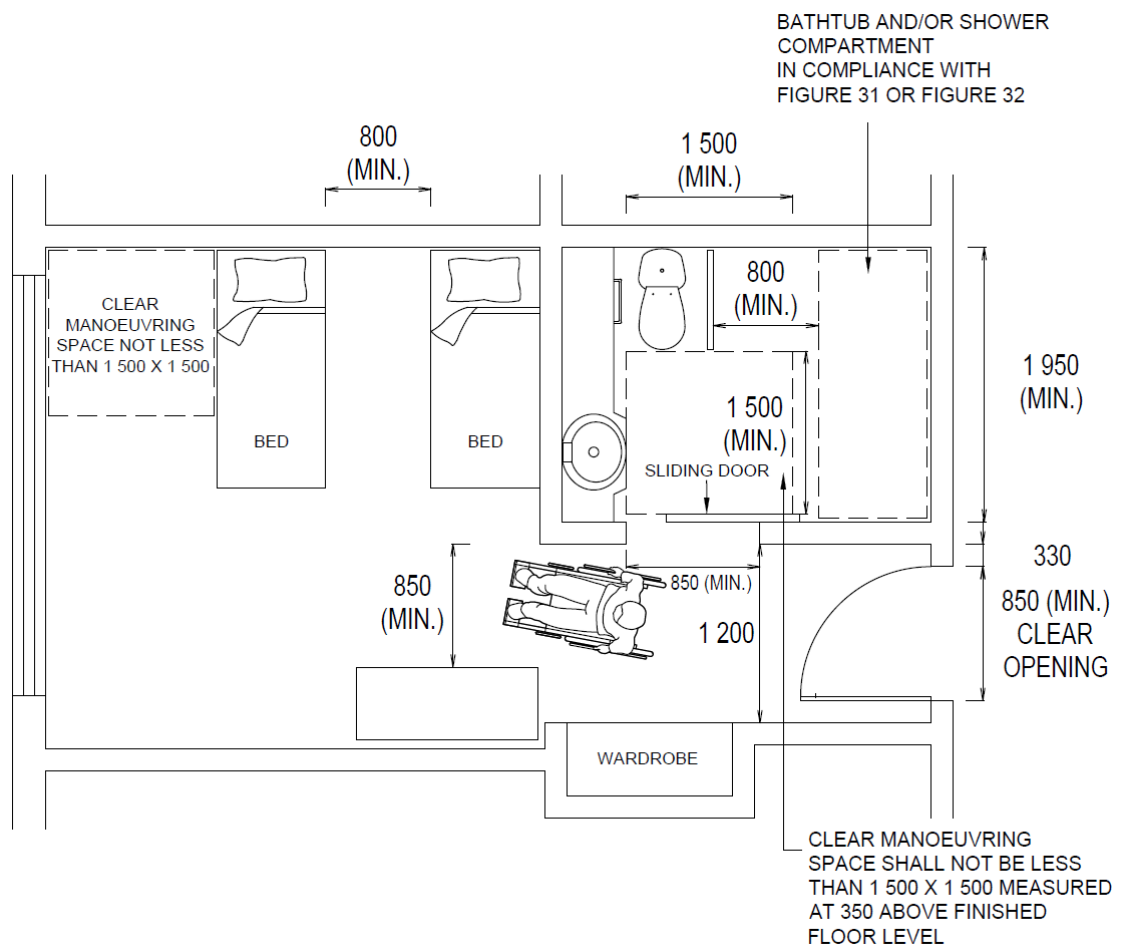
- (5) Sound enhancement system should be provided at the designated areas for persons with hearing impairment.
- (6) Visual display of subtitles should be provided at an appropriate location.
- (7) A room with a view of the performance for “audio description” should be provided for the visually impaired person.

8.5 Hotels, Hostels and Guesthouses

Provisions for Accessible Guestrooms

- (1) A typical guest room layout is shown in **Figure 28** and typical bathroom and shower compartment are shown in **Figure 31**.
- (2) For every 100 guest rooms and any part thereof, in addition to accessible guest rooms, a minimum of two guest rooms located near lift lobby should be provided with Braille and tactile fire exit maps at the back of guest room doors, and Braille and tactile information for main lamp switch controls.

- (3) Where planning allows, the accessible guest rooms required under paragraph 8.3(1) should be provided with the facilities in paragraph 8.5(2).
- (4) Where planning allows, the guest rooms in paragraph 8.5(2) above should be provided with facilities for accessible guest rooms in paragraph 8.3(1).
- (5) For floors with guest rooms provided under paragraph 8.5(2) above, Braille sign should be installed on the wall adjacent to the open side of each door leading to an exit staircase to indicate such information. The sign should be placed at 900 mm to 1 500 mm above the finished floor level.
- (6) A sign with guest room number in raised Arabic numerals/characters and Braille should be installed on the wall adjacent to the door lock of all guest rooms. The sign should be placed at 900 mm to 1 500 mm above the finished floor level.



*ALL DIMENSIONS ARE IN mm

Figure 28 - Typical Guest Room Layout

CHAPTER 9 SANITARY PROVISION

9.1 General

This Chapter aims to provide the design requirements for all people to use the sanitary facilities independently as far as practicable, including toilets, changing rooms, bathrooms and shower compartments.

MANDATORY SECTION

Obligatory Design Requirements

9.2 Toilets and W.C. Cubicles

Provision of Accessible W.C. Cubicle

- (1) There shall be at least one accessible W.C. cubicle on a floor, or in that part of a floor designed for access by the PwDs where the total number of W.C. cubicles provided on that floor or in that part of a floor is 20 or less, or 2 where the total number exceeds 20. This paragraph shall not apply to domestic buildings and the domestic parts of composite buildings or where there is no toilet provided on the particular floor.
- (2) Where the accessible W.C. cubicles are within a toilet with multiple cubicles, the minimum number of such accessible W.C. cubicles to be provided for each sex shall be based on the total number of W.C. cubicles for each sex on that floor or in that part of a floor designed for access by PwDs.
- (3) The W.C. cubicles required by this Manual shall be deemed to be included in the number of soil fitments required under the Building (Standards of Sanitary Fitments, Plumbing, Drainage Works and Latrines) Regulations (Cap. 123 sub. leg.) and Part VII of the Education Regulations (Cap. 279 sub. leg.).

Accessible Unisex Toilet

- (4) Where toilet is provided on a floor, at least one shall be designed as an accessible unisex sanitary facility for use by persons of both sexes and access to which does not necessitate traversing an area reserved for one sex only. It shall be designed for general use and include adequate circulation space for wheelchair users in accordance with the obligatory requirements as set out in this Chapter.

Location of Accessible W.C. Cubicle

- (5) W.C. cubicles shall be accessible:
- (a) directly from a public corridor; and
 - (b) where situated within a toilet containing other W.C. cubicles, through a clear space not less than 1 500 mm x 1 500 mm immediately in front of the compartment to allow manoeuvrability or by direct approach where no turning of the wheelchair is necessary.

Design of Accessible W.C. Cubicle

- (6) The accessible W.C. cubicle shall not be less than 1 500 mm x 1 750 mm in area and the clear manoeuvring space within the cubicle shall not be less than 1 500 mm x 1 500 mm measured at 350 mm above finished floor level and the cubicle shall have in it a water closet at a height not less than 380 mm and not more than 450 mm, measured to the top of the toilet seat. Water closets shall be equipped with a back support such as a seat lid and seats shall not be spring-actuated. A typical toilet is shown in **Figure 29**.

Flushing Controls

- (7) Flushing control shall be mounted on the wide side of the cubicle at a height between 600 mm and 1 050 mm above the finished floor level and shall be hand-operated or automatic. Hand-operated controls shall be capable of being operated with one hand and shall not require tight grasping, pinching or twisting of the wrist. The force required shall not be greater than 22 N.

Wash Basins

- (8) The toilet shall be provided with a wash basin mounted with the rim not higher than 750 mm above the finished floor level. A clearance of 550 mm shall be maintained from the finished floor level to the bottom of the apron.
- (9) Tap for wash basin shall be automatic or of lever control type without spring loading, subject to the approval of the Water Supplies Department. Tap shall not require tight grasping, pinching or twisting of the wrist. The operating force required shall not be greater than 22 N.

Toilet/Cubicle Doors

- (10) Door shall be installed with push-type or lever-type handles and capable of being easily opened/closed by one hand. Any door fastening shall be capable of being released from the outside in the event of an emergency. No coin box shall be affixed to the door of the cubicle.

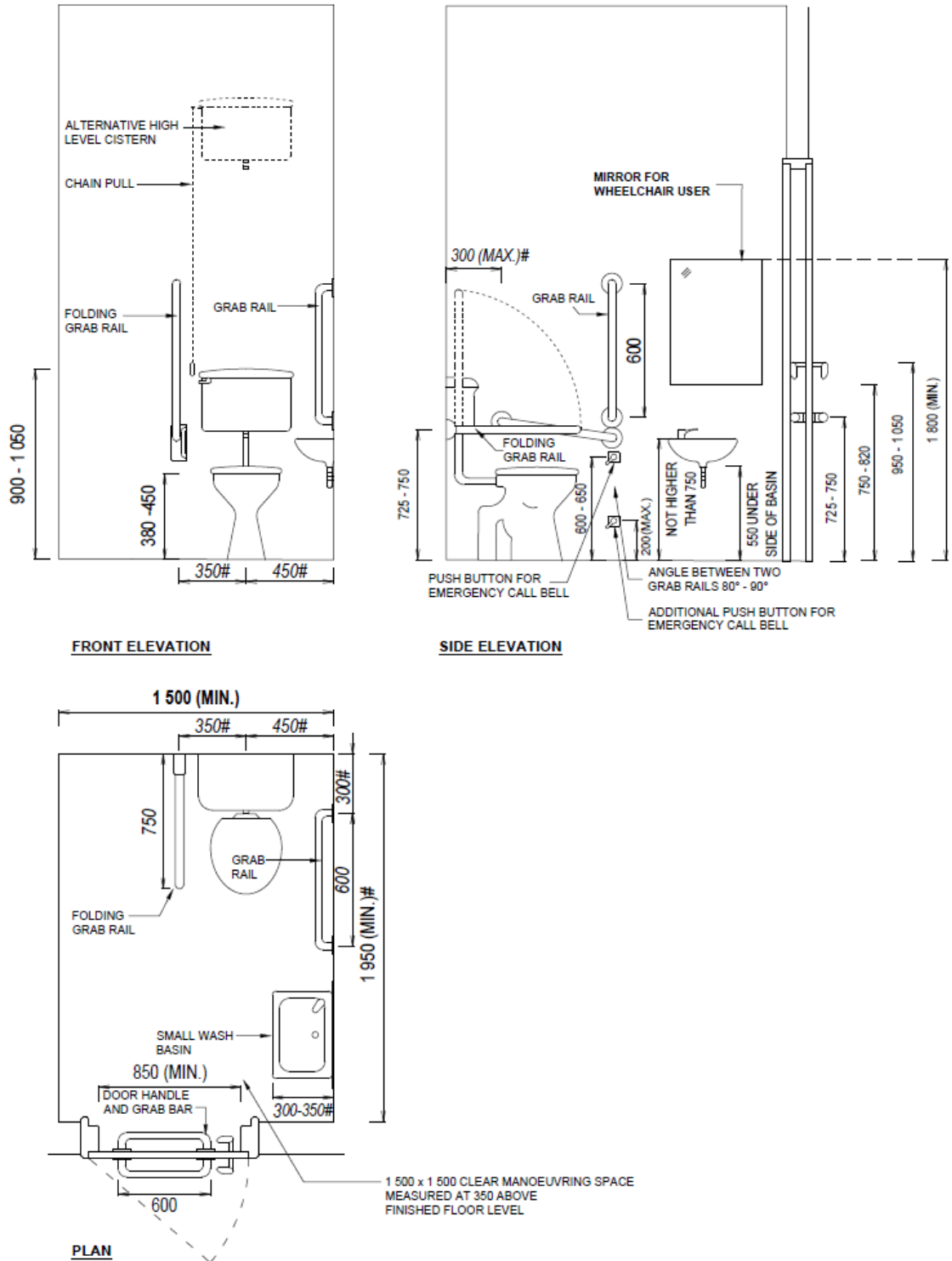
- (11) Where automatic door is provided for accessible toilet, audible signals shall be provided to indicate the opening action, closing action, closure and locking of the door.

Grab Rails

- (12) There shall be at least two grab rails which shall not be less than 32 mm and not more than 40 mm in external diameter and shall be fixed on the wall leaving a grip space of not less than 30 mm clear of the mounting wall. The two grab rails constructed in one continuous piece is acceptable. The length of grab rail shall not be less than 600 mm.
- (13) There shall be one grab rail fixed on each of both the inner and outer surfaces of the cubicle door; which shall not be less than 32 mm and not more than 40 mm in external diameter. The grab rail shall have a grip space of not less than 30 mm clear of each door surface.
- (14) There shall be one folding grab rail on the wide side of the cubicle adjacent to the water closet at a height between 725 mm and 750 mm above the finished floor level when lowered from the wall. Simple instructions in English, Chinese and Braille on how to unfold the rail should be affixed to the wall. The grab rail, folding grab rail and wash basin shall be capable of carrying a static load of 150 kg.
- (15) The grab rail shall not rotate within its fixing fittings.

Emergency Call Bell

- (16) An emergency call bell shall be provided in an accessible toilet to emit audible or visible alarm signal upon activation. The audible or visible alarm signal shall be readily noticeable for summoning assistance for the person in the accessible toilet. The alarm shall be installed outside the toilet or a buzzer shall be fitted in the caretaker's office (see **Figure 29**).
- (17) The emergency call bell shall be equipped with a weatherproof push button for activating the alarm. The push button shall be installed below the vertical grab rail inside the W.C. cubicle adjacent to the water closet at a height between 600 mm and 650 mm above the finished floor level. A notice "Emergency Call" in both English, Chinese and Braille shall be fitted next to the emergency push button.
- (18) An additional weatherproof push button shall be installed below the push button provided under paragraph 9.2(17) and shall be located not more than 200 mm above the finished floor level (see **Figure 29**).



*ALL DIMENSIONS ARE IN mm
ADVISORY DESIGN REQUIREMENTS

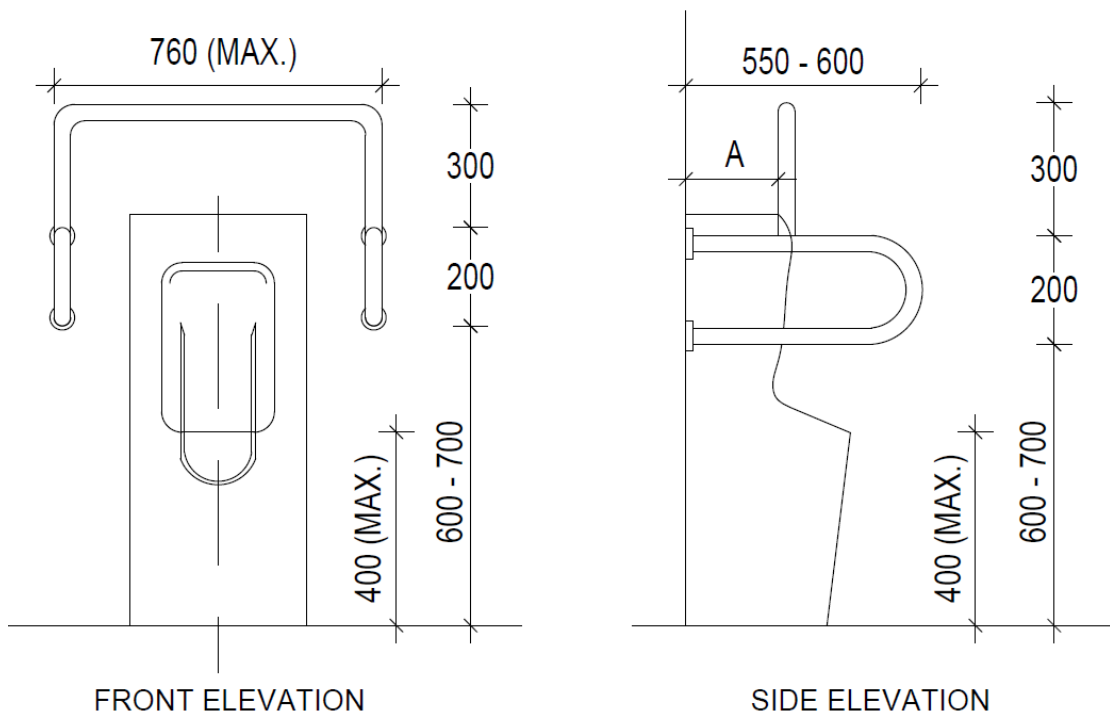
(Opposite – Handed Layout is Acceptable)

Figure 29 - Accessible Toilet

Urinals

(19) If more than one urinal is provided, at least one urinal shall:

- (a) have a clear levelled space of not less than 800 mm wide x 1 500 mm deep in front; and
- (b) be a wall hung urinal with a front rim not higher than 400 mm, and have both horizontal and vertical grab rails of not less than 32 mm and not more than 40 mm in external diameter, where the vertical grab rails shall be provided on both sides and apart from each other by a horizontal distance of not more than 760 mm, and the horizontal grab rail shall be at a height of not less than 1 100 mm and not more than 1 200 mm above the finished floor level for use by persons with ambulant disabilities (see **Figure 30**).



A DISTANCE BETWEEN THE WALL AND THE VERTICAL GRAB RAIL SHOULD COMMENSURATE WITH THE DIMENSIONS OF THE URINAL TO FACILITATE THE USE OF THE GRAB RAILS

*ALL DIMENSIONS ARE IN mm

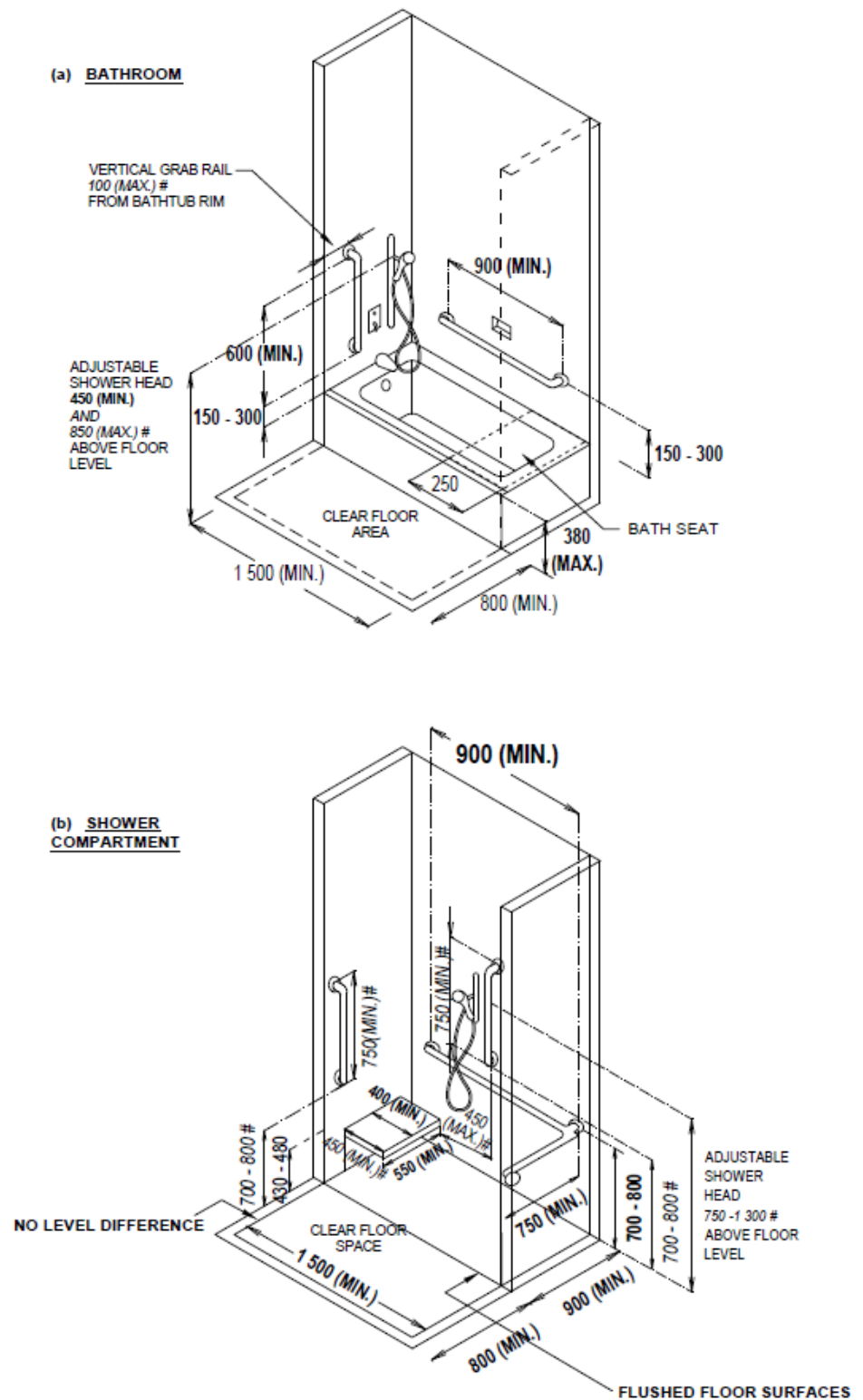
Figure 30 - Urinal for Use by Persons with Ambulant Disabilities

9.3 Bathrooms and Shower Compartments

This section is applicable to accessible bathrooms and shower compartments required under paragraph 8.3(1).

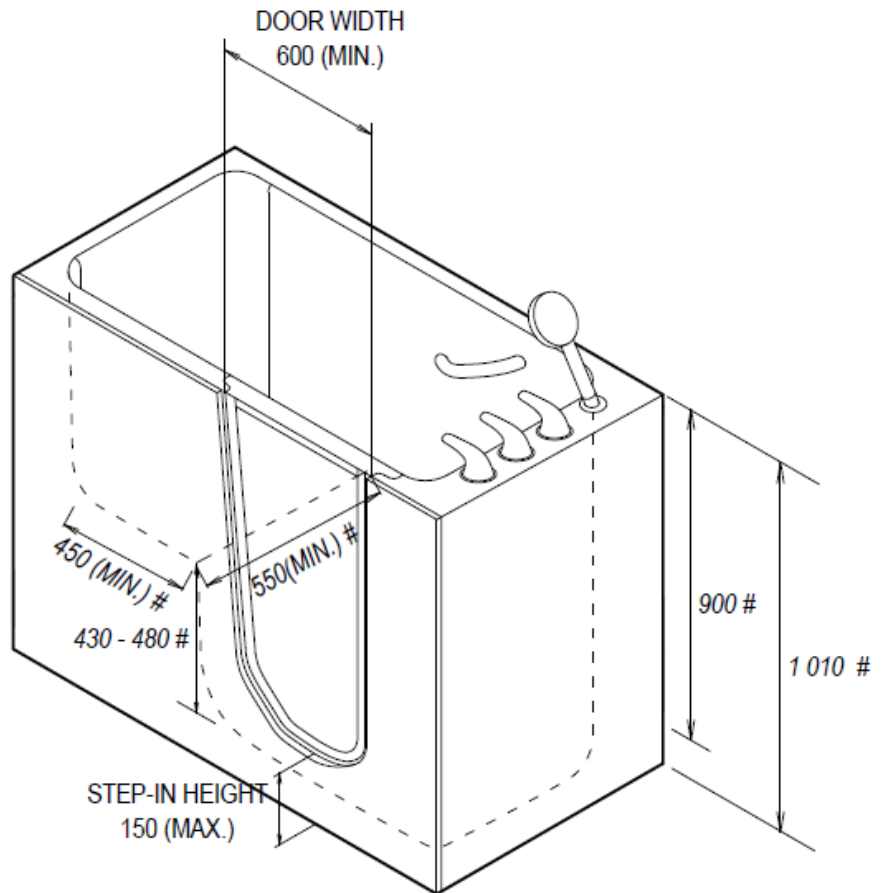
Bathtubs

- (1) There shall be a clear floor space of not less than 1 500 mm x 800 mm in front of the bathtub (see **Figure 31**).
- (2) The bathtub shall be provided with a seat of not less than 250 mm in width (see **Figure 31**).
- (3) The bathtub shall have a maximum height of 380 mm, or shall be an accessible walk-in bathtub with an outward-swinging access door of not less than 600 mm in clear width. The height between the bottom edge of the access door and the adjacent clear floor space (step-in height) shall not be more than 150 mm . A typical accessible walk-in bathtub is illustrated in **Figure 32**.



ADVISORY DESIGN REQUIREMENT

Figure 31 - Examples of Bathroom and Shower Compartment



ADVISORY DESIGN REQUIREMENT

Figure 32 - Typical Section of Accessible Walk-in Bathtub

Grab Rails for Bathtub

- (4) Grab rails shall:
- (a) not rotate within their fixing fittings;
 - (b) have a diameter between 32 mm and 40 mm and have a grip space of not less than 30 mm clear from the wall;
 - (c) be at least 900 mm long, installed horizontally or slanting at an angle not exceeding 20 degrees along the length of the bathtub and at a height between 150 mm and 300 mm above the rim of the bathtub; and
 - (d) be at least 600 mm long, installed vertically at the plug end of the bathtub adjacent to the clear floor space with the lower end 150 mm to 300 mm above the rim of the bathtub.

Taps and Controls of Bathtub

- (5) Taps and other controls shall:
- (a) have lever type handles at least 75 mm long from the centre of rotation to the handle tip;
 - (b) be installed at the plug end of the bathtub; and
 - (c) be not more than 450 mm above the rim of the bathtub.

Shower Heads

- (6) Shower heads shall:
- (a) be of the hand-held type;
 - (b) be provided with a hose not less than 1 500 mm in length; and
 - (c) be provided with a wall mounting bracket to allow use in a fixed position.
- (7) Where shower heads are mounted on a vertical bar, the bar shall:
- (a) have a minimum length of 500 mm with the lower end not less than 450 mm above the finished floor level;
 - (b) be installed so as not to obstruct the use of grab rails referred to in paragraph 9.3(4); and
 - (c) be so mounted to be able to carry a static load of 150 kg in case they are mistakenly used as a grab rail.

Shower Compartments

- (8) Shower compartments shall have internal dimensions of not less than 1 500 mm x 900 mm. The minimum clear floor space in front of the shower entrance shall be 1 500 mm x 800 mm with the 1 500 mm dimension parallel to the shower entrance.

Grab Rails for Shower Compartments

- (9) Grab rails for shower compartments shall:
- (a) comply with paragraph 9.3(4)(a) and paragraph 9.3(4)(b);
 - (b) be L-shaped or two bars arranged in L-shaped configuration and not be less than 750 mm by 900 mm in length;
 - (c) be installed at a height between 700 mm and 800 mm from the shower floor; and
 - (d) be capable of carrying a static load of 150 kg

Roll-in Compartments

- (10) The floor of the roll-in shower compartment shall be flush with the surface of the adjacent clear floor space in front of the shower entrance to ensure the smooth movement of users.

Shower Seats

- (11) The shower seats shall:

- (a) have a rounded edge and be self-draining;
- (b) be installed on the wall next to the taps and controls;
- (c) not be less than 550 mm in width and 400 mm in depth; and
- (d) be installed at a height between 430 mm and 480 mm from the top of the seat to the finished floor level.

ADVISORY SECTION

Advisory Design Requirements

9.4 Toilets and W.C. Cubicles

Provision of Accessible Toilet in Specified Premises

- (1) Accessible toilets should be provided in specified premises as defined in Chapter 2 at a minimum ratio of one accessible toilet for every 15 water closets.

Unisex Facilities

- (2) W.C. cubicles should, where possible, be unisex and accessible from a corridor so that they can be used by either sex with assistance from members of the opposite sex, if necessary.
- (3) If two or more accessible unisex facilities are provided, at least one should be of opposite handed layout to the other(s).

W.C. Cubicles

- (4) The clear distance between the water closet and the wash basin should not be more than 600 mm for the users' convenience after toileting. The clear manoeuvring space within the cubicle should not be less than 1500 mm x 1950 mm.
- (5) The angle between the two grab rails should be within the range of 80° to 90°.

Toilet/Compartment Doors

- (6) Double swing doors which open both inwards and outwards may be provided in any toilets or cubicles. Sliding door fixed with vertical grab rails on both sides of the door in accordance with paragraph below is equally acceptable provided that it is not heavy or awkward to use. The requirements of door handle and horizontal grab rail will not be applicable to sliding doors.

- (7) A power-operated door should be designed as follows:
- (a) push pad type control buttons having a minimum dimension of 20 mm should be used to open or lock the door;
 - (b) after the door is open, the door should be automatically closed after a minimum of 10-second time lapse;
 - (c) a detection device should be provided to re-open the door in the event of hitting any obstacle. The device should be positioned at a height between 300 mm and 400 mm above the finished floor level;
 - (d) a lock button should be provided inside the toilet. The locking device should be able to be released from the outside manually upon activation of an emergency break glass unit installed between 1 800 mm and 2 000 mm above the finished floor level outside the toilet. The maximum horizontal force for opening the door manually should comply with the requirements stipulated in paragraph 10.4(3);
 - (e) in case the lock button is pressed before the door-closing action is complete, the door should still be able to be locked;
 - (f) an audible message should be provided in English, Cantonese and Putonghua to inform the user after the door is locked;
 - (g) an indicator activated by the locking device signifying the toilet is being occupied should be provided both inside and outside the toilet;
 - (h) the door should be provided with vertical grab rails fixed on both sides of the door at the height of 800 mm from the finished floor level (measuring from the bottom of the grab rails) and with a grip space of not less than 30 mm clear of the door. Grab rails should not be less than 32 mm and not more than 40 mm in external diameter and not less than 600 mm in length;
 - (i) the door including control buttons should have a minimum luminance contrast of 30% with the door frame and their surrounding finishes;
 - (j) a back-up emergency power supply should be provided for at least 20 minutes in power failure situation;
 - (k) the control buttons should be installed at the wall adjacent to the door opening such that the user will not be interfered by the door movement. For a door swinging into an area with internal corner, the buttons should be located at least 600 mm from the internal corner of a room and the button should have a clear 1 000 mm minimum distance from the swing of the door;
 - (l) the control buttons should be located at not less than 900 mm and not more than 1 200 mm above the finished floor level;
 - (m) separate door open and lock buttons should be provided inside the toilet and placed together;
 - (n) Braille and tactile marking should be provided to the control buttons in compliance with the requirements stipulated in paragraph 7.2(13) and (14) as well as **Figure 33** and **Figure 34** and
 - (o) user instructions in Chinese, English and Braille on how to open and lock the door should be provided adjacent to the control buttons inside the toilet.

See examples in **Figure 33** to **Figure 36**.

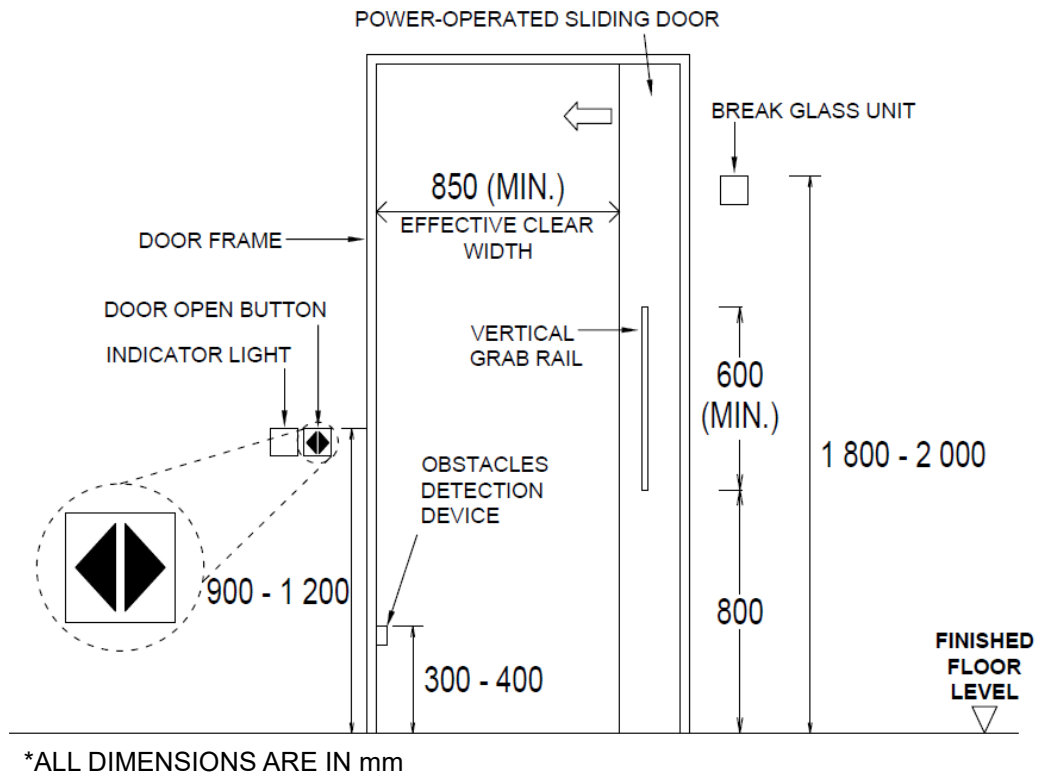


Figure 33 - Elevation of Power-operated Sliding Door (View from Outside of Accessible Toilet)

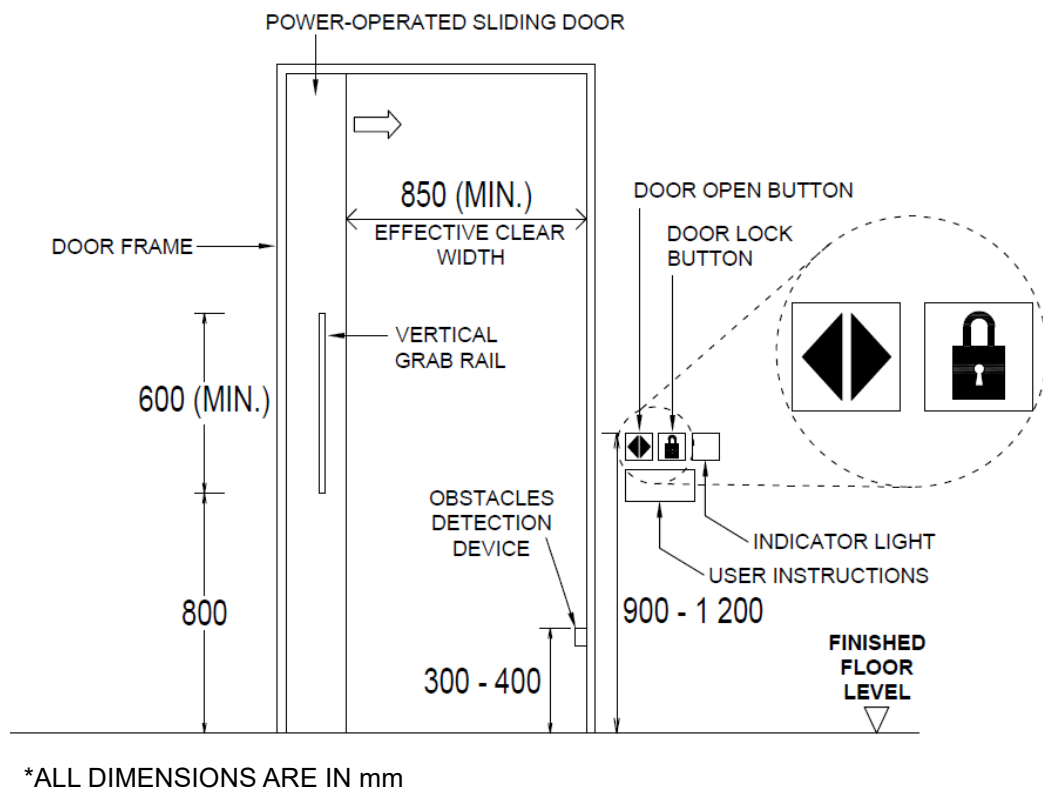
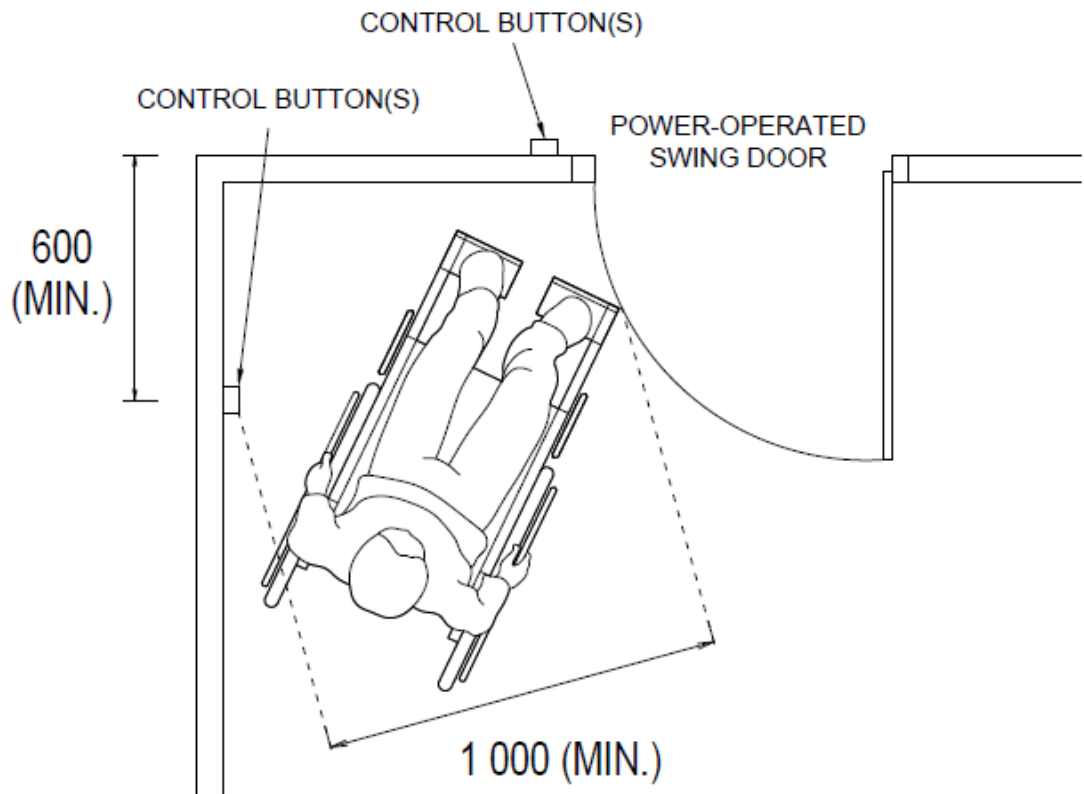
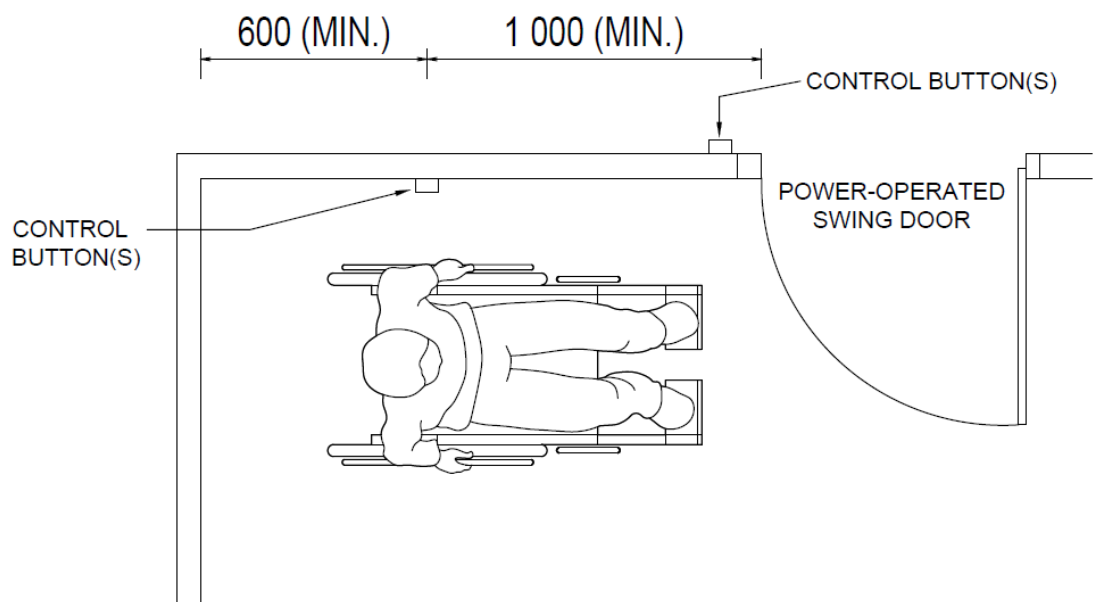


Figure 34 - Elevation of Power-operated Sliding Door (View from inside of Accessible Toilet)



*ALL DIMENSIONS ARE IN mm

Figure 35 - Plan of Power-operated Swing Door (Control button(s) on separate wall)



*ALL DIMENSIONS ARE IN mm

Figure 36 - Plan of Power-operated Swing Door (Control Button(s) on Same Wall)

Bathroom/Shower Compartment (other than that provided in accessible guest room)

- (8) The floor of the bathroom and the shower compartment should be provided with slip-resistant floor surfaces in accordance with Appendix C of this Manual.

Luminance Contrast

- (9) Wall tiling should have a minimum luminance contrast of 30% with sanitary appliances and fittings, grab rails and toilet roll holders, etc.
- (10) Toilet door and compartment door should have a minimum luminance contrast of 30% with its frame and adjacent wall.

Avoidance of Obstruction

- (11) Waste pipe, disposal bins and other fittings within the toilet should be located to avoid any obstruction or creating a tripping hazard including those under the wash basin.
- (12) Toilets and W.C. cubicles for use by PwDs should be located to avoid proximity to or obstruction by exit doors to the staircase.

Emergency Call Bell

- (13) A push button with a minimum luminance contrast of 50% with the adjoining wall surfaces should be easily identified and operated and be provided in any individual accessible toilet compartment or a water closet cubicle designed for the PwDs to summon assistance at seated position or on the floor when the person has fallen accidentally.
- (14) In addition to a position outside the compartment or cubicle, the emergency alarm should be connected to a caretaker's office or public information/service counter provided under paragraph 10.6.
- (15) Emergency call bells should be equipped with a back-up power supply.
- (16) The emergency call bell when activated should emit audible and visible alarm signal which should be readily noticeable for summoning assistance.
- (17) Visible and audible signal should be emitted inside the accessible toilet when the emergency call bell is activated

Motion Sensor

- (18) Motion sensors connecting to the building management office or the caretaker's office should be provided in an accessible toilet for detection of emergencies and prompt assistance.

Large Size Symbols

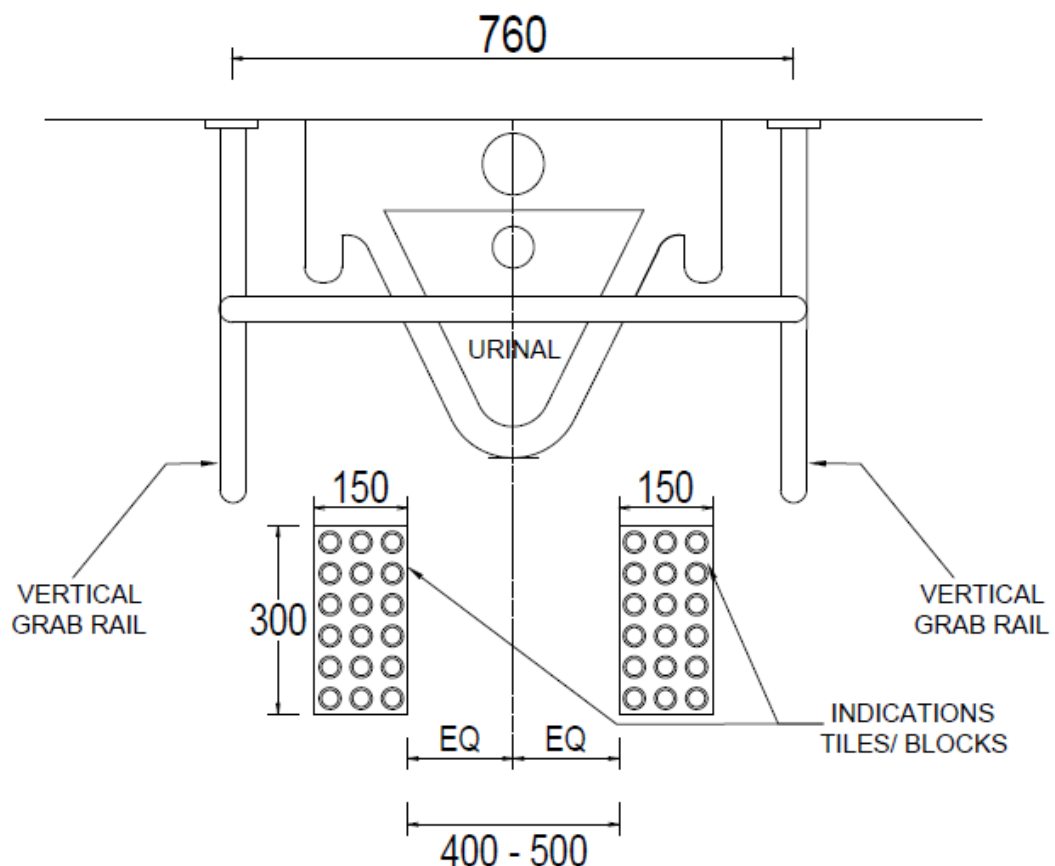
- (19) Large size symbols with luminance contrast for male and female toilets should be used.

Grab Rails

- (20) The surface of the grab rails should be slip-resistant, e.g. stainless steel with hair-line finish.

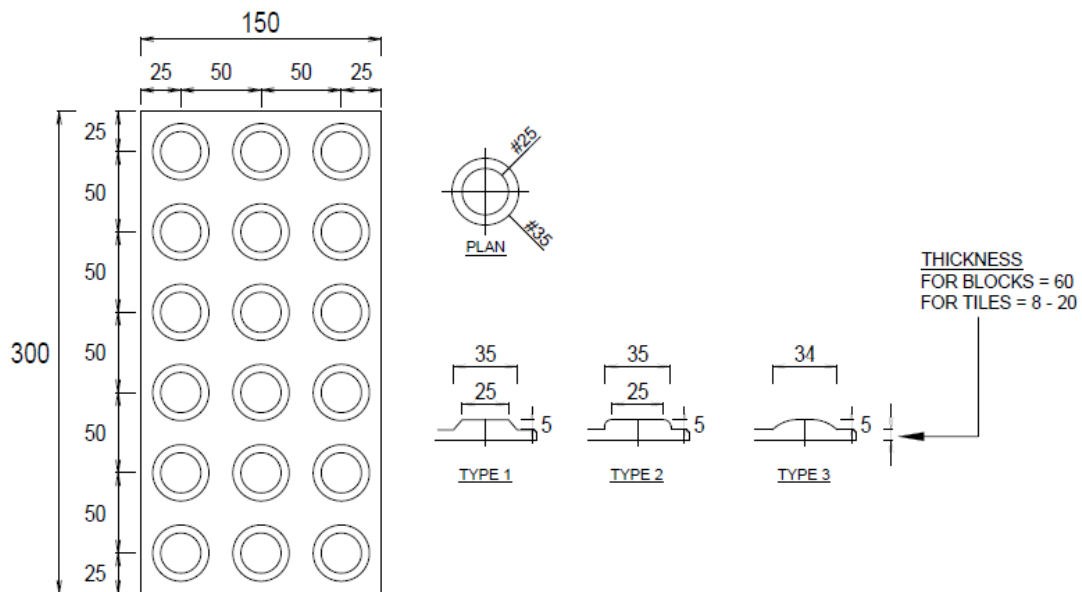
Indicating Tiles/Blocks

- (21) Two indicating tiles/blocks on the floor to facilitate the use by persons with visual impairment (see **Figure 37** and **Figure 38**) should be provided in front of the urinal for use by persons with ambulant disabilities.
- (22) The indicating tiles/blocks should have a minimum luminance contrast of 30% with the floor finishes.



*ALL DIMENSIONS ARE IN mm

Figure 37 - Indicating Tiles/Blocks for Urinal Used by Persons with Ambulant Disabilities



*ALL DIMENSIONS ARE IN mm

Figure 38 - Details of Indicating Tiles/Blocks

Washroom Accessories

(23) The manually operated controls of washroom accessories should be at a position which is not higher than 1 100 mm above the finished floor level.

9.5 Bathrooms and Shower Compartments

This section is applicable to accessible bathroom and shower compartment as required under paragraph 8.3.

Taps and Controls

(1) Taps and other controls should be installed between the centre line and the outer edge of the bathtub.

Grab Rails

- (2) The surface of the grab rails should be slip-resistant, e.g. stainless steel with hair-line finish.
- (3) Two additional vertical grab rails should be installed for shower compartments and they should:
 - (a) have a minimum length of 750 mm;
 - (b) be installed at a height between 700 mm and 800 mm measured from the bottom of the bars to the finished floor level; and
 - (c) be provided adjacent to the shower seat such that one is mounted beside the shower seat and another one is mounted not more than 450 mm from the edge of the shower seat (see **Figure 31**).

Shower Seats

- (4) The depth of shower seats in shower compartments should be 450 mm.

Emergency Call Bells in Accessible Bathrooms and Shower Compartments

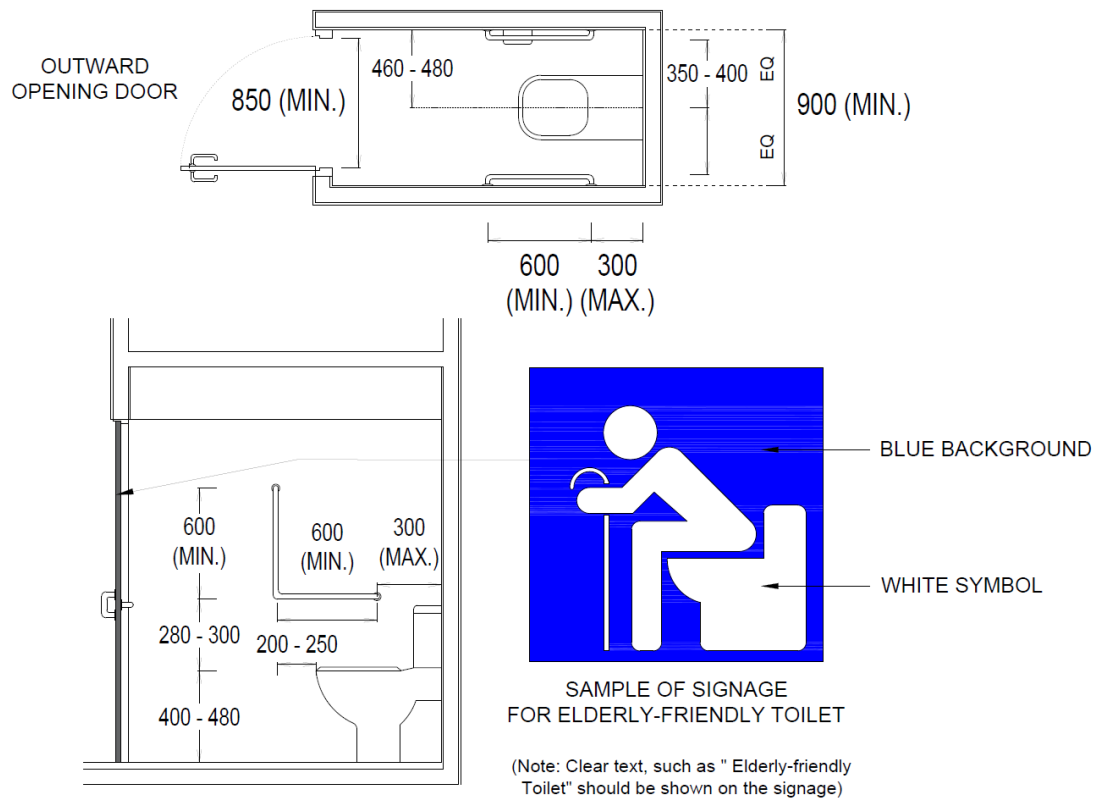
- (5) The push button of the emergency call bell should be appropriately located and conveniently accessible to all users. The emergency call bell when activated should emit audible or visible alarm signal which should be readily noticeable for summoning assistance for the person in the bathroom/shower compartment. The emergency alarm should be installed outside the bathroom/shower compartment and connected to a public information or service counter.
- (6) An emergency call bell should be equipped with a weatherproof push button for activating the alarm. The push button should be installed at the wall outside but immediately adjacent to the bathtub or shower compartment at a height between 400 mm and 600 mm above the finished floor level. A notice "Emergency Call" in English, Chinese and Braille should be fitted next to the emergency push button.

9.6 Family-friendly Facilities

Elderly-friendly Toilets / Elderly-friendly Facilities in Toilets

- (1) At least one elderly-friendly toilet or one cubicle with elderly-friendly facilities in each communal toilet should be provided in easily accessible common areas including:
- (a) Open outward/ folding door;
 - (b) Grab bars on both sides of the cubicle; one side with L-shape grab bars;
 - (c) Emergency call buttons complying with paragraph 9.2(16) to paragraph 9.2(18) in the cubicle;
 - (d) Hands-free sensor type toilet flush;
 - (e) Holding device for walking sticks and bags;
 - (f) Minimum internal width 900 mm and door width 850 mm;
 - (g) Adequate manoeuvring space of 300 mm diameter between the cubicle door and the water closet; and
 - (h) A sign installed outside showing the cubicle is equipped with elderly-friendly facilities.

A sample is shown in **Figure 39**.



*ALL DIMENSIONS ARE IN mm

Figure 39 - Sample of Elderly-friendly Toilet

Babycare Room

- (2) The babycare room for single user should have minimum dimensions of not less than 2 200 mm x 2 200 mm. The size of a babycare room for a single user should be not less than 5 m² and the room should provide a clear manoeuvring space of not less than 1 500 mm x 1 500 mm measured at 350 mm above finished floor level for wheelchair user. For a babycare room for multiple users, the size of the room should be not less than 12 m² with separate breastfeeding cubicle(s), one of which should cater for wheelchair user satisfying with the requirement under [paragraph 9.6\(3\)](#) and [paragraph 9.6\(4\)](#).

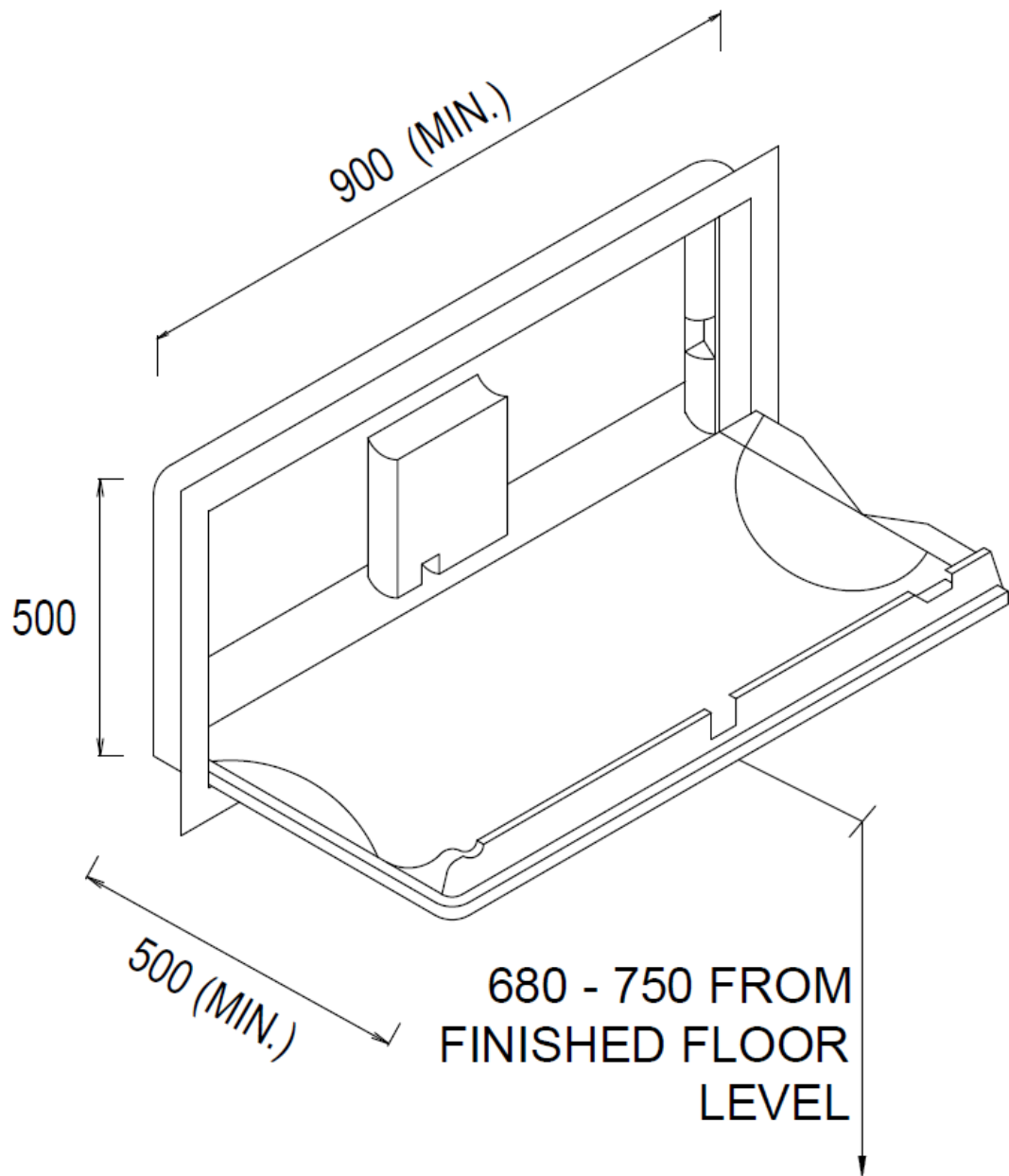
Lactation Room

- (3) The lactation room(s) should be not less than 7.5 m² and can accommodate at least 2 users at a time with their privacy protected and at least one for use by a wheelchair user with the provision of a clear manoeuvring space of not less than 1 500 mm x 1 500 mm. The room should provide separate cubicles with door for users.

- (4) The lactation area should:
- (a) be equipped with a lockable door;
 - (b) not be located within toilets;
 - (c) be well-illuminated and ventilated;
 - (d) be equipped with mirror;
 - (e) be equipped with a table to place breast pumps and personal items;
 - (f) be equipped with seat; and
 - (g) be equipped with socket.

Diaper-changing Facilities

- (5) Diaper-changing facilities should be provided in all toilets unless there is no sufficient space. The facilities should be:
- (a) located not less than 680 mm but not more than 750 mm above the finished floor level when the changing bench is in pull-down position;
 - (b) with a minimum length of 900 mm;
 - (c) with a minimum depth of 500 mm;
 - (d) with a minimum certified weight capacity of 22 kg; and
 - (e) with sufficient safety measures to prevent the risk of fall.
(see **Figure 40** for an example of a typical diaper changing station)



*ALL DIMENSIONS ARE IN mm

Figure 40 - Flat-type Diaper Changing Station

Family-friendly Toilet in Specified Premises

- (6) A family-friendly toilet, which is a unisex toilet serving a family, should be provided in specified premises as defined in Chapter 2 such that users can be accompanied by children, family members or caregivers. Such family-friendly toilet should be provided in addition to any accessible W.C. cubicle.
- (7) Diaper-changing facilities satisfying with the requirements under paragraph 9.6(5) should be provided within a family-friendly toilet.

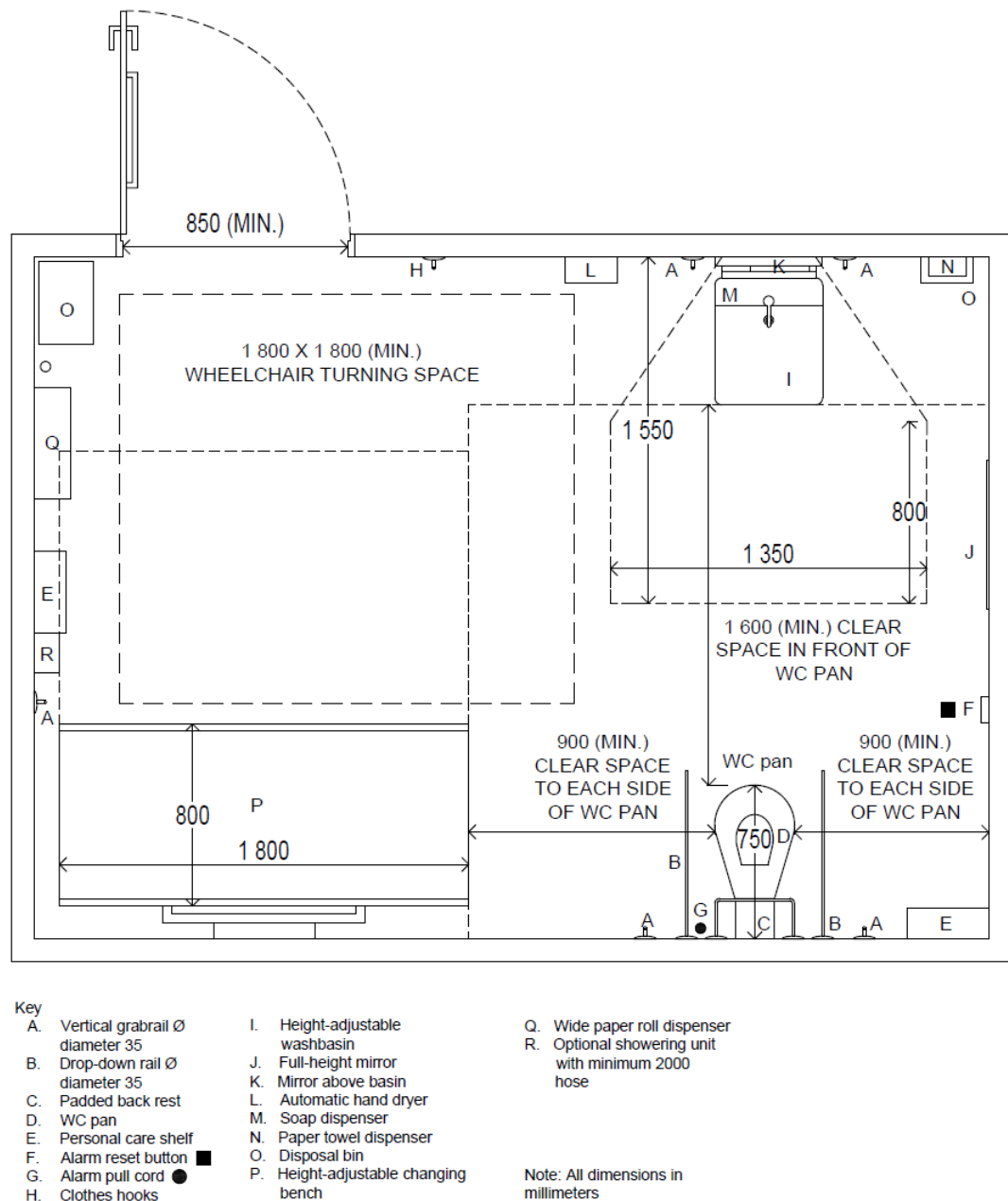
- (8) Child protection seats should be provided within a family-friendly toilet with:
 - (a) sufficient safety measures to prevent risk of fall;
 - (b) a minimum certified weight capacity of 22 kg;
 - (c) a height of not less than 400 mm but not more than 500 mm above the finished floor level; and
 - (d) 450mm radius clearance from any obstruction.
- (9) Child-sized wash basin and water closet should be provided in family-friendly toilet:
 - (a) The surface of the wash basin should be not more than 550 mm above the finished floor level; and
 - (b) The distance between the front edge of the basin and the water tap should be less than 350 mm.
- (10) Child-friendly urinal should be provided in male toilet or family-friendly toilet with a front rim not more than 400 mm above the finished floor level.

Changing Places Toilet in Specified Premises

- (11) A changing places toilet, which is an accessible water closet cubicle integrated with a hoist and adult-sized changing bench for use by PwDs, should be provided in specified premises as defined in Chapter 2.
- (12) The changing places toilet should be provided with:
 - (a) an entrance door with a minimum clear width of 850 mm, and where practicable, be enlarged to 1000 mm to accommodate large, wheeled mobility devices;
 - (b) a turning space of 1 800 mm x 1 800 mm inside and outside the changing places toilet adjacent to the door to accommodate large, wheeled mobility devices users and their caregivers;
 - (c) an emergency call button in compliance with paragraph 9.2(16) to paragraph 9.2(18);
 - (d) one grab rail, which should not be less than 32 mm but not more than 40 mm in external diameter, fixed on each of both the inner and outer surfaces of the cubicle door. The grab rail should have a grip space of not less than 30 mm clear of each door surface;
 - (e) an accessible water closet with toilet lid not less than 380 mm but not more than 450 mm above the finished floor level;
 - (f) a folding grab rail or wall-mount grab rail on both sides of the water closet complying with the requirements in paragraph 9.2(12) to paragraph 9.2(15);
 - (g) a wash basin complying with the requirements in paragraph 9.2(8) and paragraph 9.9(9);
 - (h) a wall-mounted changing bench:
 - (i) with a changing surface of 1 800 mm in length by 800 mm width with a minimum certified weight capacity of 200 kg,

- (ii) with adjustable height from 450 mm and 900 mm, which should be mains powered; and
- (i) a hoist:
 - (i) provided with full-ceiling coverage track within the changing places toilets that conforms to ISO 10535:2021 or other equivalent standards; and
 - (ii) with a minimum certified safe working load of not less than 200 kg.

Figure 41 shows an example of Changing places toilet.



*ALL DIMENSIONS ARE IN mm

Figure 41 - Changing Places Toilet

CHAPTER 10 BUILDING COMPONENTS AND EQUIPMENT

10.1 General

This Chapter aims to provide detailed design requirements for various building components and equipment to assist any persons, regardless of their ages, sizes, abilities or disabilities to access and use the facilities safely and independently.

MANDATORY SECTION

Obligatory Design Requirements

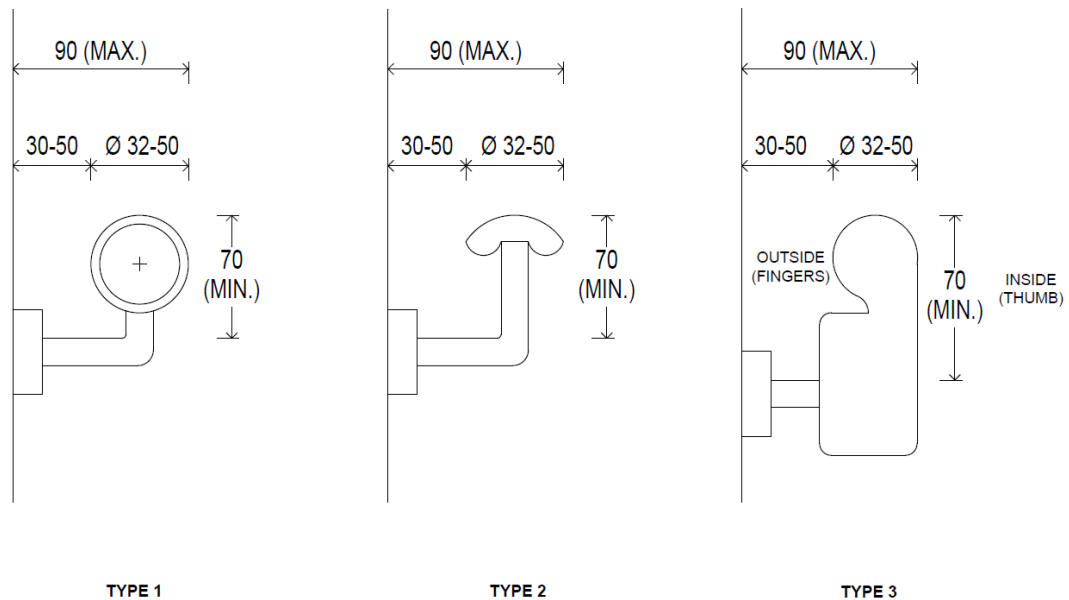
10.2 Handrails

Dimension and Shape of Handrail

- (1) Handrail to ramp and step shall be fixed not less than 30 mm and not more than 50 mm clear of wall and with a clear height of 70 mm from the top of the bracket to the top of the handrail.
- (2) Subject to paragraph 10.2(5) below, the top of handrail shall be at a height of not less than 850 mm and not more than 950 mm above any nosing, floor or landing.
- (3) Handrail shall be:
 - (a) tubular, not less than 32 mm and not greater than 50 mm in external diameter; or
 - (b) in other shapes that can provide the user a grip similar to that specified in the case of tubular handrails.
- (4) Handrail shall extend horizontally not less than 300 mm beyond the first and last nosing of every flight of steps or beyond the ends of a ramp and terminate into a closed end which shall turn down or return fully to end post or wall face and which shall not project into a route of travel. Where a door opening is in place, a shortened extension of not less than 100 mm shall be permitted.
- (5) For ramps and main circulation staircases referred in paragraph 6.2 and paragraph 6.4, in addition to handrails complying with paragraph 10.2(1) to paragraph 10.2(4), lower handrails shall be provided in accordance with the following requirements:
 - (a) The top of the lower handrail shall be at a height of not less than 700 mm and not more than 800 mm above any nosing, floor, or landing;

- (b) The lower handrail shall comply with the requirements under paragraph 10.2(1), paragraph 10.2(3) and paragraph 10.2(4); and
- (c) The vertical clearance between the top of the upper handrail and the top of the lower handrail shall be not less than 150 mm.

Typical handrail sections are shown in **Figure 42**.



*ALL DIMENSIONS ARE IN mm

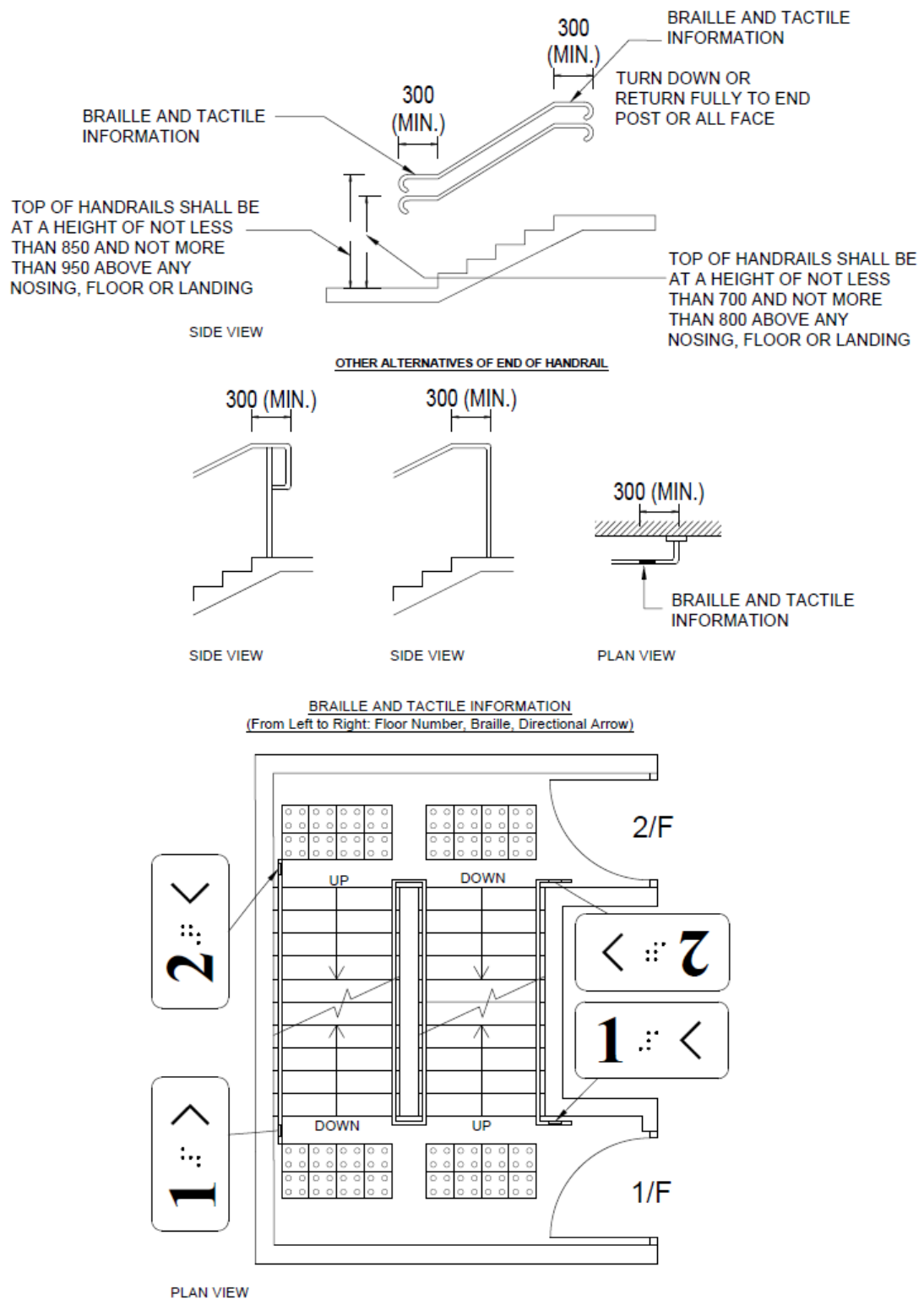
Figure 42 - Typical Handrail Sections

Loading

- (6) Handrail shall be installed to resist a load of not less than 1.3 kN applied vertically or horizontally.
- (7) Handrail shall not rotate within its fixing fittings.

Braille and Tactile Information

- (8) Braille and tactile information on directional arrow and floor number shall be provided on handrail on every floor at a designated location as illustrated in **Figure 43** to facilitate persons with visual impairment. Where a directional sign exists on handrails, Braille and tactile information shall also be provided as illustrated in **Figure 17**.

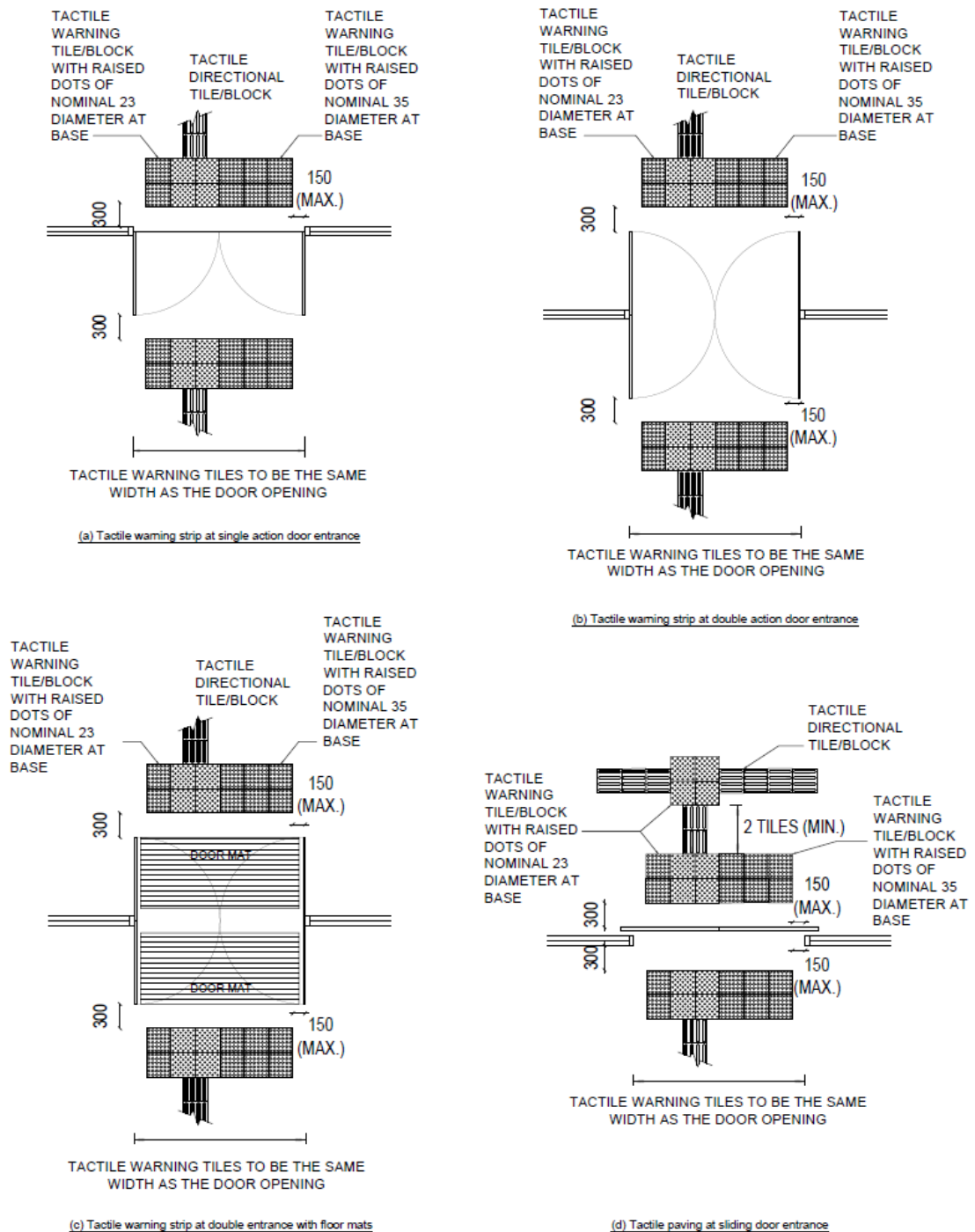


*ALL DIMENSIONS ARE IN mm

Figure 43 - Handrails of Staircase

10.3 Tactile Guide Path

- (1) All tactile guide paths required in Table 2 shall be provided with slip-resistant surfaces in accordance with Appendix C of this Manual.
- (2) Tactile guide path shall be installed from a point of access at the lot boundary to the main entrance of the building and from the main entrance to lift zone, the nearest accessible toilet, public information/service counter, Braille and tactile floor plan and staircase. The distance from the ending position of tactile guide path to these areas or facilities shall be not more than 300 mm.
- (3) Tactile guide path shall be installed in straight or curving path generally following the pedestrian flow of the public.
- (4) Examples of the design and specifications of tactile guide path are illustrated from **Figure 44** to **Figure 49**.



*ALL DIMENSIONS ARE IN mm

Figure 44 - Tactile Guide Paths at Building Entrances Linking up with Initial Access on the Lot Boundary and Interior Facilities

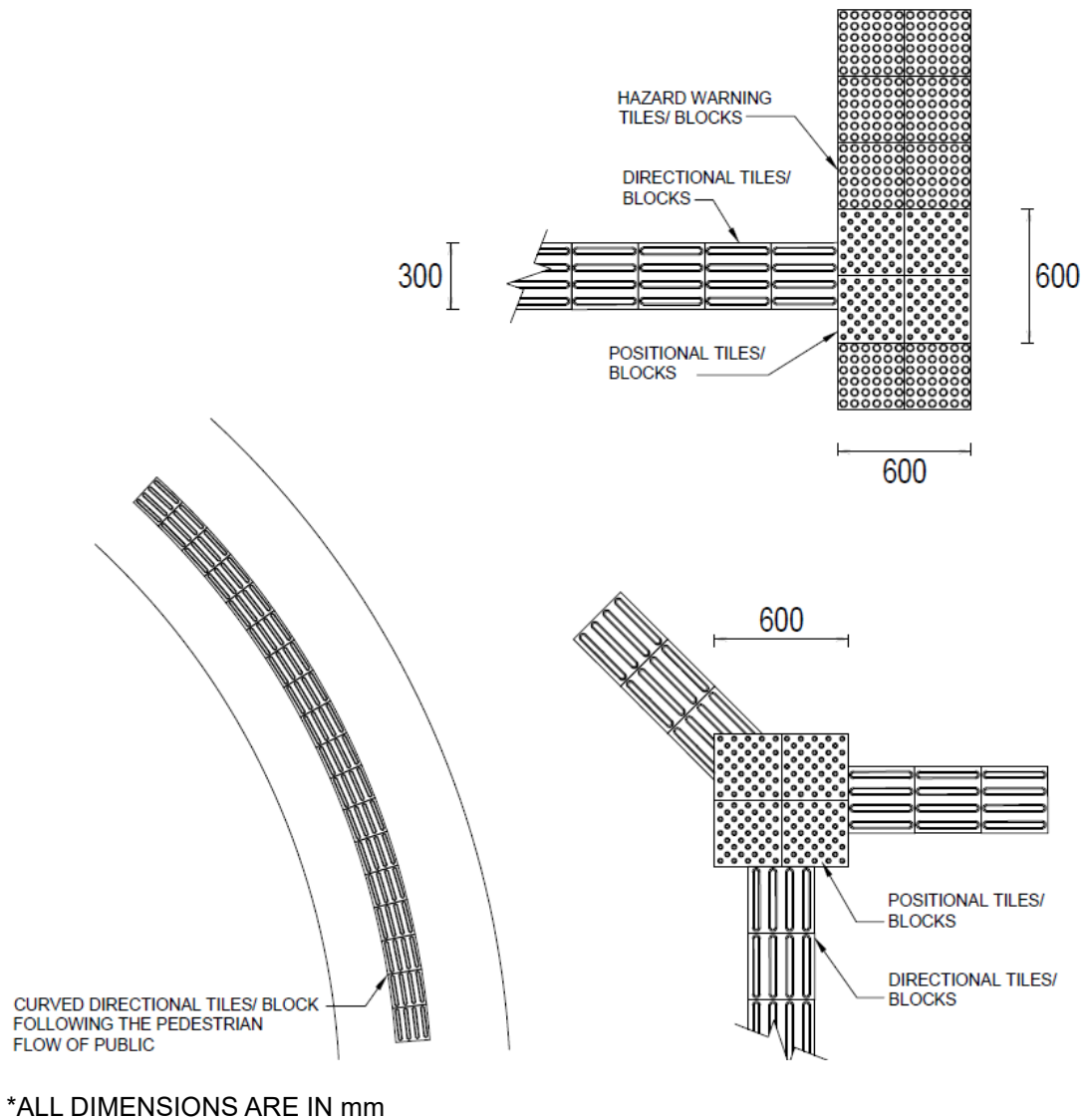
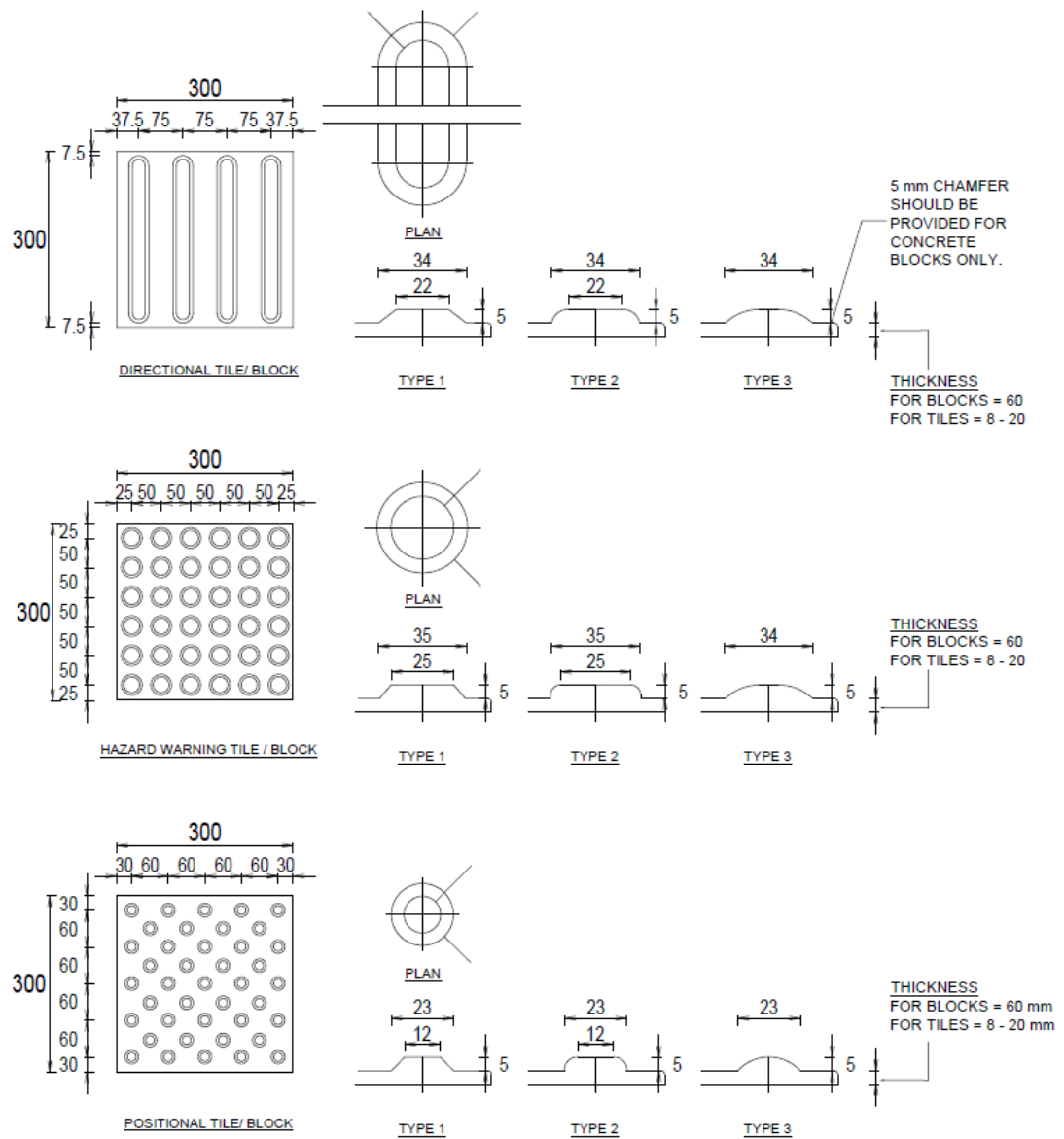
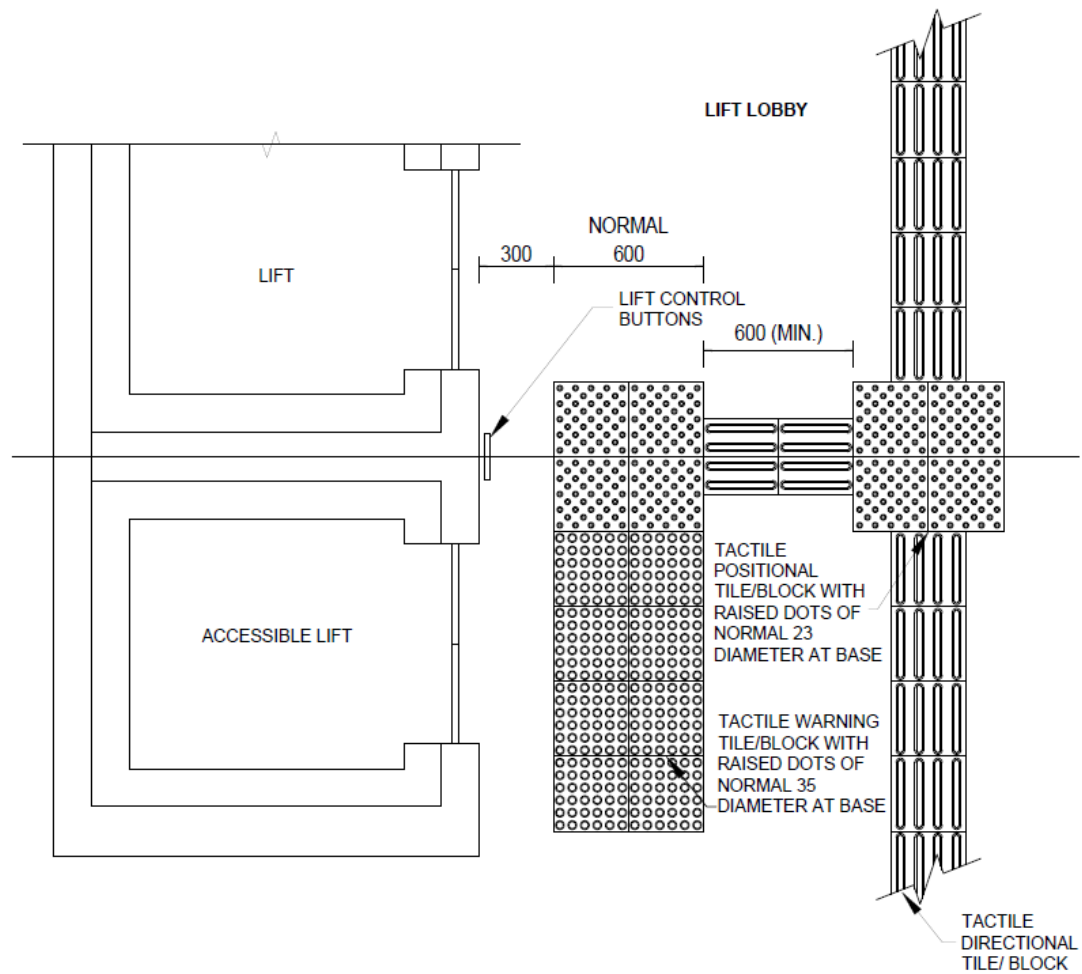


Figure 45 - Typical Tactile Guide Path Junction



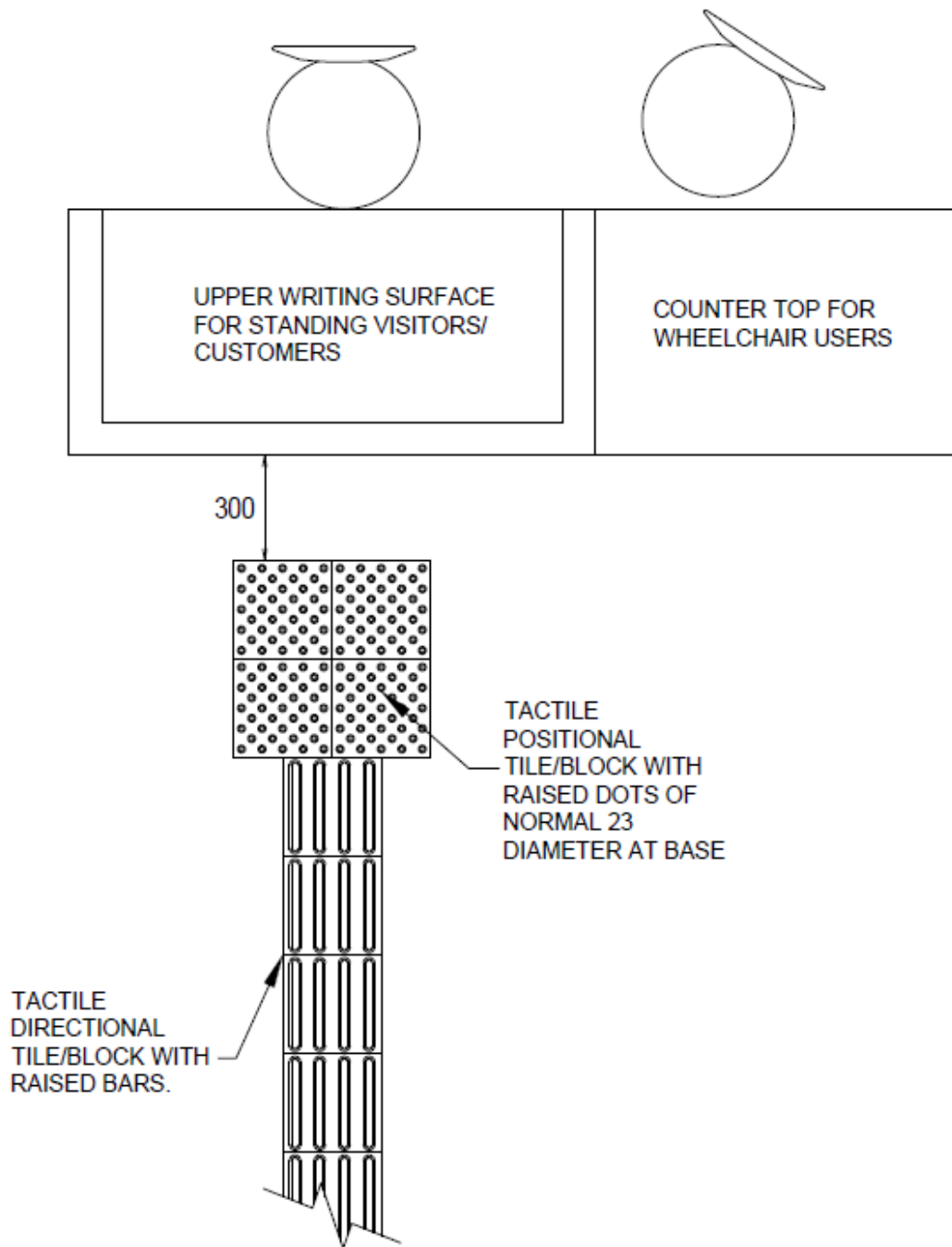
*ALL DIMENSIONS ARE IN mm

Figure 46 - Example of Details of Tactile Warning Tiles/Blocks



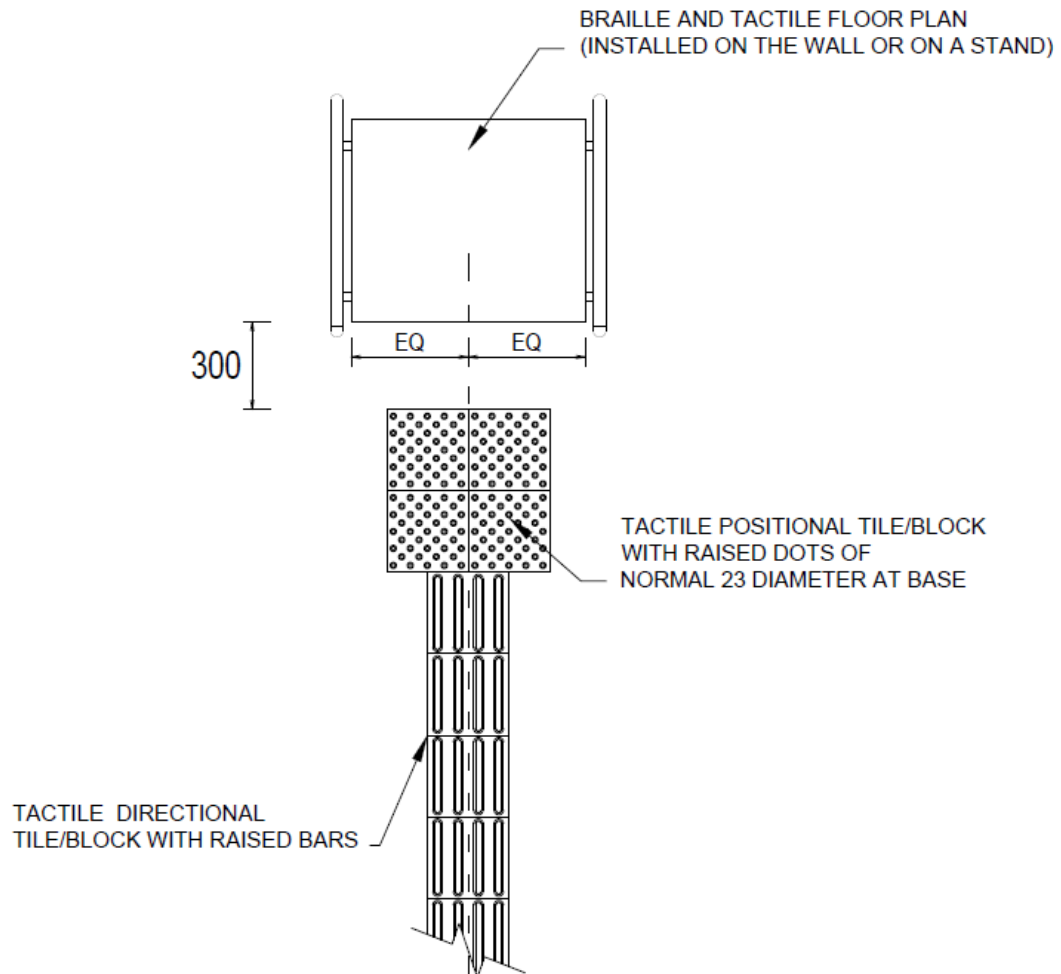
*ALL DIMENSIONS ARE IN mm

Figure 47 - Tactile Guide Path to Lift Zone



*ALL DIMENSIONS ARE IN mm

Figure 48 - Tactile Guide Path to Information/Service Counter



*ALL DIMENSIONS ARE IN mm

Figure 49 - Tactile Guide Path to Tactile/Braille Directory Map/Floor Plan

10.4 Door

Double-action Self-closing Doors

- (1) Double-action self-closing door shall have a check mechanism to prevent the door swinging beyond the closed position and a transparent vision-panel with a bottom edge not more than 1000 mm above the floor and the top edge not less than 1500 mm above the finished floor level.

Handles

- (2) Door handle shall not be less than 950 mm and not more than 1050 mm above the finished floor level, measured from the top surface of the grip.

Door Closing Devices

- (3) Door closing devices shall be designed to allow exterior and interior doors to be opened with horizontal force of not more than 30 N and 22 N respectively. Door required to have fire resistance period installed along accessible routes shall be opened with horizontal force of not more than 30 N. Closer for interior door shall have a closing period of at least 3 seconds measured from an open position of 70° to a point 75 mm from the closed position measured from the leading edge of the door. Door closing devices include door closer, spring hinge and floor hinge.

Frameless Glass Doors

- (4) If frameless glass door is used, it shall be prominently marked so as to make it visible. The marking shall be placed across on the glass door such that at least a portion of the marking is placed between 900 mm and 1500 mm above the finished floor level. The colour marking shall also be provided to glass panel adjacent to the glass door.

10.5 Illumination

Illumination Level

- (1) The following designated areas of a building, where reasonably expected to be used by the public or occupants of the building during the period of normal use or occupation or when required, shall have minimum illumination level of 120 lux measured at the finished floor level along the centre line of the passageway:
 - (a) Ground floor entrance lobby;
 - (b) Lift;
 - (c) Accessible route;
 - (d) Step and staircase;
 - (e) Accessible toilet; and
 - (f) Public information or service counter required under paragraph 10.6.

Illumination Level for Signs

- (2) Signs provided under this Chapter shall have illumination level on the sign surface of not less than 120 lux.

Means of Provision of Illumination

- (3) The illumination may be provided through automatic or manual switching devices.

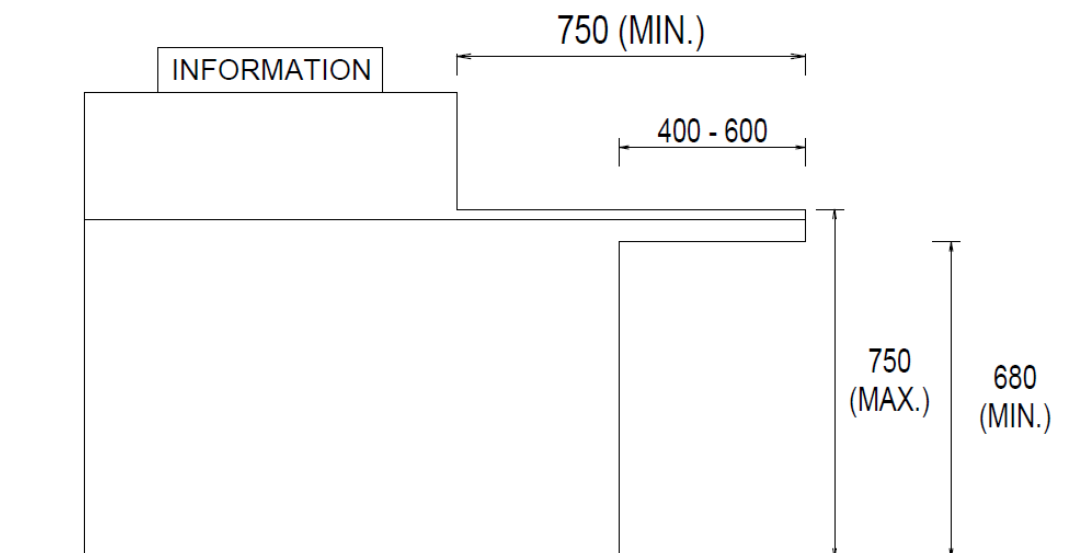
10.6 Public Information or Service Counters

Provision of Public Information or Service Counters

- (1) Public information/service counters shall be provided to various categories of buildings as specified in Table 2.
- (2) There shall be at least one public information/service counter built with a portion not higher than 750 mm above the finished floor level and not less than 750 mm wide to assist wheelchair users (see **Figure 50** and **Figure 51**).

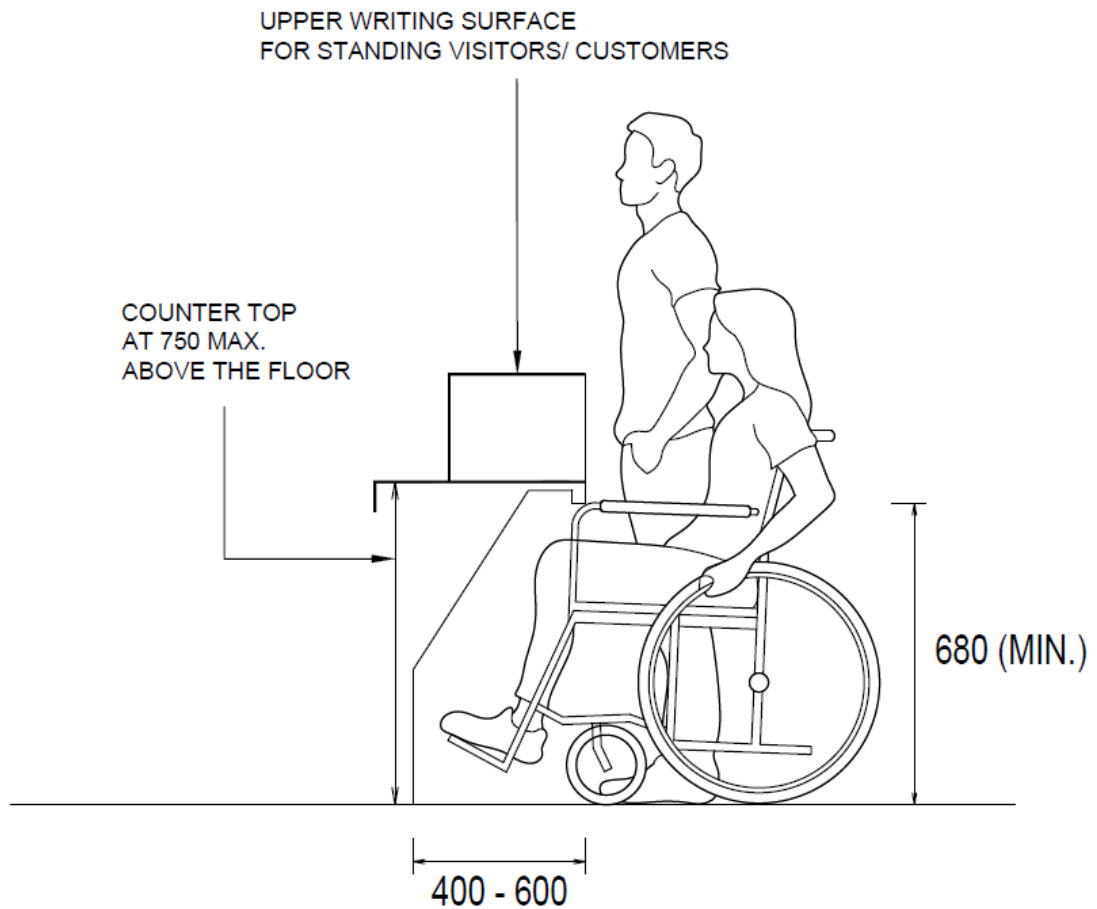
Design of Counter

- (3) Where public information/service counters are provided, they shall be accessible and easily identifiable from a building entrance by all persons with or without a disability.
- (4) A groove with a width of 50 mm and a depth of 75 mm shall be installed at the edge of the counter; or an alternative effective design for securing canes, walking sticks or other assistive tools shall be provided (see **Figure 52**).
- (5) Leg space of a depth between 400 mm and 600 mm and a height of not less than 680 mm above finished floor level shall be provided.



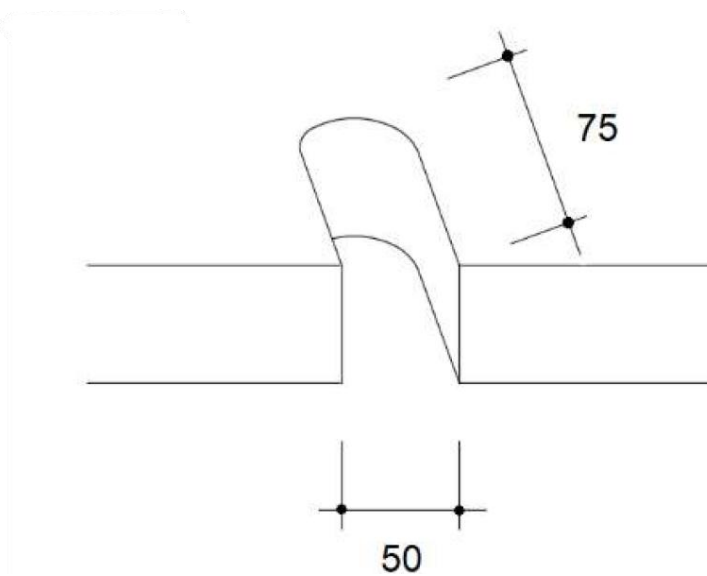
*ALL DIMENSIONS ARE IN mm

Figure 50 - Front View of Information/Service Counter for Wheelchair Users



*ALL DIMENSIONS ARE IN mm

Figure 51 - Key Heights of Counters and Reception Desks



*ALL DIMENSIONS ARE IN mm

Figure 52 - Groove at Public Information/Service Counter

10.7 Assistive Listening System

- (1) An assistive listening system, which enables sound signals to be transmitted to persons with hearing impairment without interference from background noise or excessive reverberation (such as an induction loop system or an infrared system), shall be provided in:
 - (a) the main public information/service counter, and at least one of the other public information/service counters, if any, for various categories of buildings as specified in Table 2; and
 - (b) the hall and auditorium, if any, of such a building.

10.8 Fire Alarm System

Visual Alarm

- (1) Subject to paragraph 10.8(2), visual alarm signal shall be provided to form part of the fire alarm system in the form of a flashing red light, labelled “Fire Alarm 火警” in both English and Chinese. It shall be indicated on separate plate affixed nearby or engraved on the light cover. The alarm shall be installed at a prominent location which shall be readily noticeable when activated in the categories of buildings as specified in Table 2.
- (2) The provision of visual alarm shall not apply to all exit staircases and smoke lobbies as required under the Code of Practice for Fire Safety in Buildings, domestic units and areas exempted under paragraph 3.2.1 and paragraph 3.2.2.

Positioning of Call Points

- (3) The manual fire alarm call points or activation controls, e.g. break glass units, in places which are intended to be accessible to wheelchair users, shall be positioned at prominent and accessible locations and installed in accordance with the Code of Practice for Minimum Fire Service Installations and Equipment and the Code of Practice for Inspection, Testing and Maintenance of Installations and Equipment or Fire Services Department Circular Letter.

Audible Alarm

- (4) Audible alarm shall comply with the requirements of the Code of Practice for Minimum Fire Service Installations and Equipment and the Code of Practice for Inspection, Testing and Maintenance of Installations and Equipment published by the Director of Fire Services.

Exceptional Cases

- (5) Audible and visual alarm may not be appropriate for use in hospitals and some specialised medical facilities, such as operating rooms, where it is not intended to alert or alarm patients who are incapable of independent evacuation. The sudden strobe flash might disrupt a surgical operation in progress which could be catastrophic. For such facilities, the requirements for visual and audible alarm may be modified to suit the norm of industry-accepted practices.

10.9 Signs

Signs shall be erected at prominent and conspicuous positions inside and outside a building to indicate clearly the exact locations of facilities that are available for use by PwDs. To design an effective signage system, the needs of different types of users in a building and the complexity of the building layout must be considered.

The International Symbol of Accessibility

- (1) The international symbol of accessibility shall be the wheelchair figure in white on a blue background or, when placed on dark-coloured surfaces, a blue wheelchair figure on a white background, and is to be provided at conspicuous location as shown in **Figure 53** for the purposes of identifying/advertising/signifying:
 - (a) accessible entrance to the building;
 - (b) accessible exit from the building;
 - (c) reserved car parking facilities for PwDs;
 - (d) the location of toilets for PwDs;
 - (e) usable vertical circulation facilities;
 - (f) usable cloakroom facilities; and
 - (g) The availability of special services of information/service counter and telephone in the building.

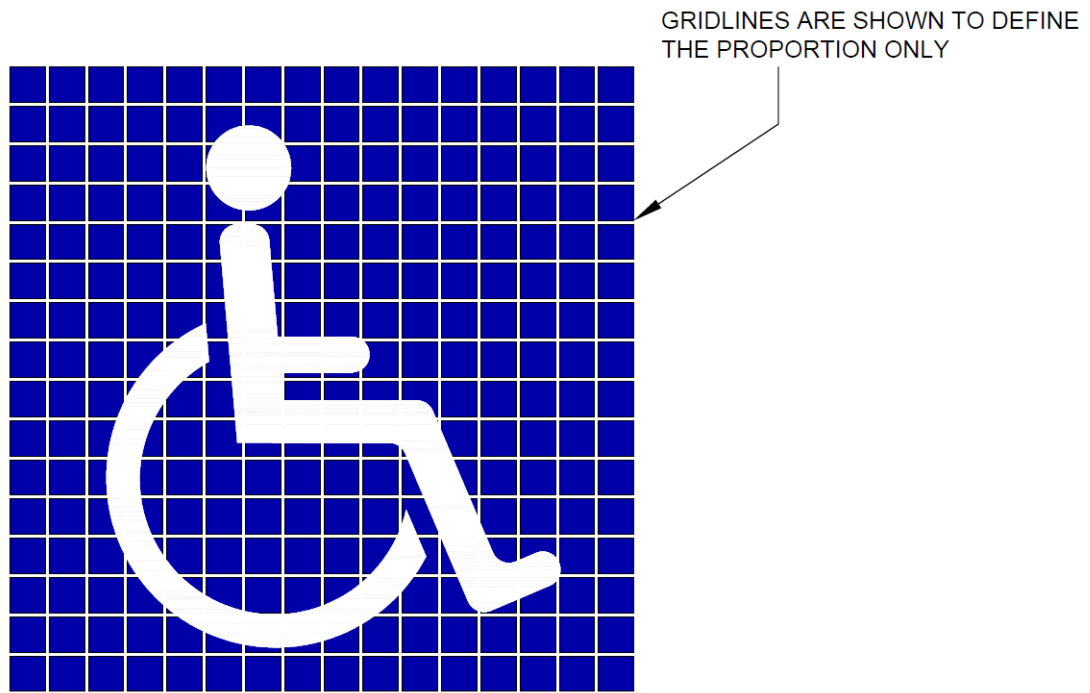
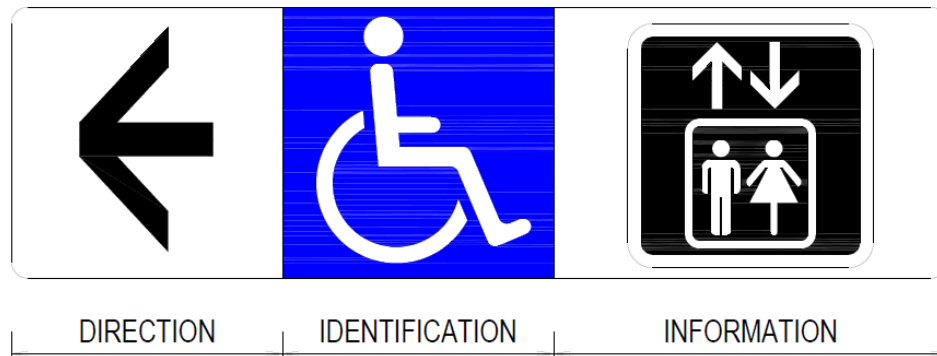


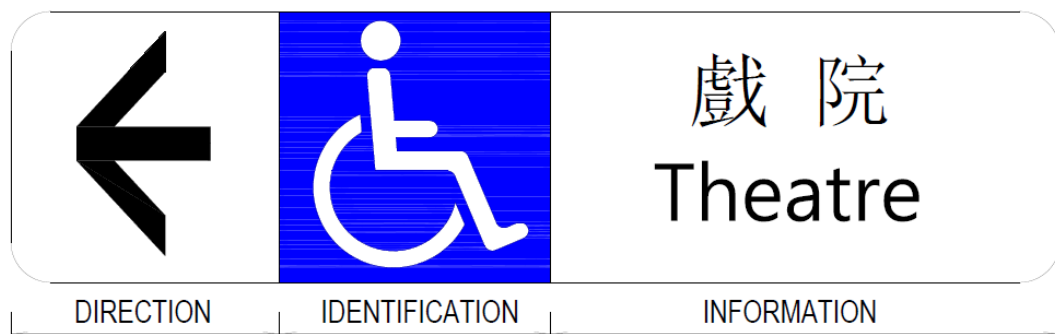
Figure 53 - Proportional Layout for International Symbol of Accessibility

Directional Signs

- (2) Directional arrows and visual information shall be provided at conspicuous location in conjunction with the international symbol for accessibility to guide PwDs to the exact locations of the accessible facilities (see **Figure 54**).



(A) EXAMPLE OF A SIGN IDENTIFYING A FACILITY AND INDICATING ITS
DIRECTION USING A STANDARD SYMBOL [LIFT]



(B) EXAMPLE OF A SIGN IDENTIFYING A FACILITY AND INDICATING ITS
DIRECTION USING A WORD

Figure 54 - Directional Sign

Size

- (3) The height of signs shall be not less than the following:
- (a) 60 mm for doors;
 - (b) 110 mm for corridors; and
 - (c) 200 mm for external use.

Signs for Persons with Hearing Impairment

- (4) International symbol of access for hearing loss as shown in **Figure 55** shall be provided if there is an assistive listening system provided for persons with hearing impairment.

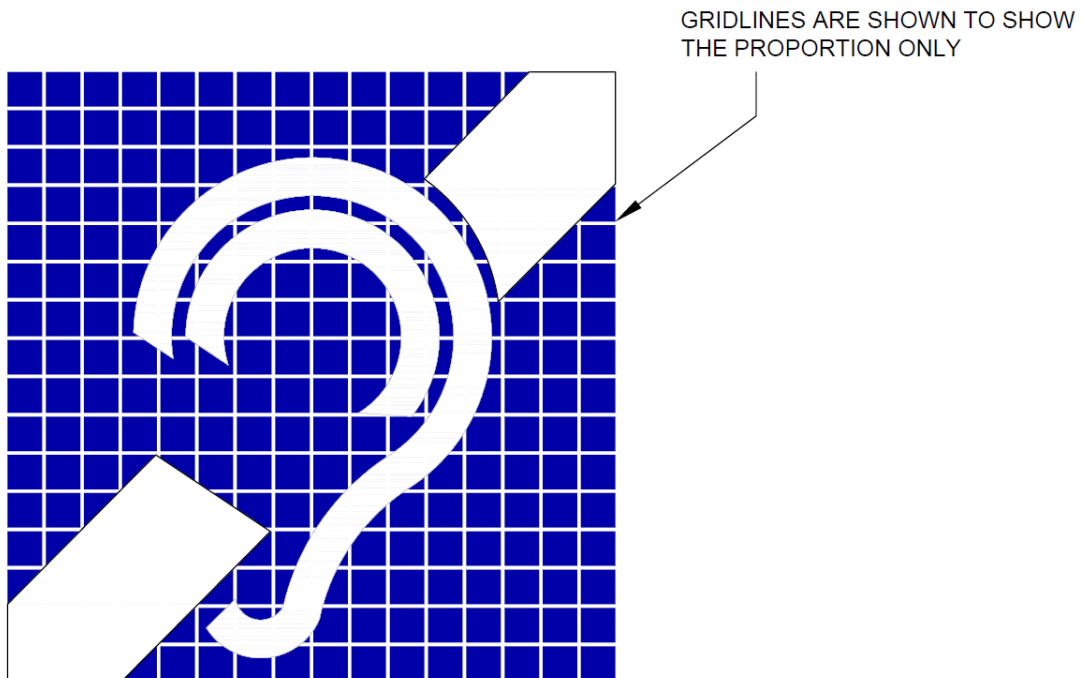
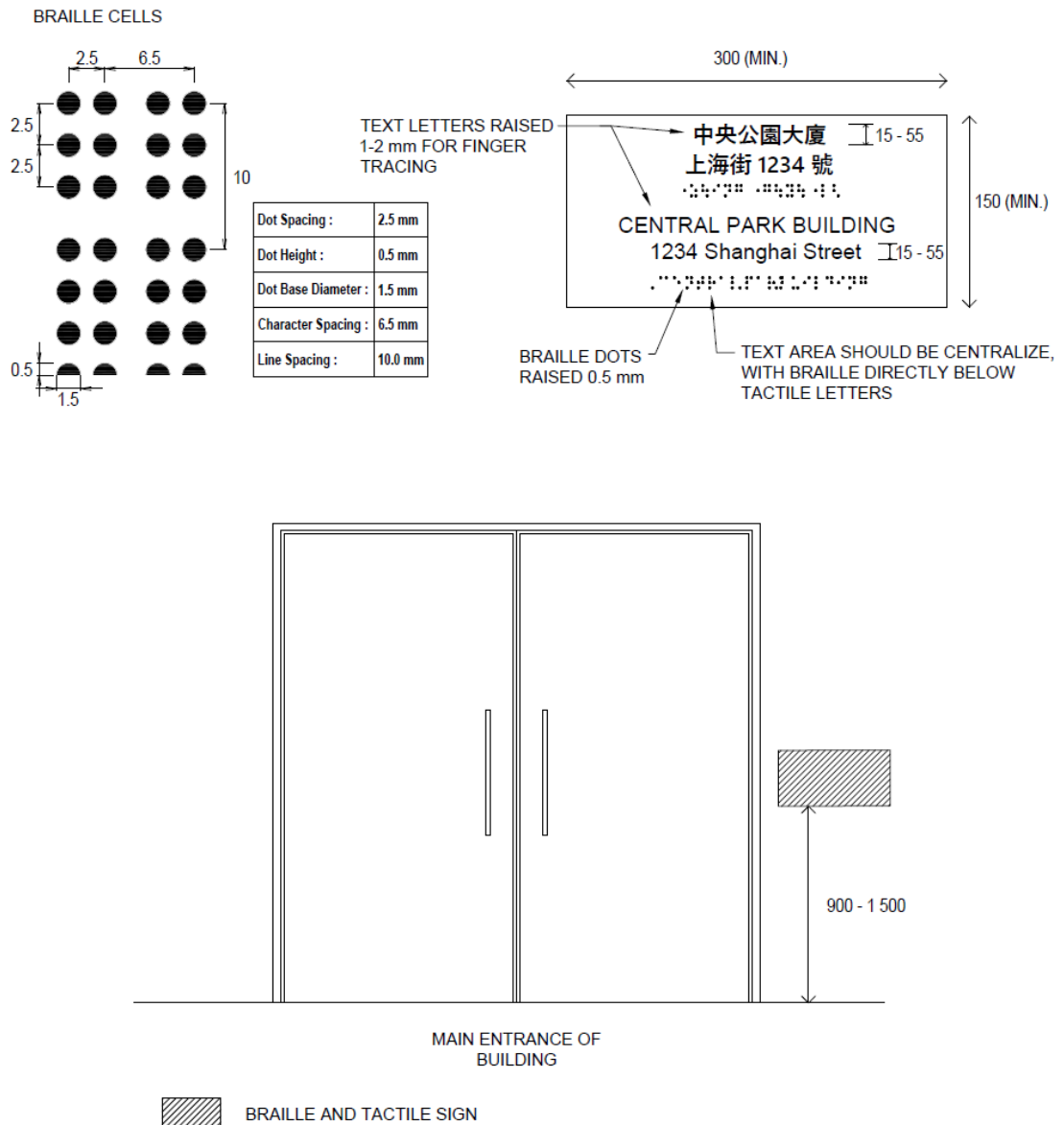


Figure 55 - Proportional Layout for International Symbol of Access for Hearing Loss

Braille and Tactile Sign

- (5) Braille and tactile signs indicating the Chinese and English building name and address (i.e., street name with number), or an audible device shall be provided on the side of the main entrance door of a building at a height between 900 mm and 1 500 mm above the finished floor level. The Braille and tactile sign shall be horizontal with text and Braille aligned for easy reading. A sample sign with guidelines and specifications of Braille cells are shown in **Figure 56** below.

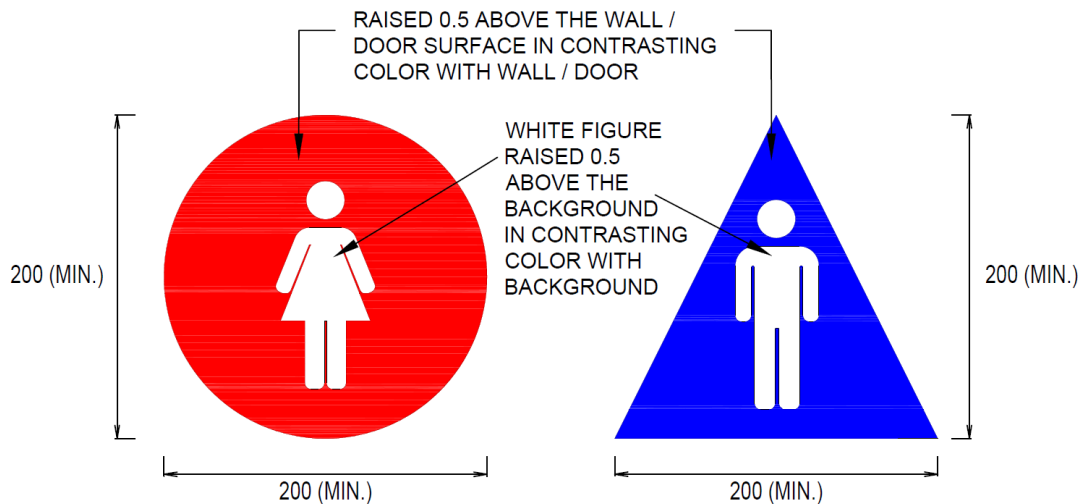


*ALL DIMENSIONS ARE IN mm

Figure 56 - Braille and Tactile Sign at Main Entrance of Building and Specifications of Braille Cells

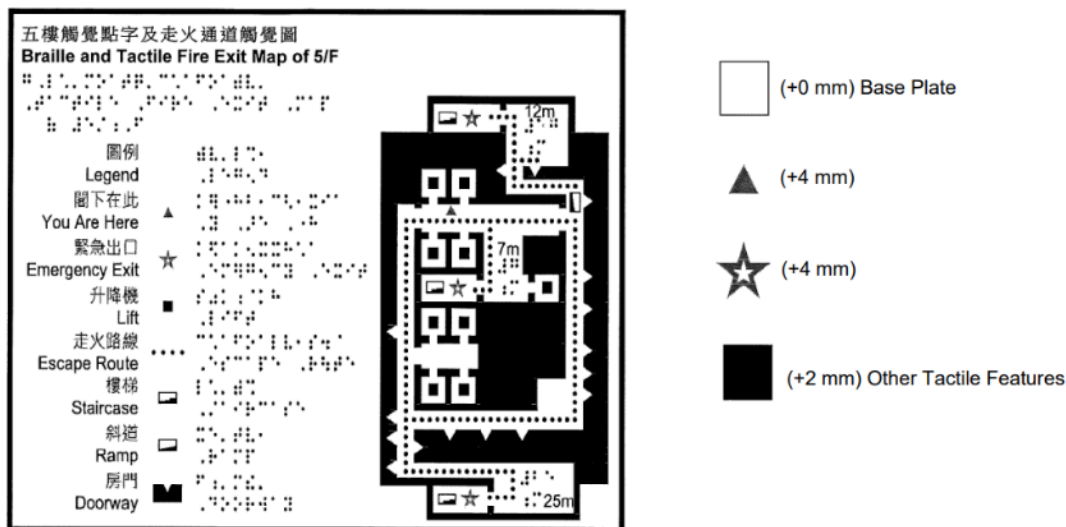
- (6) The audible device provided in accordance with paragraph 10.9(5) shall be clearly labelled (e.g., with a tactile button) and provide clear spoken information. The frequency shall be 300 Hz minimum to 3 000 Hz maximum for verbal annunciator. It shall be easily activatable (e.g. manual button) by persons with visual impairment. The audible information shall be provided in both English, Cantonese and Putonghua with a minimum volume of 65 dB(A) at a 1-meter distance.

- (7) Braille and tactile signs with information in English and Chinese or symbols shall be installed on adjacent wall or door of public toilet to indicate whether the toilet is for male, female or unisex. The signs shall be placed at 900 mm to 1500 mm above the finished floor level. Specifications of Braille cells are shown in **Figure 56**. A sample of tactile toilet signs is shown in **Figure 57**.
- (8) If there is no door, the sign shall be provided on the wall in front of the toilets.
- (9) A Braille and tactile fire exit map as shown in **Figure 58** shall be provided directly above the call button of the accessible lift in the lobby of the accessible lift in a building if a fire exit map for the use of the public is provided. The map shall be placed at 800 mm to 1200 mm above the finished floor level. Audible devices complying with paragraph 10.9(6) above should be provided.



*ALL DIMENSIONS ARE IN mm

Figure 57 - Sample of Tactile Toilet Signs



*ALL DIMENSIONS ARE IN mm

Figure 58 - Braille and Tactile Fire Exit Map

Luminance Contrast

- (10) Luminance contrast of not less than 50% shall be provided to differentiate the signs from the background.

10.10 Special Design Requirements to Assist Persons with Visual/Hearing Impairment

Braille and tactile floor plan

- (1) If a floor plan for the use of the public is provided, Braille and tactile floor plan with audible information showing the main entrance, public toilet and major common facilities shall be provided in building at a place which is conspicuous to persons with visual impairment.

Visual display board

- (2) If visual display board (such as LED) is provided, it shall be able to display the essence of the information broadcasted by the public address system in the building. Examples include display boards to inform travellers of the arrival and destination of the MTR trains.

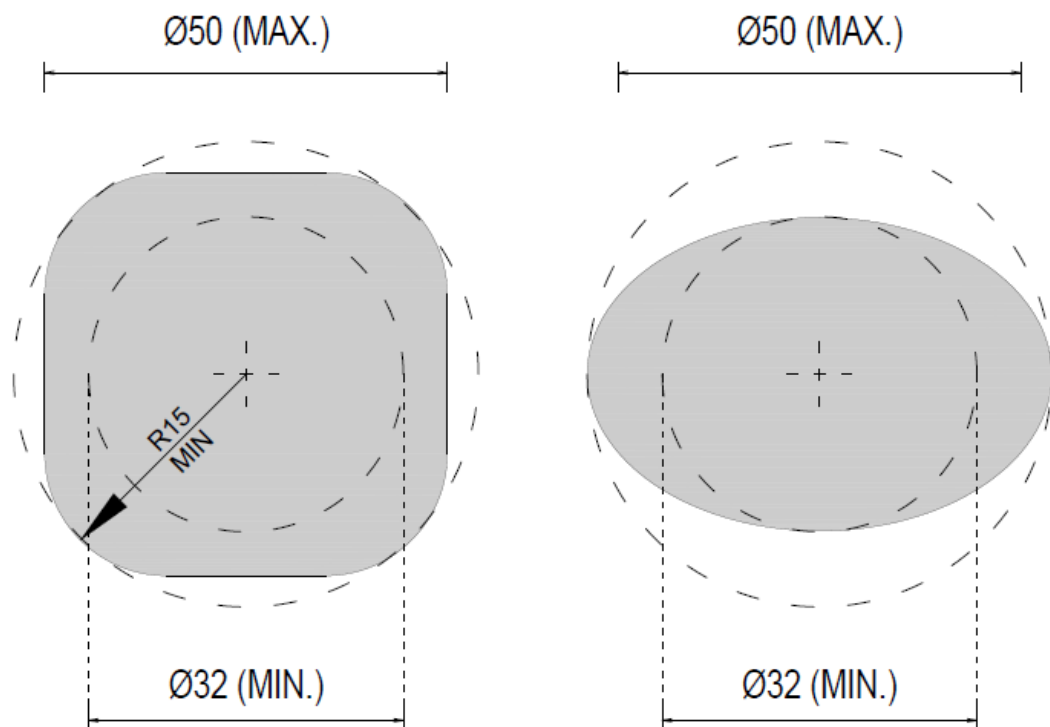
ADVISORY SECTION

Advisory Design Requirements

10.11 Handrails

Handrail Design

- (1) Handrail should be designed to provide easy, firm and comfortable grip to all users and should have no obstruction when people slide their hands along the handrail. Handrail in shape other than tubular should provide user a grip similar to tubular handrail and adequate space should be provided for installation of the Braille and tactile information as stated in [paragraph 10.2\(8\)](#). Examples of acceptable handrail cross section profiles (other than tubular shape) are shown in **Figure 59**.

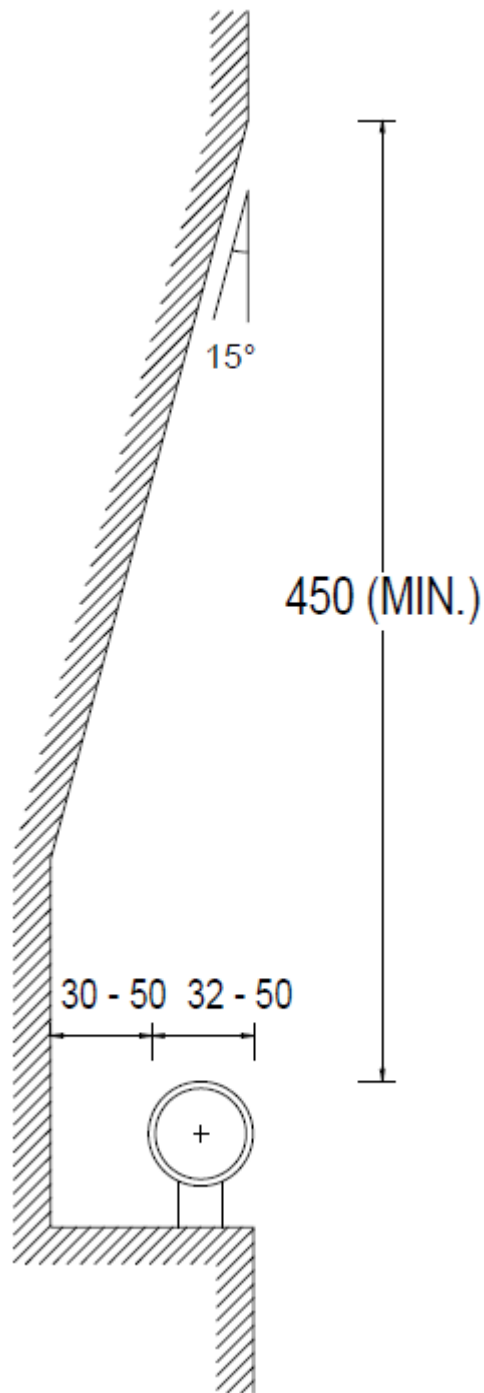


*ALL DIMENSIONS ARE IN mm

Figure 59 - Examples of Acceptable Handrail Cross Section Profiles (other than tubular shape)

Clear Space

- (2) Where the wall has a rough surface, the clear space should be not less than 45 mm between the handrail and the wall.
- (3) A recess containing a handrail should extend not less than 450 mm above the top of the handrail as illustrated in **Figure 60**.



*ALL DIMENSIONS ARE IN mm

Figure 60 - Handrail in Recess

Lower Handrail

- (4) One more lower handrail should be provided in accordance with the requirements under paragraph 10.2(5) for schools and places of public entertainment.

Luminance Contrast

- (5) Handrails should have a minimum luminance contrast of 30% with the surrounding wall surfaces.

Slip-resistant

- (6) The surface of the handrails provided under paragraph 10.2 should be slip-resistant, e.g. stainless steel with hair-line finish.

Braille and Tactile Information for Exit of Building

- (7) For landings of staircase leading to the exit of a building, Braille and tactile information on exit of building (see **Figure 61**) should be provided on handrails of staircase as illustrated in **Figure 17**.

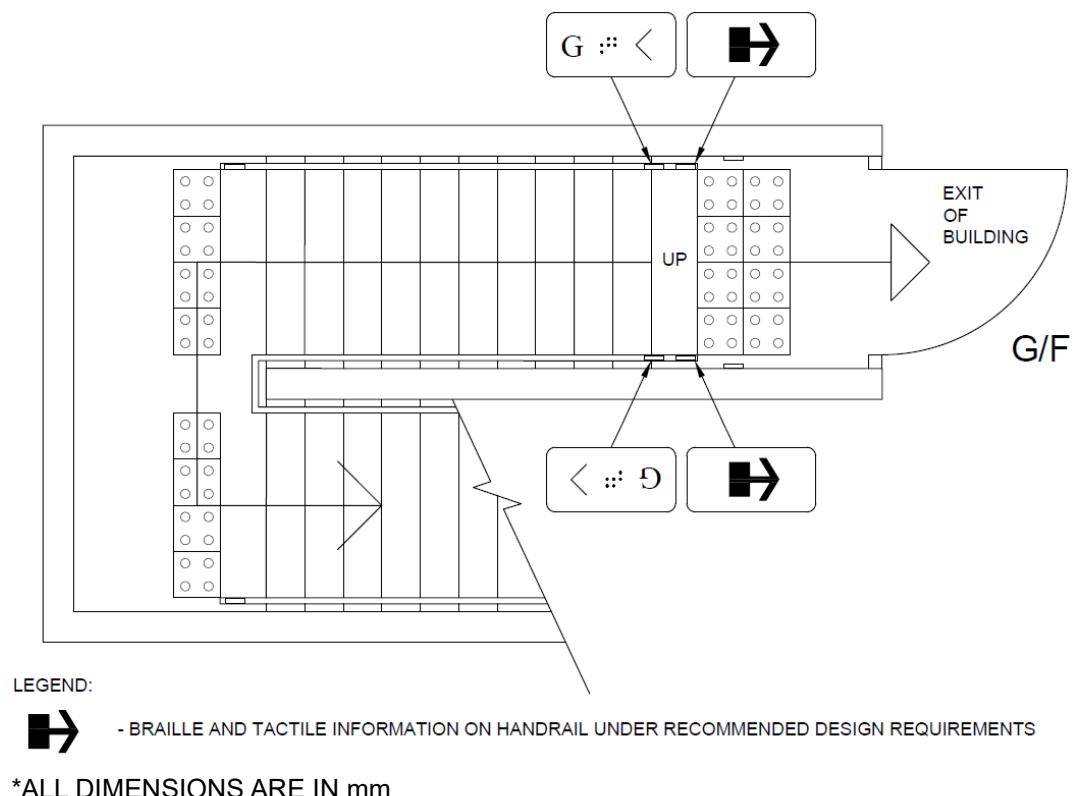


Figure 61 - Braille and Tactile Information on Handrail of Required Staircase

10.12 Tactile Guide Path

- (1) For floor area of more than 200 m² where the cues of physical edges such as walls and handrails are not present, tactile guide paths should be used to facilitate orientation of persons with visual impairment.

- (2) A minimum clearance between directional tiles/blocks of tactile guide path and an obstruction should be maintained as follows:

Clear Width of Accessible Route	Minimum Clearance
1 200 mm – less than 1 350 mm	450 mm
1 350 mm – less than 1 500 mm	500 mm
1 500 mm or above	600 mm

- (3) On each accessible floor, tactile guide paths should lead to at least two escalators with audible signal indicating the direction of travel.
- (4) The tactile guide path should be constructed from durable materials capable of withstanding pedestrian traffic and, where exposed to outdoor conditions, should be weatherproof to maintain long-term tactile detectability and slip resistance.
- (5) Tactile guide path should have a minimum luminance contrast of 70% with the floor surfaces.
- (6) Tactile guide path at the point of access at the lot boundary of a private development as required under [paragraph 10.3\(2\)](#) should be designed and installed to enable connection to the tactile guide path outside the lot boundary so as to achieve a smooth and continuous access from a public street to the private development.

10.13 Door

External Doors

- (1) Sliding/folding/double-action doors should be provided as far as practicable, with door handles or grab rails should be positioned between 950 mm and 1 050 mm above the finished floor level (measured from the top surface of the grip).
- (2) External door should be single-action and open outwards to obviate high tension in spring closers in sustaining wind pressure.

Latched Doors

- (3) Where door is latched, lever-type handle should be used.

Kick-plates

- (4) All doors which allow the passage of wheelchairs should have kick-plates of not less than 200 mm high fitted on the face which swings away.

Vision Panels

- (5) Transparent vision-panel should be provided to door in between accessible path. The vision-panel should be installed with bottom edge not more than 1 000 mm and top edge not less than 1 500 mm above the finished floor level

Glass Doors

- (6) A sign should be provided or the leading edge should be marked to indicate glass.
- (7) The marking provided at a frameless glass door should consist of at least a horizontal band with a minimum height of 100 mm and contrasting colours with the background to assist visibility. The marking including those in broken or solid lines, patterns or company logos, etc., should cover at least 10% of the glazing area within the zone between 900 mm and 1 500 mm above the internal finished floor. If the horizontal band design is not adopted, other types of marking should still be designed horizontally across the glass door. Such kind of marking should cover at least 30% of the glazing area within the zone between 900 mm and 1 500 mm above the internal finished floor.

Luminance Contrast

- (8) Door handle of manually operated doors and control switch or button of door with powered open devices should have a minimum luminance contrast of 30% with the background finishes.

Doors to Areas with Accessible Facilities

- (9) Fire rated doors leading to areas with accessible facilities should be provided with hold-open device except for doors to a required staircase forming part of a protected exit and its protected lobby.

10.14 Illumination

- (1) To enhance the illumination level of the lift lobby of upper floors, corridors, accessible paths and staircases, additional lighting to be provided may be activated by sensors.

Lighting for Steps

- (2) Lighting should be provided at the entrances and exits to provide a contrast between the treads and the risers.

Illumination Level

- (3) Uniformity of illumination level should be maintained throughout the designated areas and toilets.
- (4) Higher illumination level at the entrance doors to flats and the exit doors should be provided.
- (5) External common areas such as landscaped areas and recreational spaces should have an illumination level of not less than 20 lux measured at the finished floor level.

10.15 Public Information or Service Counters

Design of Counter

- (1) The portion of public information/service counter provided in accordance with paragraph 10.6(2) should be readily adjustable in height to suit different wheelchair users. The initial position of the desk should comply with paragraph 10.6(2) and paragraph 10.6(5).
- (2) Additional groove at the edge of the counter, or alternative effective design for securing more than one cane, walking stick or other assistive tool, should be provided in accordance with paragraph 10.6(4).

Leg Space

- (3) Leg space of a depth between 480 mm and 600 mm should be provided for public information or service counters.

10.16 Assistive Listening System

- (1) Special-purpose receivers are required by infrared and radio frequency systems while hearing aids equipped with a T-switch are capable of receiving the signal from an induction loop system. Designers should seek expert advice concerning the most appropriate system for their purposes.

Coverage

- (2) The assistive listening system should cover all floor areas of the room or space served by the system. If a system requiring the use of receivers or the like is adopted, a minimum of 2 receivers should be provided. The number of receivers provided must be not less than 1 for every 50 persons (or part thereof).

10.17 Switches and Controls

- (1) Except as otherwise provided in Chapter 7 for lifts, the controls for the operation of building services or safety devices including electrical switches, light switches, thermostats, intercom switches and card reading machines which are intended to be accessible to wheelchair users should be located between 450 mm and 1 200 mm above the finished floor level (see **Figure 62**).

Charging Points for Powered Wheelchair

- (2) Charging points for powered wheelchair should be provided in the common areas of a building.

Positioning

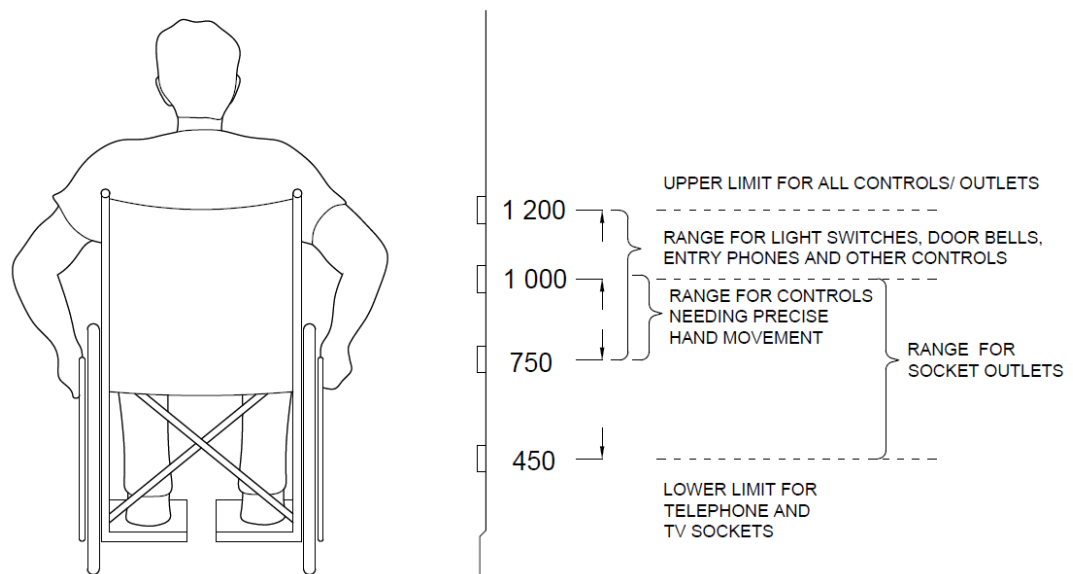
- (3) Electric sockets should be located not lower than 450 mm above the finished floor level.

Luminance Contrast

- (4) Light switches and socket outlets should have a minimum luminance contrast of 30% with their background to indicate their locations.

Controls

- (5) Controls should:
 - (a) have a minimum luminance contrast of 30% with the background finishes;
 - (b) not require tight grasping or twisting;
 - (c) be visible from a seated position; and
 - (d) not be of small button type.



*ALL DIMENSIONS ARE IN mm

Figure 62 - Heights of Switches and Controls

10.18 Remote Signage Systems

Remote Directional Signage

- (1) A remote directional signage system which transmits a voice message by means of infrared or the like detectable by a proper receiver should be provided to facilitate access and movement of persons with visual impairment.

10.19 Drinking Fountains

Provision

- (1) Where drinking fountains are provided, at least one of the drinking fountains on a given floor should have a design complying with this section.

Spout

- (2) The fountain spout should:
 - (a) have an opening located between 750 mm and 900 mm from the floor or ground surface;
 - (b) be located near to the front of the unit;
 - (c) direct the water flow in a trajectory that is parallel or nearly parallel to the front of the unit; and
 - (d) provide a flow of water at least 100 mm high to allow for the insertion of a cup or glass.
- (3) Automatic controls or controls with handles of the lever type operable with a closed fist should be provided.

Control

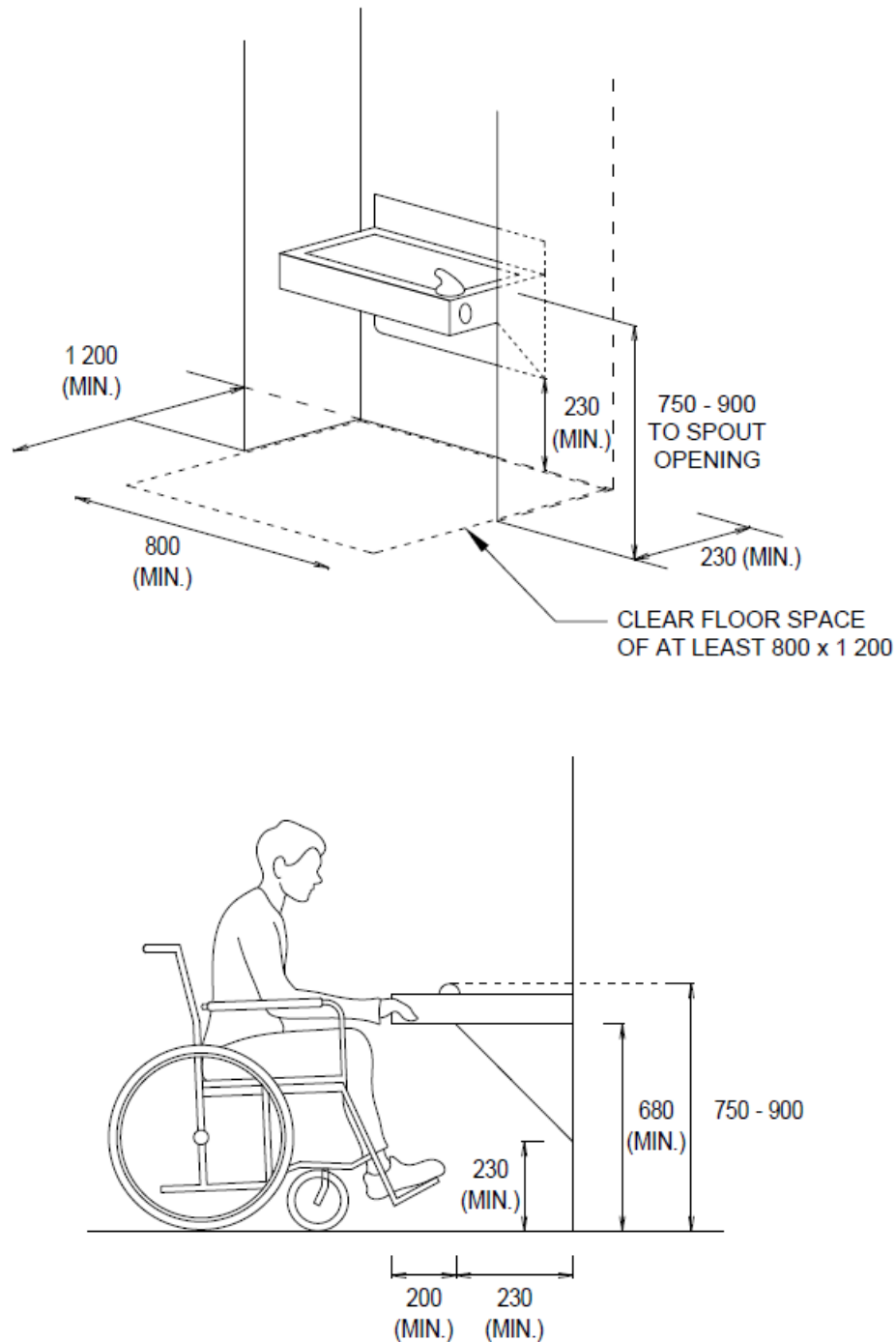
- (4) Controls should:
 - (a) be at or near to the front of the fountain;
 - (b) be operable with one hand;
 - (c) require no tight grasping, pinching, or twisting of the wrist;
 - (d) require a force of 22 N or less to activate; and
 - (e) be hand-operated control with minimum dimension of 50 mm.

Spatial Arrangement

- (5) The spatial arrangement should allow for the provision of:
 - (a) a clear floor space of at least 800 mm x 1 200 mm;
 - (b) a clear knee space of at least 800 mm wide, 200 mm deep and 680 mm high between the bottom of the apron and the floor or ground; and
 - (c) a toe space not less than 800 mm wide, 230 mm deep and 230 mm high.
- (6) All wall-mounted drinking fountains should be placed in an alcove to eliminate the hazard of collision to persons with visual impairment.
- (7) A wall guard should be installed to protect a drinking fountain that extends into a corridor and has an open space underneath.

- (8) Directional signs should be provided at prominent and conspicuous positions to guide persons to the locations of drinking fountains.

Dimensions relating to the design of drinking fountains are shown in **Figure 63**.



*ALL DIMENSIONS ARE IN mm

Figure 63 - Drinking Fountain

10.20 Signs

Braille and Tactile Sign

- (1) Each locker door in a sports complex, public swimming pool complex or public area should be provided with the locker number in Braille and raised Arabic numerals/characters*.

*The locker key holders should be provided with raised Arabic numerals/characters.

- (2) A Braille and tactile plan (see **Figure 64**) should be provided for the toilets with five or more water closets (urinal in a public male toilet and changing room is regarded as a water closet for this purpose) in a public toilet/changing room. The plan should be installed on the wall adjacent to the door of the public toilet/changing room, or their entrance if entrance door is not provided. The plan should be placed at 900 mm to 1500 mm above the finished floor level.

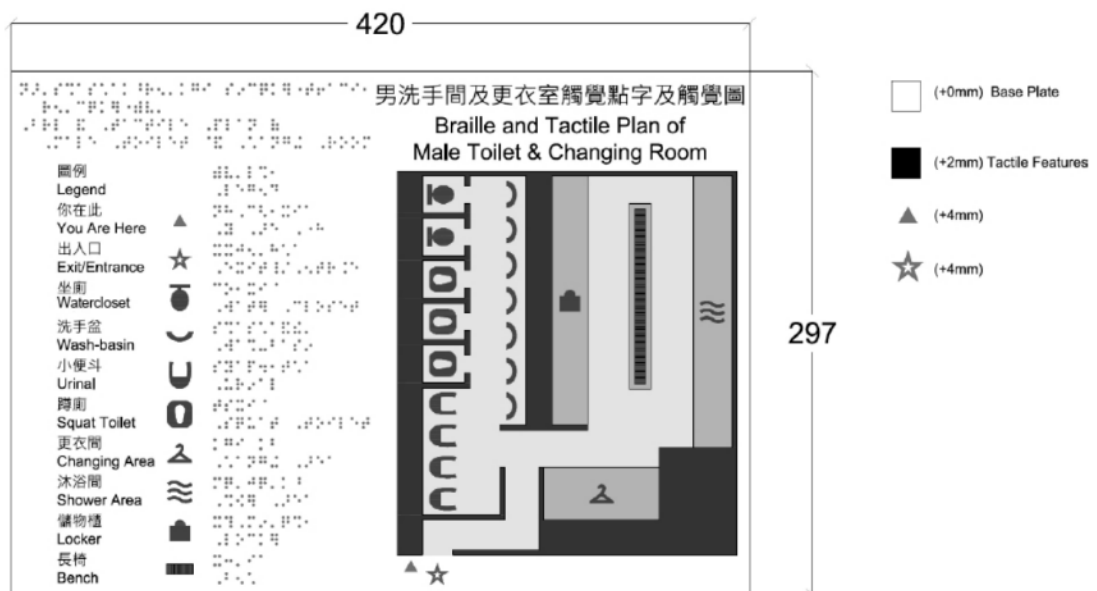


Figure 64 - Example of Braille and Tactile Plan of Toilet and Changing Room

Size of Text and Signs

- (3) Based on the viewing distance, the size of texts and signs should be provided in accordance with the following tables:

English letter	
Expected viewing distance	Recommended height of letters
1 m or below	30 mm
2 m	40 mm
3 m	90 mm
4 m	120 mm
5 m	155 mm
6 m or above	180 mm

Chinese character	
Expected viewing distance	Recommended height of letters
1 m or below	40 mm
2 m	55 mm
3 m	120 mm
4 m	160 mm
5 m	210 mm
6 m or above	245 mm

Signs	
Expected viewing distance	Recommended size
7 m or below	90 mm x 90 mm
7 m to 18 m	150 mm x 150 mm
18 m or above	300 mm x 300 mm to 600 mm x 600 mm

Design of Signs

- (4) Clear and simple language should be adopted and jargons should be avoided.
- (5) Bilingual sign (in both Chinese and English) satisfying the requirements under paragraph 10.9(5) is recommended.
- (6) In addition to paragraph 10.9(7), all signs are recommended to be provided with Braille and tactile information at a height between 900 mm and 1 500 mm above the finished floor level.

- (7) Signs should be placed at key decision points, such as entrances, exits, elevators, restrooms and other critical areas without visual obstruction.
- (8) Adequate directional signs should be provided for wayfinding.

10.21 Special Design Requirements to Assist Persons with Visual/Hearing Impairment

Location of Sign

- (1) Sign should be erected to indicate clearly the locations of accessible routes through the building.

Luminance Contrast

- (2) Luminance contrast of not less than 70% should be provided to differentiate the international symbol of accessibility from the background, either light-on-dark or dark-on-light. The commonly employed colours are white for the wheelchair figure and blue for the background.

Lettering and Colour

- (3) Lettering should be plain and legible, e.g. Helvetica (medium) using lower case letters except for initial capitals.
- (4) Corners of sign should be rounded.
- (5) Chinese characters should be legible e.g. “Haak Tai” style.
- (6) Sign should be in raised characters.
- (7) The system of sign used should be clear and consistent.

Braille and Tactile Sign/ Audible Sign

- (8) If public address system is provided to convey information to the public in a building, means of conveying the same or equivalent information to persons with hearing impairment should be provided.
- (9) The Braille and tactile sign for public toilet required under paragraph 10.9(7) should be installed on the wall adjacent to the open side of the toilet door or toilet entrance if toilet door is not provided.

Visual Sign

- (10) Visual display board (such as LED) should be provided in public waiting areas other than those required in Table 2 and where there is an announcer installed to regularly convey information to the people inside. The visual display board should be able to display the essence of the information so announced.

Braille and Tactile Floor Plan

- (11) The Braille and tactile floor plan provided in accordance to paragraph 10.10(1) should comply with the following requirements:
- (a) The content of the Braille and tactile floor plan should be positioned between 900 mm and 1 500 mm above the finished floor level.
 - (b) A “You Are Here” marker in the form of a triangle raised by not less than 4 mm should be provided. Such marker should have a luminance contrast of 30% with the background.
 - (c) The reading surface of the Braille and tactile floor plan should be inclined at an angle of not less than 45 degrees and not more than 60 degrees from the horizontal so as to face towards the user.
 - (d) Where the bottom edge of the Braille and tactile floor plan is not less than 680 mm above the finished floor level and the recess beneath such edge should not be more than 300 mm deep so as to allow detection of obstruction by a cane.
 - (e) The audible device should provide audio information including the location of the Braille and tactile floor plan, descriptions of the major common facilities and directions to the main entrance, public toilet and the major common facilities.

CHAPTER 11 ADVISORY DESIGN GUIDELINES

11.1 General

This Chapter outlines desirable features that aim to create a more inclusive built environment accessible and functional for a broader spectrum of our community. While these features are worth promoting, flexibility should be allowed for the building industry to consider adoption at its own pace.

The guidelines in paragraph 11.6 aim to provide practical alternatives, drawn from past experience, to guide alterations and additions, or improvement works in existing buildings where spatial or structural constraints exist.

11.2 Elderly-friendly Design Guidelines

General

- 11.2.1 The guidelines set out under paragraph 11.2 aim to provide a safe and comfortable living environment for the elderly who may gradually experience declining abilities; promote active ageing by enabling the elderly to achieve autonomy and independence without the help of others; and promote the elderly's well-being, in particular social interaction among the elderly and between the elderly and other age groups. Building owners and designers are encouraged to adopt these guidelines in the building design as far as practicable.

Elderly-friendly Design Guidelines

- 11.2.2 The elderly-friendly design guidelines set out as recommended design requirements are grouped into four categories, namely Mobility and Accessibility (paragraph 11.2.2(1) – A1 to A17), Design Adaptability (paragraph 11.2.2(2) – B1), Well-being (paragraph 11.2.2(3) – C1 to C9), and Gerontechnology (paragraph 11.2.2(4) – D1).

Mobility and Accessibility

- (1) The functional capacity (such as muscular strength, balancing, vision and hearing) of a person often declines gradually as ageing progresses, which may increase the risk of accidents and injuries. One crucial factor for ageing in place is to ensure safe mobility and accessibility of the built environment, both inside and beyond the residential units.

Improving mobility and accessibility can significantly enhance the safety, health and well-being, as well as the convenience of the elderly, promoting their independence and reducing the risk of falls and other health complications.

In addition, ensuring smooth circulation between various major destinations will facilitate the convenient and efficient movement of building occupants, especially for the elderly.

Location	Features	Recommended Design Requirements
Residential Units		
Bathrooms	A1. Bathrooms	(a) Adequate clear space of not less than 1 200 mm diameter should be allowed for turning of wheelchair and transfer of users. (b) Grab bars should be provided.
	A2. Doors	(a) Doorways of not less than 850 mm should be provided to allow a wheelchair to pass through. (b) Sliding doors should be provided for easy and convenient use. Where sliding doors could not be provided, doors that take up less space, such as folding or double swing doors, should be provided.
	A3. Bathtubs / Showers	(a) Bathtub should have either built-in seat at the head end of bathtub or attachable portable seat that fastens securely to the tub when needed. (b) Showers should be of size not less than 1 500 mm x 900 mm with a folding seat to allow transfer of users. (c) Grab bars should be provided.
	A4. Floor Surfaces	(a) Floor surfaces should be firm and provided with slip-resistant floor surfaces in accordance with <u>Appendix C</u> of this Manual. Shiny and reflective floors such as marble, glazed tiles and the like should be avoided.

Location	Features	Recommended Design Requirements
Others	A5. Fittings and Furniture	(a) Double-switching systems should be considered. Switches and controls should be provided at reachable level. For bedrooms, switches and controls should be provided near bedside to avoid crossing the room in the dark. (b) Lever-type controls and handles should be used. Knob handles, push operated and self-closing type faucet controls should be avoided. Examples of elderly-friendly door handles and faucets are provided in Figure 65. (c) Cupboards should not be installed at high levels. Overhead cabinets should be avoided. Pull-down shelves should be provided for cupboards at high level or overhead cabinets. (d) Drying racks and laundry poles projecting from the external walls of the building should be avoided. Flip hooks should be provided at main entrance for hanging items.

Location	Features	Recommended Design Requirements
Common Areas and Facilities		
Accessible Routes	A6. Corridors /Paths and Lobbies / Lift Lobbies, Steps /Staircases, etc.	(a) Accessible routes should be without steps, thresholds, small ramps or kerbs, wherever possible. Where changes in level are unavoidable, handrails or grab bars should be provided. The floor and wall surface along the level of difference should be in luminance contrast. (b) Handrails should be provided along common corridors. (c) Steps and staircases should be designed with wider treads and lower risers.
	A7. Handrails	(a) Subject to regulation 73 of the B(P)R, handrails to corridors, steps and staircases should be lowered to a height between 810 mm and 900 mm from the finished floor level to the top of the handrails. Handrails should be of materials such as timber or with plastic coated surfaces for easier grip.
	A8. Doors	(a) Automatic doors should be provided as far as practicable. (b) If door closing devices are installed, they should be designed to allow external and internal doors to be opened with horizontal forces of not more than 28N and 18N respectively. (c) Lever-type handles should be used. Knob handles should be avoided. Examples of elderly-friendly door handles are provided in Figure 65.

Location	Features	Recommended Design Requirements
	A9. Floor Surfaces	(a) Floor surfaces should be firm and provided with slip-resistant floor surfaces in accordance with <u>Appendix C</u> of this Manual. Shiny and reflective floors such as marble, glazed tiles and the like should be avoided. Open jointed pavers or aeration paver blocks with uneven or very rough surface should be avoided at external open spaces.
Lifts, Escalators and Passenger Conveyors	A10. Larger Lift Cars	(a) Larger lift cars with clear depth of not less than 1.5 m should be provided.
	A11. Seating within Lift Cars	(a) Resting facilities such as folding seats or lean on railings should be provided within the lift cars except fireman's lifts.
	A12. Lift Control Panels or Buttons	(a) Control panels or buttons at the lift lobbies should be located at 1 050 mm above the finished floor level. They should be easy to operate and have a minimum luminance contrast of 30%. (b) Lift control buttons including the numbers on floor numbering buttons in the lift cars should be back-lit. (c) Audio indication helping the elderly to locate themselves should be provided in the lift cars
	A13. Escalators and Passenger Conveyors	(a) Escalators and passenger conveyors should be designed with slower speed.

Location	Features	Recommended Design Requirements
Toilets and WC Cubicles	A14. Toilet and WC Cubicles	(a) Doors of toilets and WC cubicles should open outward or both ways. Locks and latches should be of larger sizes and be able to open from outside with a coin. (b) Lever-type controls and handles should be used. Knob handles, push operated and self-closing type faucet controls should be avoided. Examples of elderly-friendly door handles and faucets are provided Figure 65. (c) Emergency alarm system should be provided. Emergency call bell should be provided in accordance with <u>paragraph 9.2</u>. Should pull-cord be installed, they must be extended to floor-level.
Car Parks	A15. Vehicular Entrances	(a) Audio and visual warning signal should be provided at vehicular entrances of car parks.

Location	Features	Recommended Design Requirements
Others	A16. Signs and Decorations	(a) Signage Bright primary colours should be used as contrast or highlight for easier differentiation ¹. Monotone colour should be avoided. Colour combination of signs such as yellow figure on black background or white figure on blue background is recommended. In addition to colour contrasts, clear signs with bigger font size should be provided for elderly-friendly facilities and floor numbering. (b) Wayfinding Different colours for different building blocks, floor levels, zonings or areas of different functional purposes are recommended to aid wayfinding. Different floor surfaces or colours should be considered for tactile / visual cues for navigation. Noticeable features, artworks, landmarks such as sculptures, planters, or fountains should be provided at different entrances, zonings and facilities for easy identification.

¹ In general, elderly may become less sensitive to colours having shorter wavelengths (blues, greens and violets) as their lenses thicken and yellow with age. Warm colours with longer wavelengths (reds, oranges and yellows) are easier for them to differentiate.

Location	Features	Recommended Design Requirements
	A17. Lighting Arrangement	(a) All common areas of a building should have an illumination level of not less than 120 lux measured at the finished floor level, and uniformity of illumination level should be maintained for any space (b) Where additional lighting activated by sensors is provided to enhance the illumination level of the lift lobby of upper floors, corridors, accessible paths and staircases, the length of the corridors and paths and the walking speed of the elderly should be considered in setting the duration of such sensors. Alternative or stand-by light sources should be provided to illuminate any spaces in case of power failure. Consideration should be given to ensure gradual transition of lighting levels from one place to another. The use of natural lighting or ambient artificial white light should be encouraged. The use of wall-mounted light or peripheral lighting from floor lamps are superior to a central ceiling source as the formation of shadows can be avoided. Bare light bulbs producing glare which would cause pain to the aging eyes should be avoided. All interior spaces should be lit at a consistent and even level, from floor to ceiling and from wall to wall. High contrast

Location	Features	Recommended Design Requirements
		between shadow and light creating confusion and disorienting patterns should be avoided.

Design Adaptability

- (2) A person's living needs and preferences may change throughout his/her lifespan. Allowing the flexibility in modifying a residential unit without the need for structural alteration is a crucial aspect to facilitate ageing in place. Providing design adaptability in a home environment can accommodate evolving needs and preferences over time while obviating the need for relocation at old age. This enables the elderly to stay in the same and familiar environment and neighbourhood as they age, thus maintaining their family and community connections.

Location	Features	Recommended Design Requirements
Residential Units		
Individual Flats	B1. Convertibility without Structural Alteration	(a) To suit the needs of the elderly at different ages, the residential units should be convertible without the need for making structural alterations so as to accommodate the following elderly-friendly designs: <u>To reduce possible safety risks</u> (i) Raised deck or similar to overcome level difference between indoor living area and the balcony; (ii) Handrail / grab bars in toilet / shower area; (iii) Shower seat; and (iv) Low threshold and curbless walk-in shower. <u>To cater for the elderly on wheelchair</u> (v) Lower door viewer for main entrance (see Figure 66); (vi) Wider corridor within units with a clear width of not less than 950 mm; Large and / or two-way switches and controls at reachable level; (vii) Large and / or two-way switches and controls at reachable level;

Location	Features	Recommended Design Requirements
Residential Units		
		(viii) Manoeuvring area of not less than 1200 mm diameter in main entrance, kitchen, toilet and bedroom; (ix) Wider internal door with a clear width of not less than 850 mm; and (x) Knee space under kitchen sink and wash basin. <u>To improve convenience</u> (xi) Counter-top in kitchen with a depth of not more than 600 mm; and (xii) Adequate drainage provisions to enable adaptive design. See Sample Layout of Elderly-friendly Adaptive Design in Figure 67.

Wellbeing

- (3) Design features relating to well-being aim to promote a sense of contentment, encourage social participation and enhance intergenerational interaction and harmony. They reflect the important role that the built environment plays as they cater to the physical, mental and emotional needs of the elderly. Convenient and easy access to recreational facilities with equipment/ facilities that are designed for the elderly and a greater connection with nature can significantly improve the health and quality of life of the elderly, promote physical activity, social engagement, and sense of well-being.

These amenities may also serve “multi-generation” users which will in turn promote intergenerational harmony.

Location	Features	Recommended Design Requirements
Residential Units		
Habitable Space	C1. Windows	(a) Larger windows with lower window cills should be provided to increase the amount of natural light and allow views of the outside for the elderly when seated or on wheelchair (see Figure 68). This can have positive impact on moods and reduce feelings of depression and anxiety. Nevertheless, requirements of protective barrier should be complied with. Other concerns such as privacy and fear of height should also be considered. Sliding windows should be adopted to enhance daily use by the elderly.
Common Areas and Facilities		
Corridor / Paths and Lobbies / Lift Lobbies	C2. Lighting and Ventilation	(a) Large windows at the end of long and dark corridors that may cause glare should be avoided. Cross ventilation should be optimised in common areas such as corridors and lift lobbies.

Location	Features	Recommended Design Requirements
	C3. Resting Facilities	(a) Resting places with resting facilities, such as seats (including fold-down seats) or lean on railings, should be: (i) provided on stair landings; (ii) provided in corridors more than 50 m in length; or (iii) provided in typical lift lobbies. (b) Long corridors and passageways within transport stations, interchanges and passenger terminals, and public passages within a building connecting with transport stations, interchanges and passenger terminals should be provided with resting places with resting facilities. The resting facilities should be provided in recessed areas at maximum intervals of 50 m. Resting facilities for items (a) and (b) above should not reduce the statutory requirements on the clear width of access and manoeuvring space, and should not cause obstruction to access, circulation and exit routes.
Toilets and WC Cubicles	C4. Elderly-friendly Toilets	(a) Elderly-friendly toilets should be provided and located in easily accessible common areas or within a toilet with multiple cubicles complying with requirements under <u>paragraph 9.6(1)</u> of this Manual. See Sample of Elderly-friendly Toilet in Figure 39.

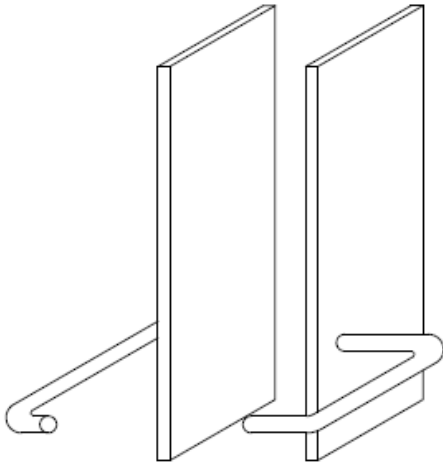
Location	Features	Recommended Design Requirements
	C5. Elderly-friendly Changing Station / Facilities in Accessible Toilets	(a) Elderly-friendly changing station / facilities with an adult sized, height adjustable changing bench, either wall mounted or free standing in accessible toilets should be provided.
External Recreation Spaces	C6. Elderly-friendly Resting Facilities	(a) Resting places with resting facilities, such as seats (including fold-down seats), lean-on railing, fixed chairs, tables and benches, steps, planter edges for sitting preferably with shelters or at covered areas should be provided at outdoor open space.
	C7. Elderly-friendly Fitness Equipment	(a) Elderly-friendly fitness equipment such as twister and stepper, pull-down equipment, stepping platforms, chest press equipment, tai-chi wheels and areas with safety flooring system should be provided.
	C8. Elderly-friendly Planters for Community Farming	(a) Knee spaces should be provided under the planters for community farming which can help to improve focus, memory and cognitive function (see Figure 69).
Others	C9. Choice of Material	(a) Sound-absorbing materials should be used for floors and walls to avoid echoes. Non-glare or low gloss finishes on floors, matt paint or textured wallpaper on walls should be considered to help reduce glare. Glass or reflective material should be avoided.

Gerontechnology

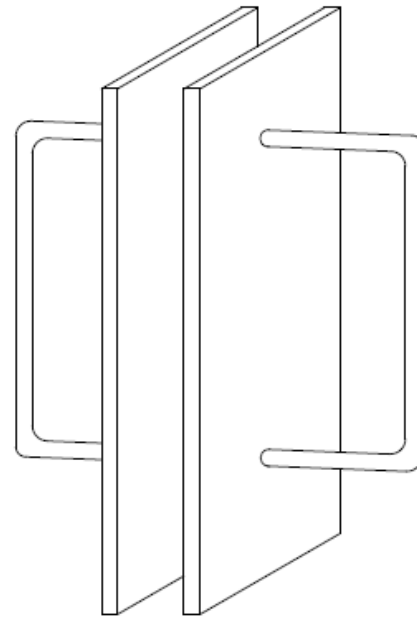
- (4) There are new technologies that can enhance safety and convenience, facilitate ageing in place, and foster confidence and independence of the elderly. Effective use of technology is not only a solution to elderly care but also an enabler for the elderly to lead a more independent life.

Gerontechnology helps the elderly with declining capacities to support themselves. Hardware and software applications to assist the elderly and/or their carers are under rapid development, and the elderly are becoming more technologically literate. By integrating assistive technologies into daily life, the elderly can overcome certain physical limitations, maintain cognitive function, and access vital services more easily. Some intelligent devices may also help detect accident of an unattended elderly. The adoption of gerontechnology requires a robust and accessible infrastructure that supports both technological and social integration. This includes high-speed and stable internet access in residential units and common areas of the buildings.

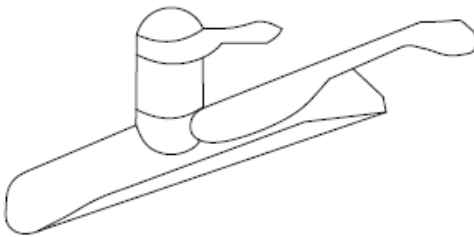
Location	Features	Recommended Design Requirements
Residential Units and Common Areas	D1. Network	(a) Enabling works for high-speed and stable internet provision (e.g. 5G) / Global Positioning System / Radio Frequency Identification Technology to support the use of gerontechnology and Internet of Things (e.g. smart card system for main entrance; water flow sensor; call caring services; motion sensor; window opener; artificial intelligence and robotic support; remote gas heater control; and door sensor) should be provided in residential units and common areas of the buildings.



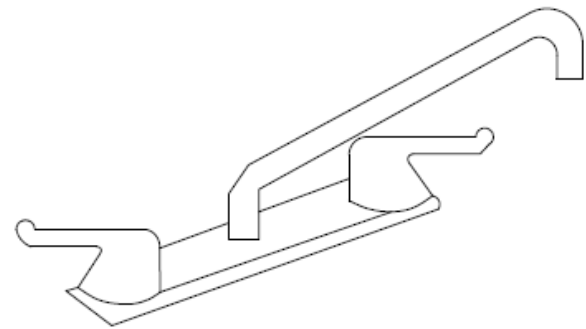
LEVER HANDLES



PUSH/PULL PLATE DOOR PULL

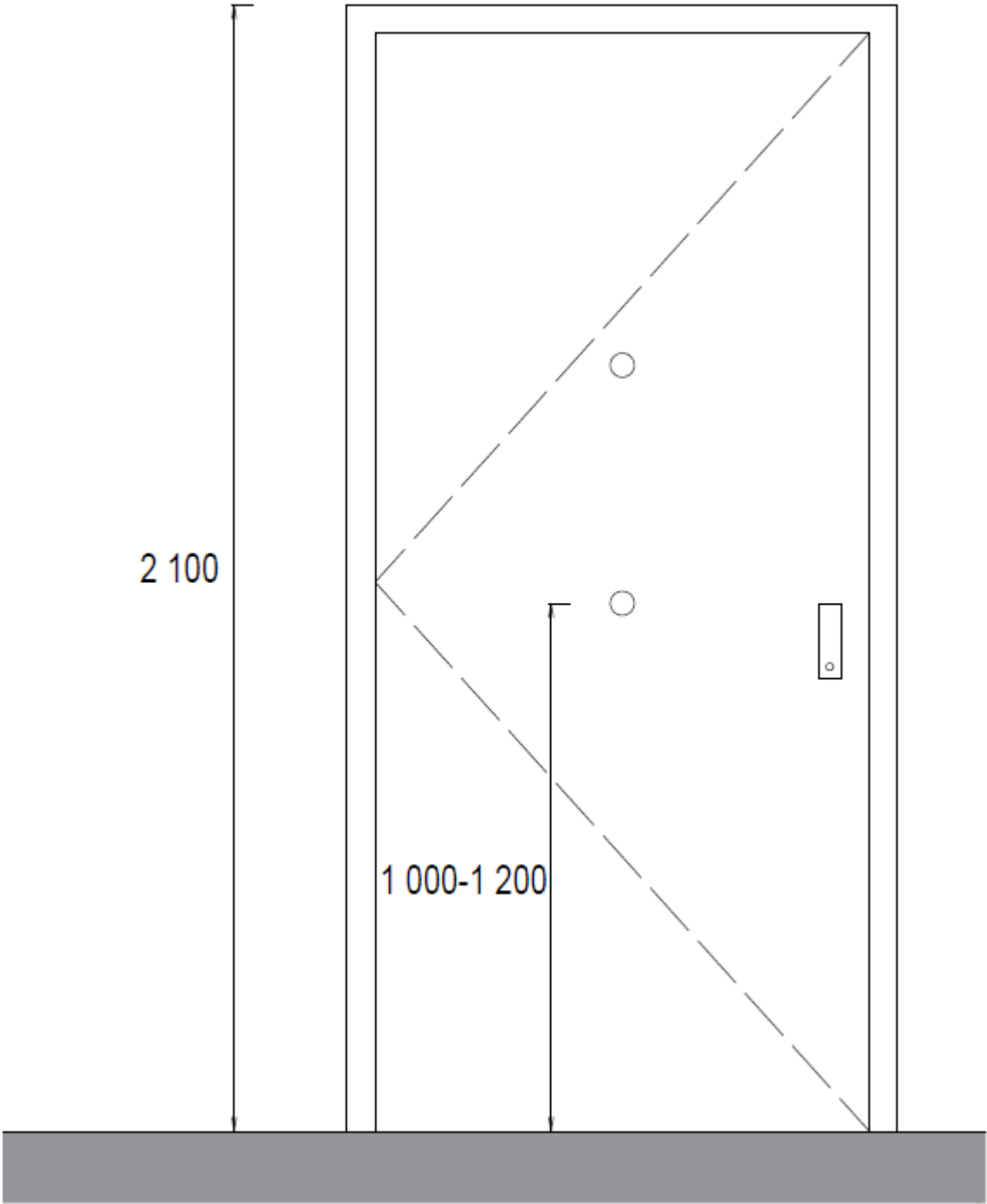


LONG LEVER HANDLE
(RECOMMENDED)



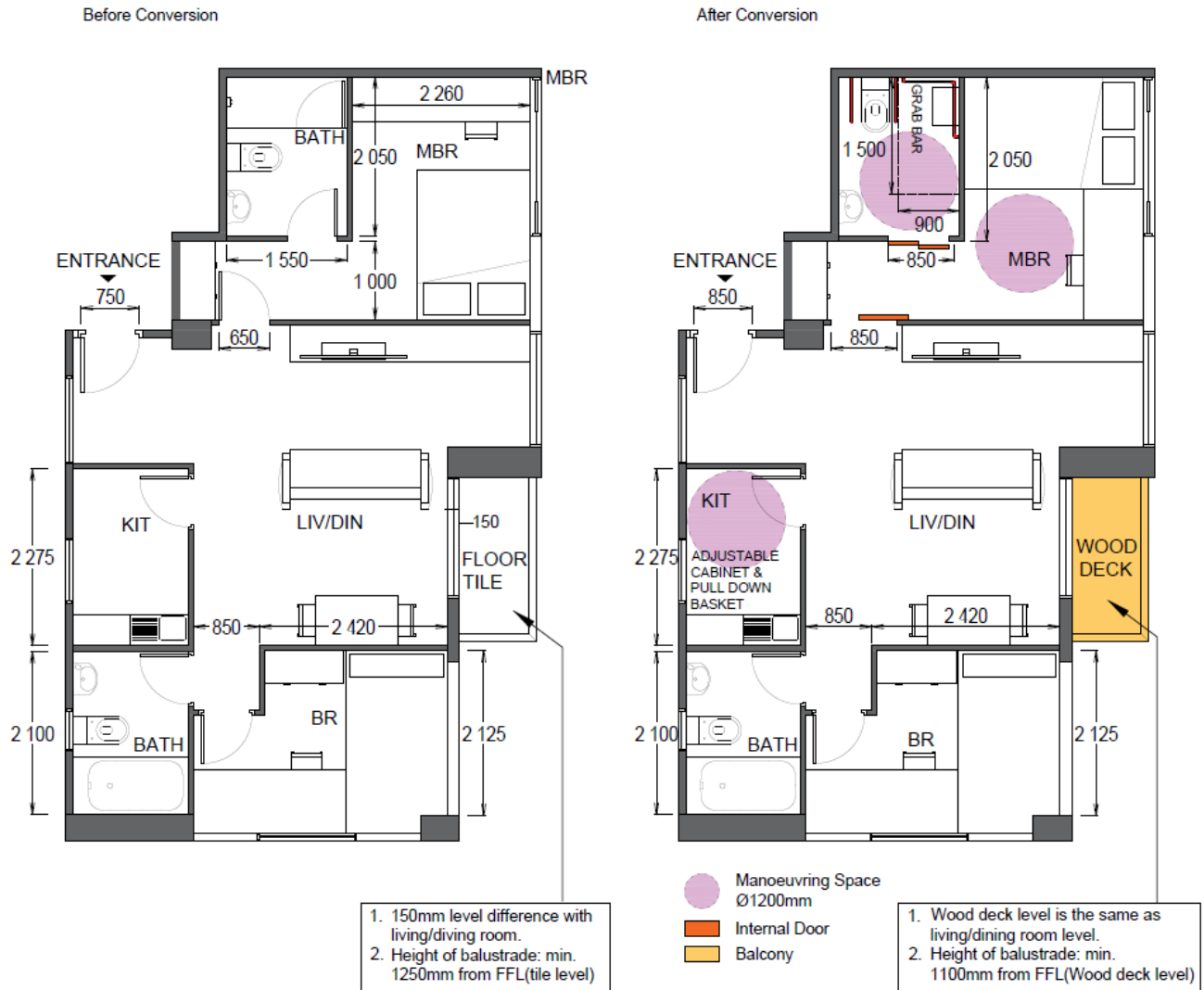
SINGLE LEVER HANDLE
(RECOMMENDED)

Figure 65 - Examples of Elderly-friendly Door Handles and Faucets



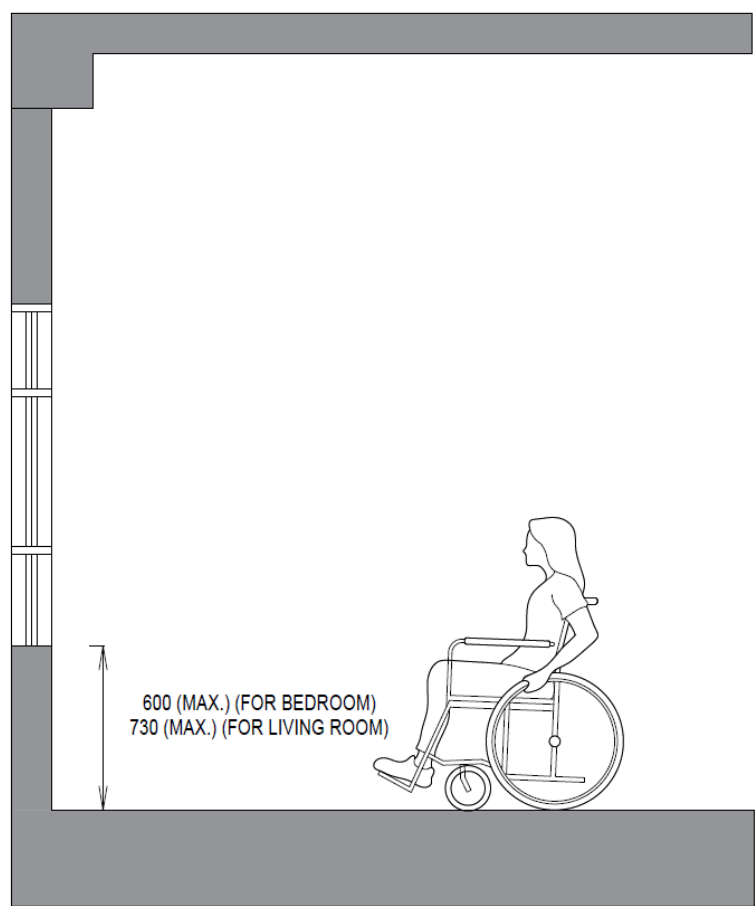
*ALL DIMENSIONS ARE IN mm

Figure 66 - Sample of Lower Eye Viewer



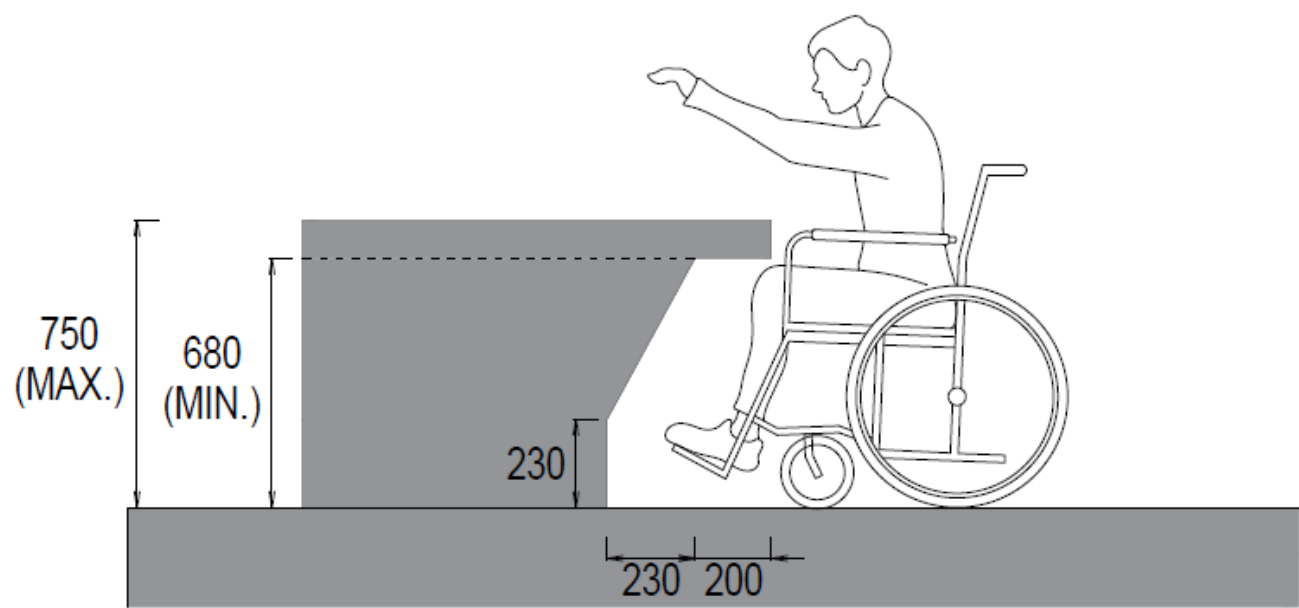
*ALL DIMENSIONS ARE IN mm

Figure 67 - Sample Layout of Elderly-friendly Adaptive Design



*ALL DIMENSIONS ARE IN mm

Figure 68 - Example of Lower Window Level



*ALL DIMENSIONS ARE IN mm

Figure 69 - Knee Space under Planter for Community Farming

11.3 Design Guidelines for Dementia

General

- 11.3.1 Designing for dementia and inclusion aims to create built environment that provide accommodations and supports, regardless the neurological makeup, as well as the caregivers.

By incorporating principles of simplicity, predictability, sensory considerations, and flexibility in the built environment, designers can create inclusive spaces that enhance well-being, productivity and overall satisfaction. Implementing these guidelines not only enhances independence, safety and overall well-being, but also fosters a more inclusive and supportive built environment.

Considerations

- 11.3.2 Designing buildings for dementia requires addressing their distinct sensory, cognitive, and physical needs. Dementia involves progressive cognitive decline, particularly in memory and navigation. The following key principles can enhance independence, safety, and well-being.

Key Principles

- 11.3.3
- (1) Simplicity and Clarity
 - (a) Avoid unnecessary distractions and keep designs straightforward.
 - (b) Use clear and simple language in instructions and signages.
 - (c) Minimise the number of steps required to complete tasks.
 - (2) Predictability
 - (a) Ensure user flows and interactions are intuitive and easy to follow.
 - (b) Maintain consistency in design elements and layout.
 - (c) Provide clear expectations and avoid sudden changes.
 - (3) Sensory_Considerations
 - (a) Be mindful of sensory inputs like bright colors, noises, and vibrations.
 - (b) Offer options to control sensory stimulation, such as adjustable lighting and sound levels.
 - (c) Use materials and finishes that reduce glare and noise.
 - (4) Flexibility
 - (a) Offer options for interaction with the environment to suit various needs and preferences.
 - (b) Provide adaptive spaces that can be customised by individual users.
 - (c) Include features that support different styles and preferences.

Dementia-friendly Environment

- 11.3.4 (1) Design principles illustrated in **Figure 70** and the following design requirements are recommended in creating an environment that support cognitive clarity for the individual with dementia:
- Use color and tonal contrasts between walls, floors, and critical features (e.g., doorways, benchtops) to aid visual perception.
 - Avoid shiny or reflective surfaces and contrasting patterns in flooring to prevent confusion.
 - Ensure sufficient lighting levels to reduce anxiety and support orientation.
 - Incorporate familiar objects and sensory cues (e.g., objects with specific graphics and textures) to trigger memory.

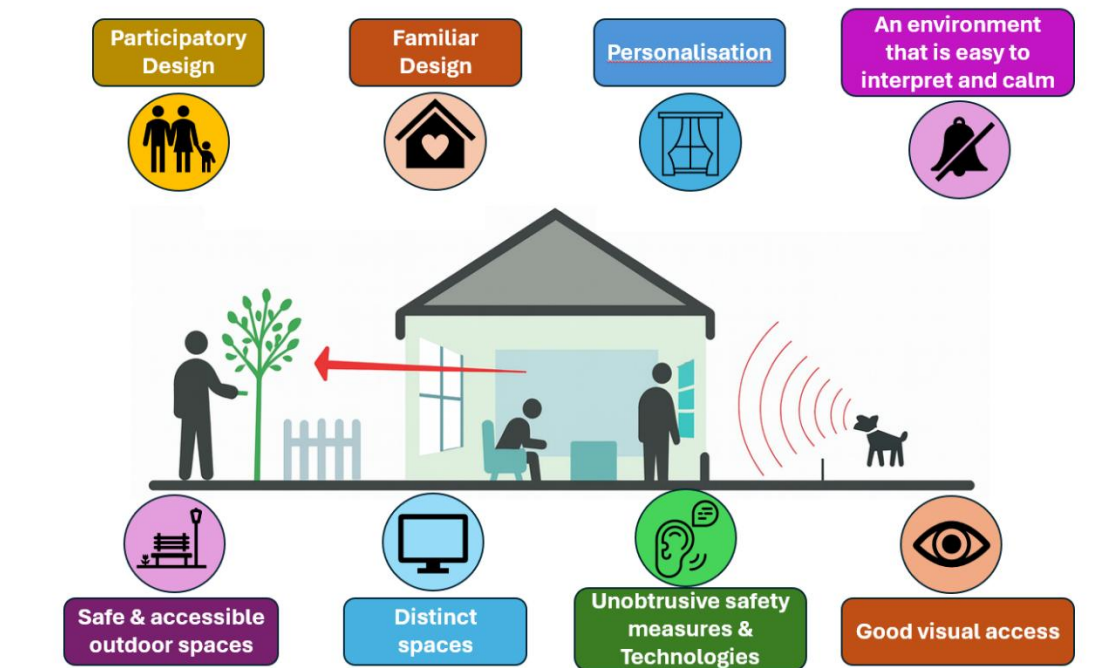


Figure 70 - Design Principle for Dementia-friendly Building Design

Wayfinding and Navigation

11.3.5

(1) Common Areas

- (a) Maintain a logical sequence of spaces and travel (e.g., Main entrance → Lobby → Lift → Corridor → Unit door → living room → bedroom).
- (b) Adopt Elderly-friendly Design Guidelines in this Manual that are inclusive of dementia-sensitive and neurodiverse users.
- (c) Provide shorter corridors/paths within buildings with natural lighting and ventilation.
- (d) Lifts should be directly visible from entrance of units.
- (e) Provide covered walkways/passageways connecting main entrance of buildings.
- (f) Use signage and other cues, such as sound, touch, smell to reinforce wayfinding (e.g. plants with distinct smell). Examples of design and visual cues are provided in **Figure 71** and **Figure 72**.

(2) Residential Units

- (a) Provide open plan layouts to allow free movement without obstruction.
- (b) Provide adequate storage cabinets (with clear labels) to avoid clutter.
- (c) Provide personalised signages and familiar objects or markers for orientation and wayfinding.
- (d) Provide clear and direct view of bedroom to bathroom and kitchen.
- (e) Provide handrails and non-slip shock absorbent floor surfaces in all rooms.
- (f) Avoid locks on doors inside the home, except the main entrance door lock which may need specialised security alarm sensors.
- (g) Provide CCTV, GPS tracking and motion sensors for observation by caregivers.



Figure 71 - Dementia-friendly Building Design in Residential Building



Figure 72 - Dementia-friendly Wayfinding Design

Outdoor Areas

- 11.3.6
- (1) Provide accessible paths to outdoor areas, such as gardens, courtyards, and verandahs, for access to sunlight, breezes, and natural sounds.
 - (2) Provide sensory gardens with different sensory elements to stimulate memory and five senses, including sight, smell, sound, touch and taste. An example of design with sensory elements is provided in **Figure 73**.
 - (3) Provide shaded, covered outdoor access to encourage engagement.



Figure 73 - Dementia-friendly Design in Outdoor Areas

Safety and Security

- 11.3.7
- (1) Provide wide corridors to support mobility and reduce confusion.
 - (2) Minimise non-essential doors in corridors to simplify navigation.

Acoustic Environment

- 11.3.8
- Adopt acoustic glazing and sound-absorbing materials (e.g. cork, softwood) to minimise noise, especially in small rooms without windows.

Lighting

- 11.3.9 (1) Provide natural lighting to enhance orientation and reduce agitation, while avoiding glare from reflective surfaces.
- (2) Ensure sufficient lighting in all areas.

Material and Texture

- 11.3.10 Contrasting colours between door handles and doors, walls and floors, handrails and wall, doors and walls, switches and walls, taps and basins, etc. should be adopted. An example of dementia-friendly material and texture is provided in **Figure 74**.

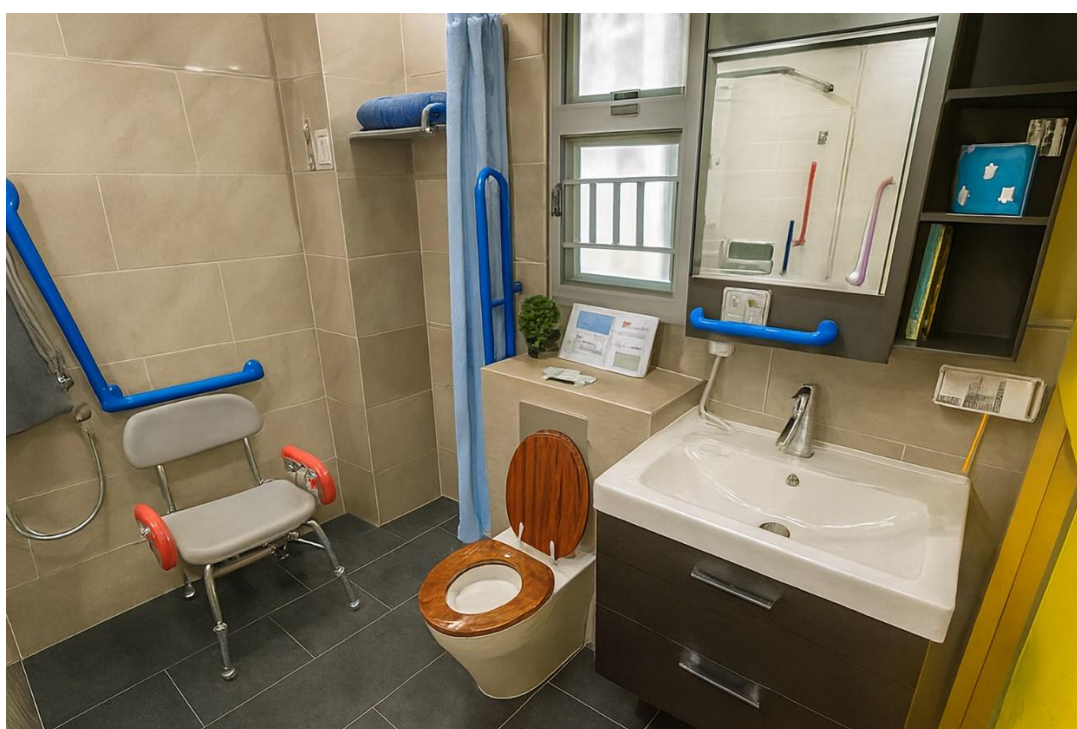


Figure 74 - Dementia-friendly Materials and Textures

11.4 Design Guidelines for Outdoor Areas

General

- 11.4.1 Accessibility should be implemented in outdoor areas to accommodate various needs of users. Trails should be designed with varying levels of difficulty, and incorporating distinct characteristics such as surface types, widths, slopes, cross-slopes, lengths, edge treatments, and the frequency of rest areas.

Outdoor Accessible Routes

- 11.4.2 Accessible routes should be provided to the outdoor facilities and features.

Floor Surfaces

- 11.4.3 (1) Outdoor accessible routes, including all connecting pathways and resting areas, should be firm, stable, slip-resistant, and properly drained.
- (2) For routes constructed with concrete, asphalt, or board materials, any surface obstacles should not exceed 6 mm in height when measured vertically at their highest point.

Clear Width

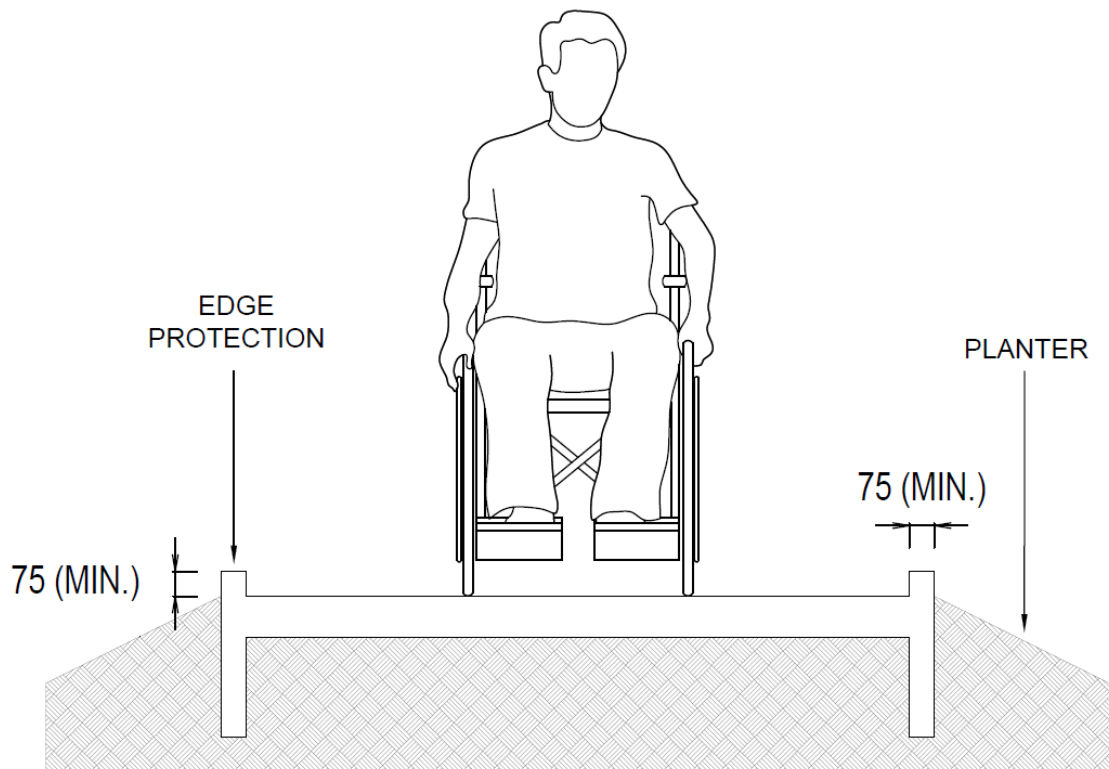
- 11.4.4 Outdoor accessible routes should be at least 1 800 mm wide without obstructions.

Passing Space and Protruding Objects

- 11.4.5 Passing space should be allowed for elements on outdoor accessible routes and protruding objects should comply with paragraph 5.4(2).

Edge Protection

- 11.4.6 If a route has a drop of level of more than 200 mm in height, an upstand edge protection should be installed with at least 75 mm in height and 75 mm in width, as shown in **Figure 75**.



*ALL DIMENSIONS ARE IN mm

Figure 75 - Edge Protection for Accessible Route

Illumination

- 11.4.7 Outdoor facilities and activity areas, including shelters, playgrounds, and fitness zones, should be illuminated to an average of 20 lux measured at finished floor level.

Outdoor Steps and Staircases

- 11.4.8
- (1) Outdoor steps and staircases should have a minimum clear width of 1 200 mm.
 - (2) All outdoor steps and staircases should have treads not less than 300 mm in width and risers not more than 150 mm in height.
 - (3) The maximum vertical rise between landings should not exceed 1 500 mm.
 - (4) All landings should have a minimum length of 1 500 mm.
 - (5) All step nosings should:
 - (a) be constructed with permanent contrasting materials for visibility; and
 - (b) have highlight at tread edges of not less than 50 mm deep and across the full width of the tread.

- (6) Detectable warning surfaces should be provided at both top and bottom of all stair flights.

Sensory Gardens

- 11.4.9 (1) Sensory gardens are recommended to be provided and designed with interactive facilities with key sensory elements including:
 - (a) physical touching;
 - (b) sound;
 - (c) sight; and
 - (d) smell
- (2) The facilities therein should be accessible by the wheelchair users.

Resting Areas

- 11.4.10 (1) Resting areas should be provided at 50-meter intervals along the route, with an area of not less than 1 800 mm in length and 1 200 mm in width. These areas should be able to accommodate wheelchairs.
- (2) Seating should include armrests and backrests to provide proper support.
- (3) Shelters should be provided to the resting areas and to accommodate mobility aids, including wheelchairs and pushchairs.

Sheltered Shuttle Bus Stops

- 11.4.11 Seating and shelters should be provided for shuttle bus stops.

Swimming Pools

- 11.4.12 (1) Notwithstanding paragraph 3.2.2, accessible water entry/exit such as ramp and platform lift may be provided to swimming pools as appropriate.
- (2) Ramp entry should:
 - (a) have slip-resistant floor surfaces;
 - (b) have a maximum gradient of 1:14;
 - (c) have handrails complying requirements in paragraph 10.2; and
 - (d) extend to a depth of not less than 900 mm and not more than 1 100 mm below the stationary water level.
- (3) Platform lift entry should be:
 - (a) capable of being operated from the swimming pool surround, within the swimming pool and on the platform;
 - (b) located where the water depth is not more than 1 300 mm; and

- (c) designed with a minimum certified weight capacity of not less than 160 kg and be capable of sustaining a static load of not less than 1.5 times the rated load.

Designated Outside Toileting Areas for Guide Dogs

11.4.13 A clean and convenient toileting area should be provided for guide dogs as follows:

- (1) It should be located in an easily identifiable area, such as near the entrance of a building or near the accessible toilets, which is accessible by PwDs.
- (2) A clear signage should be provided for leading to and at the entry of the toileting area. The signage should be provided with luminance contrast in compliance with paragraph 10.9(10) and with raised letters and Braille.
- (3) Ground surface should be provided with natural grass.
- (4) It should have a minimum area of not less than 3 m x 3 m provided with fence of not less than 1 200 mm in height.
- (5) Garbage bin should be provided inside or immediately next to the designated toileting area.

Directional Signs

- 11.4.14
- (1) Directional signs should be placed at least 600 mm away from the outdoor recreation accessible routes or trails. The outer limit of the sign should not overlap an outdoor recreation accessible route or trail.
 - (2) The headroom clearance of the directional signs should be at least 2 200 mm above the ground surface.
 - (3) The height of text used on the sign should be not less than 20 mm.
 - (4) Directional signs should be illuminated at a level of at least 200 lux.

11.5 Management and Maintenance of Buildings

General

- 11.5.1 Effective management and maintenance of accessible routes and facilities are essential to creating an inclusive environment for all individuals. Building owners and property managers bear the responsibility of ensuring that approved facilities for PWDs remain functional, intact, and free from unauthorised alterations. To safeguard accessibility, proper measures should be adopted, including routine inspections, systematic accessibility checks, and mechanisms that encourage users to report suspected removals or modifications.

The design guidelines in paragraph 11.5 aim to strengthen accessibility through proactive management and continuous oversight, thereby ensuring that barrier-free facilities remain compliant and supportive of equal access.

Maintaining Accessibility Facilities during Temporary Events

- 11.5.2 (1) During temporary events, such as pop-up store in common areas of shopping mall or temporary closure of accessible or barrier free facilities due to maintenance, the following measures should be taken to ensure accessibility.

Auxiliary Aids

- (2) Provide auxiliary aids such as portable ramps complying with paragraph 6.2 of this Manual when the permanent accessible ramp is not available.

Accessible Facilities

- (3) In case a facility is temporarily unavailable (e.g. accessible toilet), supplementary information, including a floor plan with Braille and tactile information that clearly indicates the location of alternative facilities nearby, should be provided at such facilities with directional signs indicating the accessible route to the alternative facilities.
- (4) If alternative facilities is not available, contact number of management office should be provided for assistance.

Tactile Guide Paths

- (5) The tactile guide paths should be remained clear at all times. If it is temporary obstruction is unavoidable, an alternative temporary tactile guide path, compliant with this paragraph 10.3 of this Manual, should be provided in advance.

Temporary Cable ramp

- (6) Measures should be taken to avoid cables from being laid on floor whenever possible. Cable protectors with gradients complying with the ramp requirement in paragraph 6.2 of this Manual should be provided in all accessible areas to ensure safety and accessibility.

Regular Inspection for Accessible Routes and Facilities

- 11.5.3
- (1) Regular inspection to the accessible and barrier free accesses and facilities should be conducted. Proper repair and replacement should be carried out to ensure these accesses and facilities are maintained in good condition.

Doors and Hardwares

- (2) Check whether the doors, door closers and other hardwares are in good working condition.

Accessible Toilets

- (3) Check whether the operation of the accessible toilets is maintained in good function. All essential controls and fittings are in order, including but not limited to:
 - (a) Power operated door (if any);
 - (b) Folding grab rail; and
 - (c) Emergency call bell.

Flooring and Surfaces

- (4) Check whether the floor surfaces, matting, carpets, etc. are in good condition.

Hearing enhancement systems

- (5) Test and commission the hearing enhancement systems to ensure their functionality.

Sanitary fitments

- (6) Check whether the sanitary fitments, including the fixing of toilet seats or shower seats, taps for wash basins and flushing taps, sensor-type controls, are in good condition and functionable.

Emergency Systems

- (7) Test and commission the emergency assistance pull cords and alarm systems to ensure their functionality.

Grab Rails and Facilities

- (8) Check whether the fixing and mechanism of grab rails and bars are in good condition.

Accessible Lifts and Hoists

- (9) Test and commission the accessible lifts and hoists including their audio and visual indication to ensure their functionality.

Floor Maps and Signs

- (10) Verify whether the maps and signs are properly updated.

Tactile Guide Path

- (11) Check the integrity of the tactile guide path, including:
 - (a) the height of raised dots/bars;
 - (b) the thickness of tiles/blocks; and
 - (c) the surface condition (free of cracks and breakage).
- (12) Check and ensure that the tactile guide path and the surrounding clearance are free from any obstruction.

Improving Accessibility in Existing Buildings

- 11.5.4
 - (1) The level difference between different areas should be minimised as far as practicable. Addition of accessible lifts, vertical lifting platform as well as portable ramps may be required depending on circumstances.
 - (2) Tactile guide paths complying with paragraph 10.3 for assisting persons with visual impairment should be provided.
 - (3) At least one cubicle in each male and female toilets complying with the requirements of Elderly-friendly Toilets in item C4 of Chapter 11.2.
 - (4) Charging points for powered wheelchair in the common areas of the buildings should be provided.

11.6 Alterations and Additions or Improvement Works in Existing Buildings

General

- 11.6.1 The requirements set out in this Manual apply to all alterations and additions to existing buildings. For buildings that were originally constructed without access provisions for PwDs, every practical measure should be taken to provide suitable barrier-free access.

It is recognised, however, that structural and spatial constraints may sometimes prevent full compliance with the mandatory design requirements. In such cases, these limitations may constitute unjustifiable hardship for the project proponent and even hinder the implementation of improvement works.

In this connection, the design guidelines in paragraph 11.6 aim to provide practical alternatives, drawn from past experience, to guide alterations and additions, or improvement works in existing buildings where spatial or structural constraints exist.

Where technical difficulties, site limitations, or claims of unjustifiable hardship are substantiated on the merits of individual cases, the compensatory measures outlined in the scenarios below are considered acceptable alternative means of achieving the objectives of the obligatory design requirements.

Examples on alternative means are provided below for reference.

Non-provision of Initial Access

- 11.6.2 (1) Circumstance
- (a) Accessible ramp as initial access (including the associated landing and horizontally projecting handrail) occupies more than one-third of the area or the frontage of the premises, and/or the premises with structural and/or topographical constraints.
- (2) Required mitigating measures
- (a) Intercom device for connection to the management office/ personnel with adequate notice should be provided at a level of 900 mm to 1 200 mm from the floor level of the access point;
 - (b) Internal floor and the adjoining pavement/level should be provided with contrasting colour;
 - (c) User operable vertical lifting platform (VLP) should be provided to overcome the level difference unless such VLP would occupy more than one-third of the frontage;
 - (d) Movable ramp with designated attendant should be provided for cases where provision of VLP was impossible; and
 - (e) Stairlift with designated attendant should be provided for cases where provision of both VLP and movable ramp were impossible.

Insufficient Clear Width of Accessible Routes

- 11.6.3 (1) Circumstance
- (a) The clear width of accessible route is not less than 1 050 mm but less than 1 200 mm.
- (2) Required mitigating measures
- (a) Additional lighting should be provided in accordance with paragraph 10.14(1) and activated by sensors;
 - (b) Luminance contrast of not less than 30% should be provided between wall, floor and door surface;
 - (c) No protruding objects should be located along the corridors, lobbies and paths;
 - (d) A recess area of not less than 1 500 mm x 1 500 mm should be provided at a maximum interval of 50 m; and
 - (e) A management undertaking on offering assistance to wheelchair users should be provided.

Insufficient Size of Accessible Lifts

- 11.6.4 (1) Circumstance
- (a) The internal car dimension is not less than 1 400 mm x 1 100 mm but less than 1 400 mm x 1 200 mm.
- (2) Required mitigating measures
- (a) Non-breakable mirror or mirror-like surface should be provided inside the lift car in addition to that required under paragraph 7.2(4);
 - (b) The lift door should remain fully open at a landing for at least 3 seconds;
 - (c) Keypad complying with paragraph 7.5(10) should be provided;
 - (d) An additional detection device of lift door complying with paragraph 7.5(15) should be provided;
 - (e) Intercom device connecting to the management/caretaker office with adequate notice at a level of 900 mm to 1 200 mm from the finished floor level should be provided at the lift lobbies of accessible floors; and
 - (f) A management undertaking on offering assistance to wheelchair users should be provided.

Insufficient Clear Width of Lift Entrance

- 11.6.5 (1) Circumstance
- (a) The clear width of lift entrance is not less than 750 mm but less than 850 mm.
- (2) Required mitigating measures

- (a) Two no. of infra-red detection devices positioned at a height of 200mm and between 300 to 400 mm above the floor of the lift car should be provided in addition to the mechanical safety edge of the lift door;
- (b) Thresholds at the lift entrances which may hinder the access for PwDs should be modified to a slight counter slope not steeper than 1:20;
- (c) One additional set of essential lift control buttons of keypad design complying with paragraph 7.5(4) should be provided;
- (d) Components of the lift should not have sharp-edges and the lift jambs should have non-slippery surfaces which could be grabbed hold of by wheelchair users to facilitate their ingress or egress;
- (e) No carpet which may hinder the movement of wheelchairs should be placed either inside the lift car or at the lift lobbies on all floors; and
- (f) A management undertaking on offering assistance to wheelchair users should be provided.

Insufficient Width and Height of Steps of Main Circulation Staircases

11.6.6 (1) Circumstance

- (a) The treads are not less than 225 mm but less than 300 mm in width and the risers are not more than 175 mm but more than 150 mm in height.

(2) Required mitigating measures

- (a) Non-slip nosing having a minimum luminance contrast of 30% with the adjoining surfaces should be provided; and
- (b) Treads of staircase should have a minimum luminance contrast of 30% with the walls.

Insufficient Clear Width of Door

11.6.7 (1) Circumstance

- (a) The subject building(s) have been rated Grade 3 historic or above;
- (b) door or door opening has been identified as character defining elements;
- (c) for double-leaf door, the clear opening should be not less than 850 mm when both leaves were open; and
- (d) for single-leaf door, the clear opening should be not less than 750 mm.

(2) The required mitigating measures for a door being a double-leaf door with clear opening not less than 850 mm or a single-leaf door with clear opening not less than 750 mm should be provided as follows, where:

- (a) Under Scenario A, the door would be kept close during operating hours and not provided with automatic opening devices, and;
- (b) Under Scenario B, the door would be kept open during operating hours or provided with automatic opening device.

Item	Required Mitigating Measures	Scenario A	Scenario B
(i)	Sufficient notice and Braille and tactile maps	✓	✓
(ii)	Accessibility information covering the deficiency in widths of passages at the website (if any) of the operator	✓	✓
(iii)	Glass portions of the doors, if not constructed of tempered glass, to be strengthened by film backing	✓	NA
(iv)	Hooks at the back of doors for securing door leaves to walls	✓	✓
(v)	Smaller wheelchairs suiting the passages with inadequate widths and personnel trained to transfer PwD between wheelchairs	✓	✓
(vi)	Clear acrylic kicking plates for protecting the lower part of the doors which are character defining elements	*	NA
(vii)	Intercom connecting between the management office and the first point which the visiting PwDs would be confronted with the passage with substandard width	*	*
(viii)	Assistance from the management office to PwDs for gaining access	*	*
(ix)	Sharp edges of the doors, frames or the nearby walls to be rounded, cushioned or non-abrasive as applicable	*	*
(x)	Floors on either sides of the doorway to be made flat	*	*

✓ required
* where appropriate

APPENDIX A ANTHROPOMETRICS

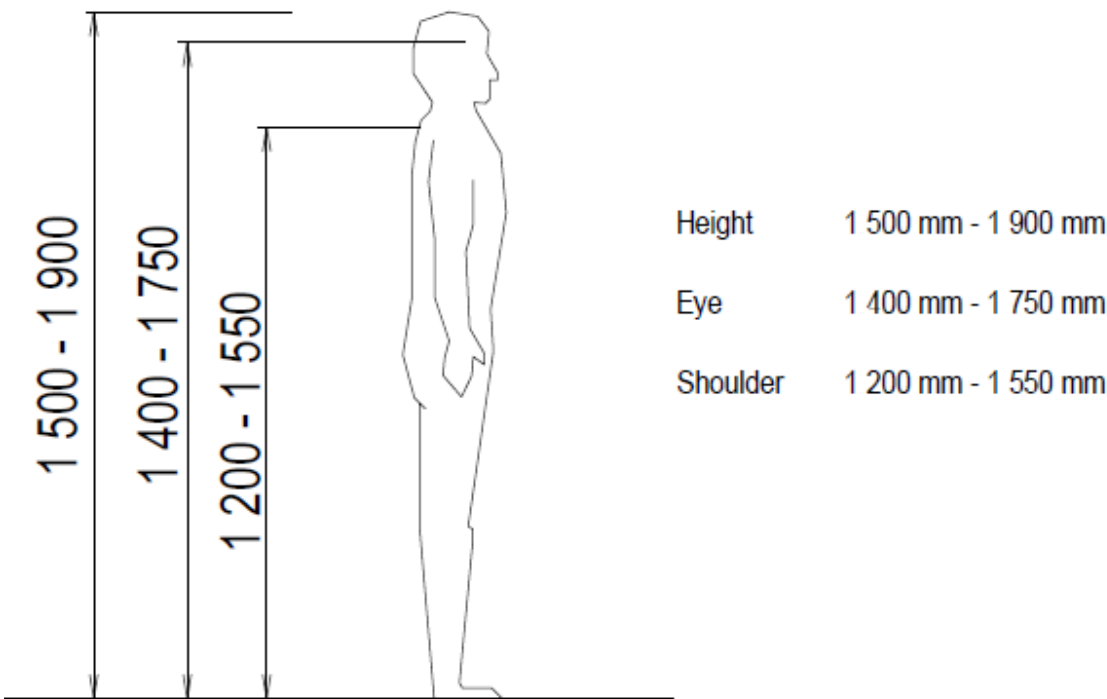
A.1 Objective

This Appendix contains dimensional data which can be used for guidance when designing facilities and equipment for use by PwDs.

A.2 Considerations

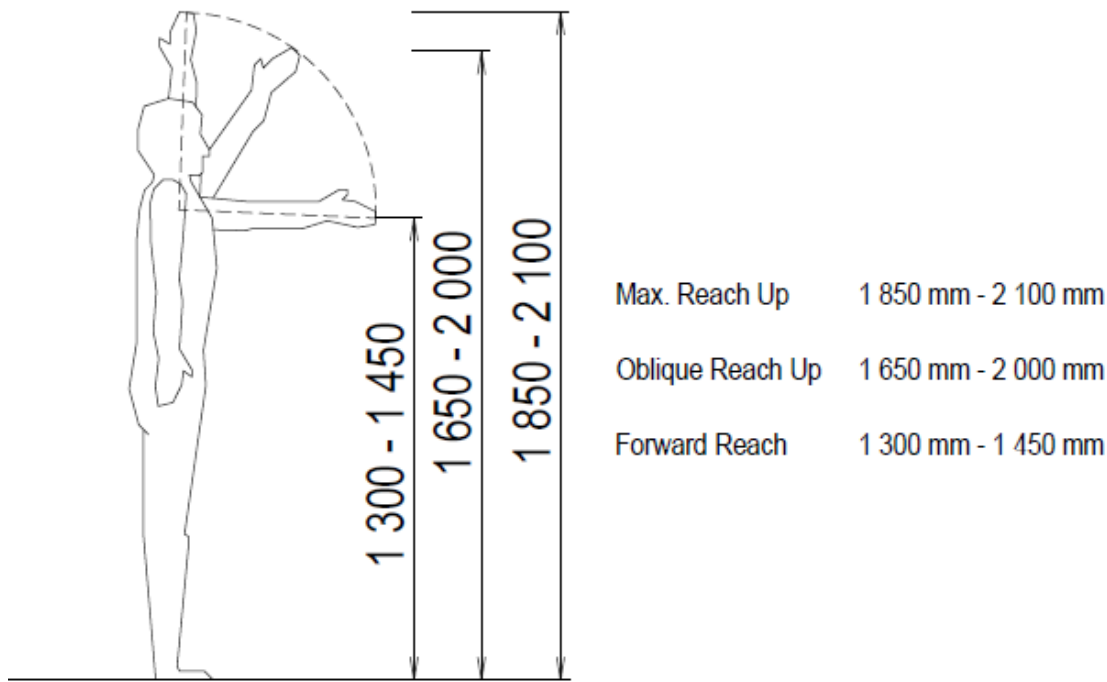
The dimensions of the individual human being vary with time and from one person to another. In addition, the average dimensions vary from one country to another.

When carrying out detailed design, consideration should be given to size variation between males and females as well as between different ages (see **Figure 76** to **Figure 79**).



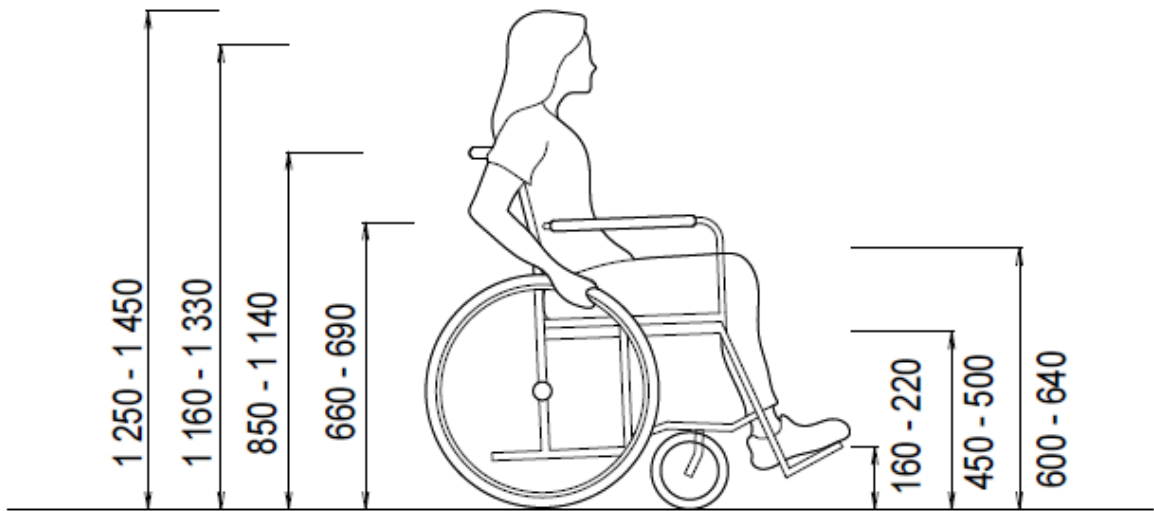
*ALL DIMENSIONS ARE IN mm

Figure 76 - Dimensional Data of an Average Person



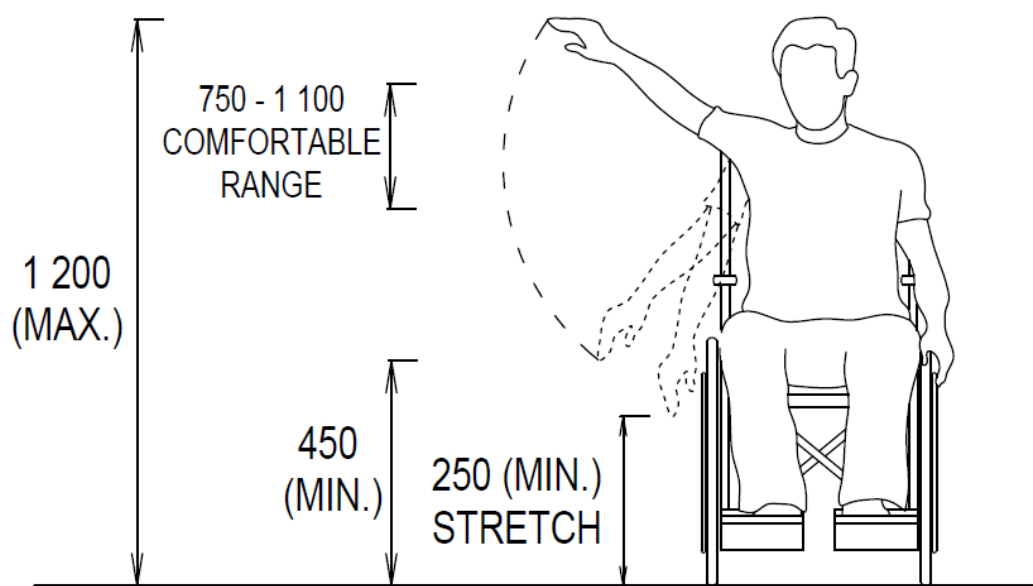
*ALL DIMENSIONS ARE IN mm

Figure 77 - Reaching Zones of an Average Person



*ALL DIMENSIONS ARE IN mm

Figure 78 - Dimensional Data of a Wheelchair User



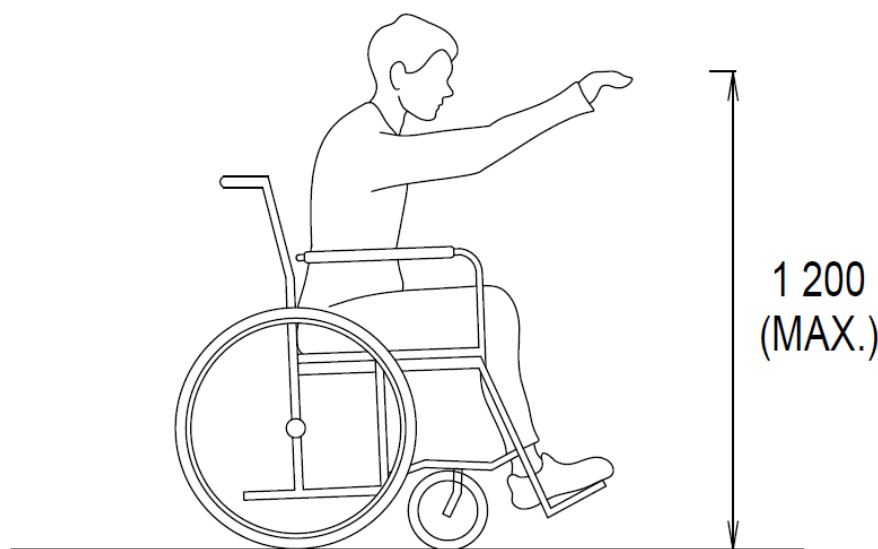
*ALL DIMENSIONS ARE IN mm

Figure 79 - Vertical Reaching Zones of a Wheelchair User

A.3 Forward reach of a wheelchair user

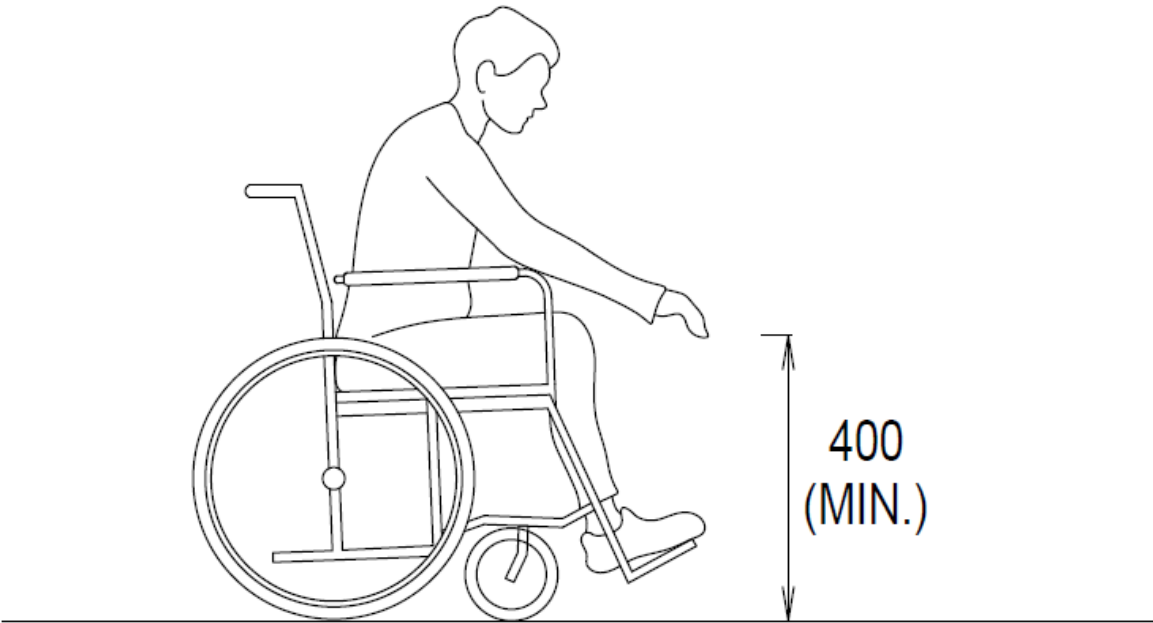
The maximum forward reach, without obstruction, is 1 200 mm from the floor and the minimum forward reach is 400 mm from the floor as shown in **Figure 80** and **Figure 81**.

The maximum forward reach over an obstruction 400 mm deep is 1 100 mm from the floor as shown in **Figure 82**.



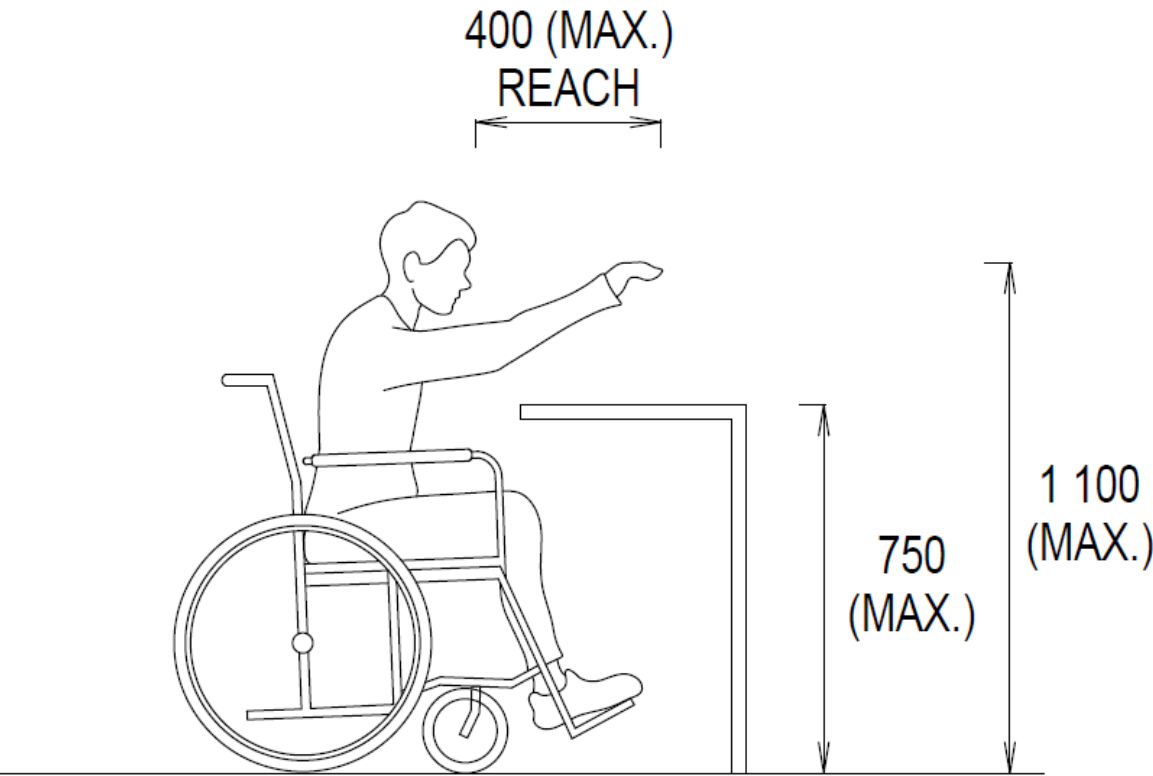
*ALL DIMENSIONS ARE IN mm

Figure 80 - Maximum Forward Reach without Obstruction



*ALL DIMENSIONS ARE IN mm

Figure 81 - Minimum Forward Reach without Obstruction



*ALL DIMENSIONS ARE IN mm

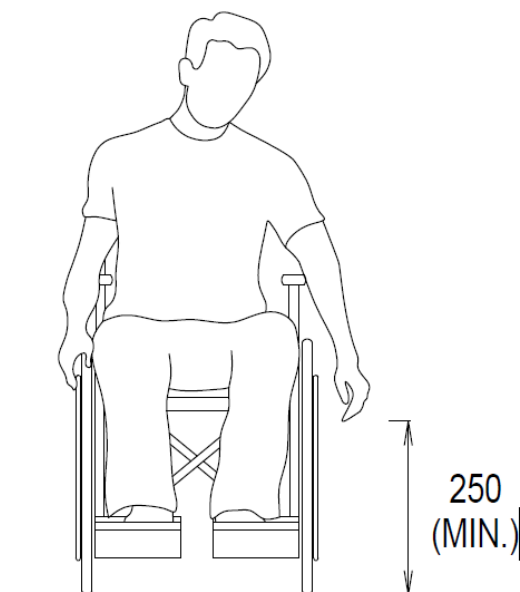
Figure 82 - Maximum Forward Reach over Obstruction

A.4 Side reach

The maximum side reach, without obstruction, is 1 300 mm from the floor and the minimum side reach is 250 mm from the floor as shown in **Figure 83** and **Figure 84**.

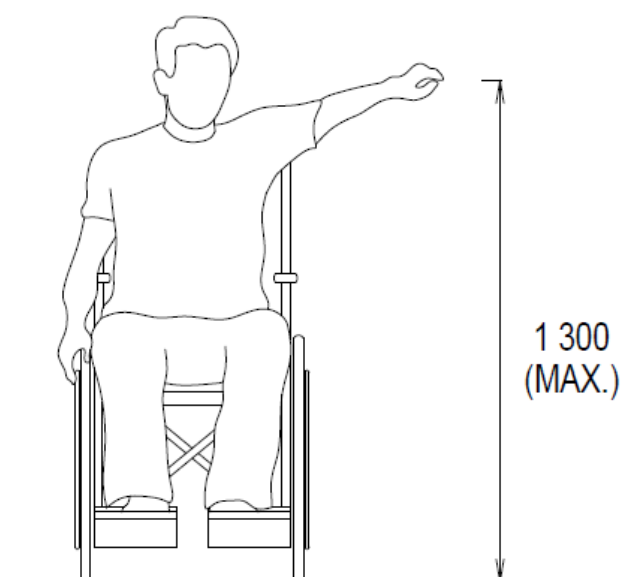
The maximum side reach over an obstruction 860 mm high by 500 mm deep is 1 200 mm from the floor as shown in **Figure 85**.

The dimension required for general pathways is also provided in **Figure 86**.



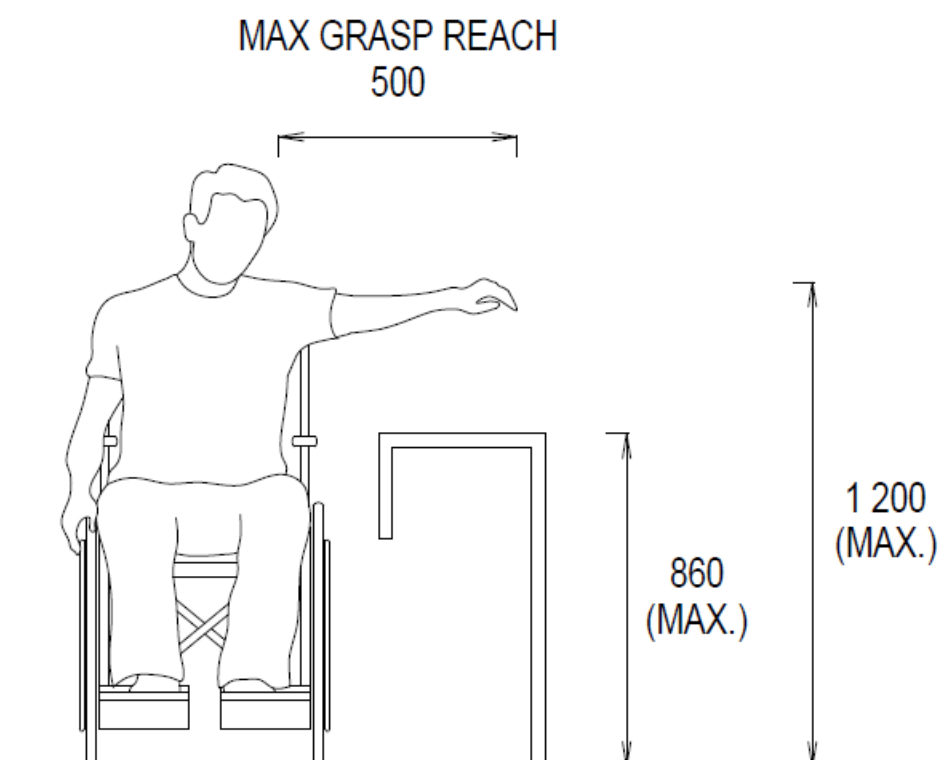
*ALL DIMENSIONS ARE IN mm

Figure 83 - Minimum Side Reach without Obstruction



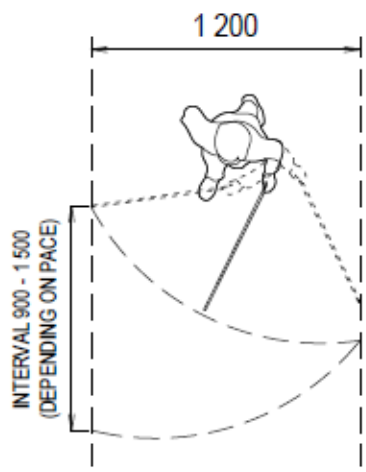
*ALL DIMENSIONS ARE IN mm

Figure 84 - Maximum Side Reach without Obstruction

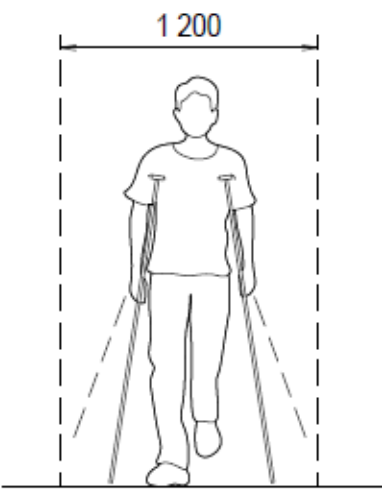


*ALL DIMENSIONS ARE IN mm

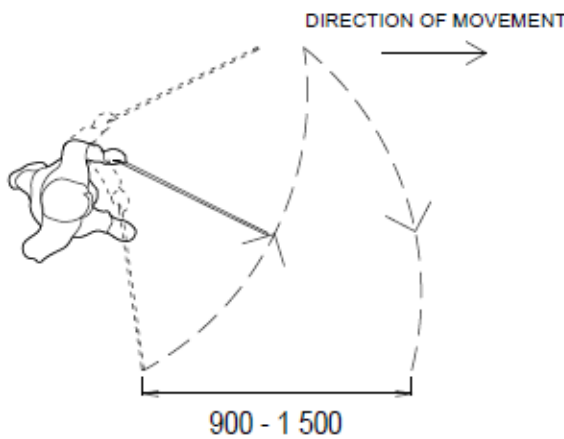
Figure 85 - Maximum Side Reach over Obstruction



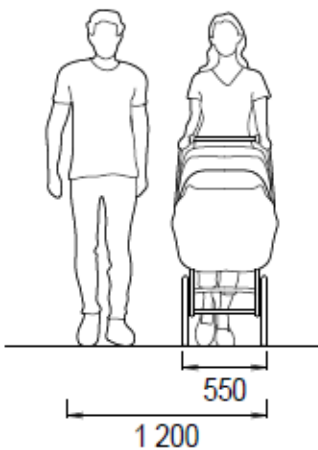
Persons with Visual impairment using
a long white cane



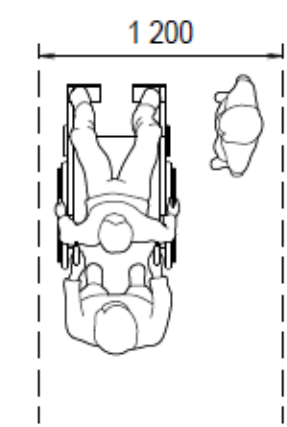
Persons using crutches



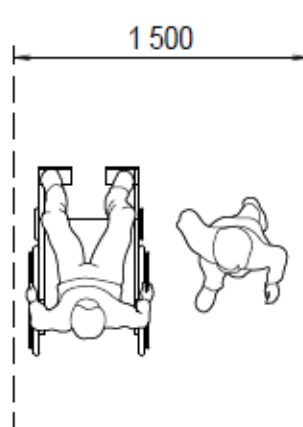
Detectable path dimension



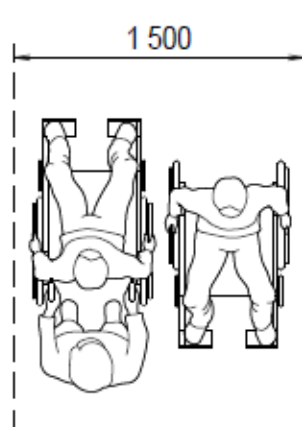
Person with baby carriage



Wheelchair and a
person facing sideways



Wheelchair and a
person passing



Two wheelchairs passing

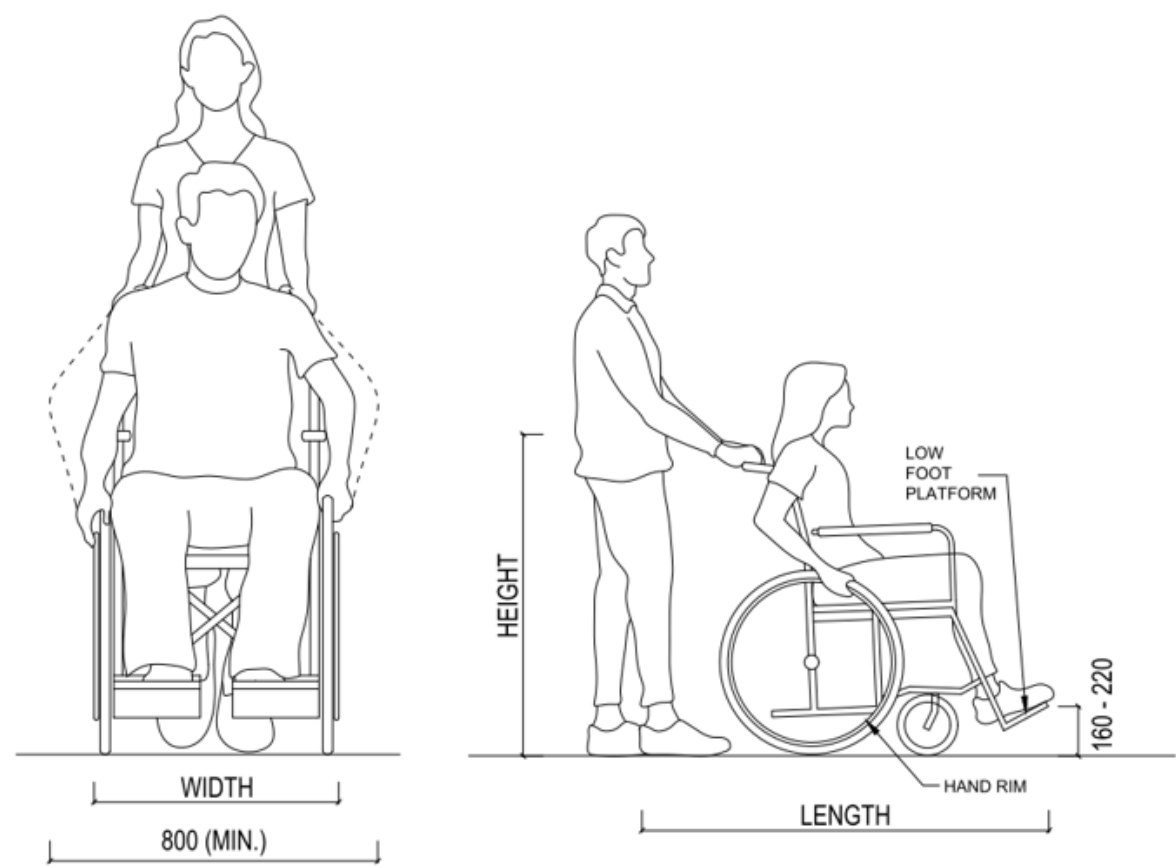
*ALL DIMENSIONS ARE IN mm

Figure 86 - Dimensions Required for General Pathway

APPENDIX B GUIDELINES FOR WHEELCHAIR TRANSFER AND
MOVEMENT

B.1 Dimensions of Wheelchairs

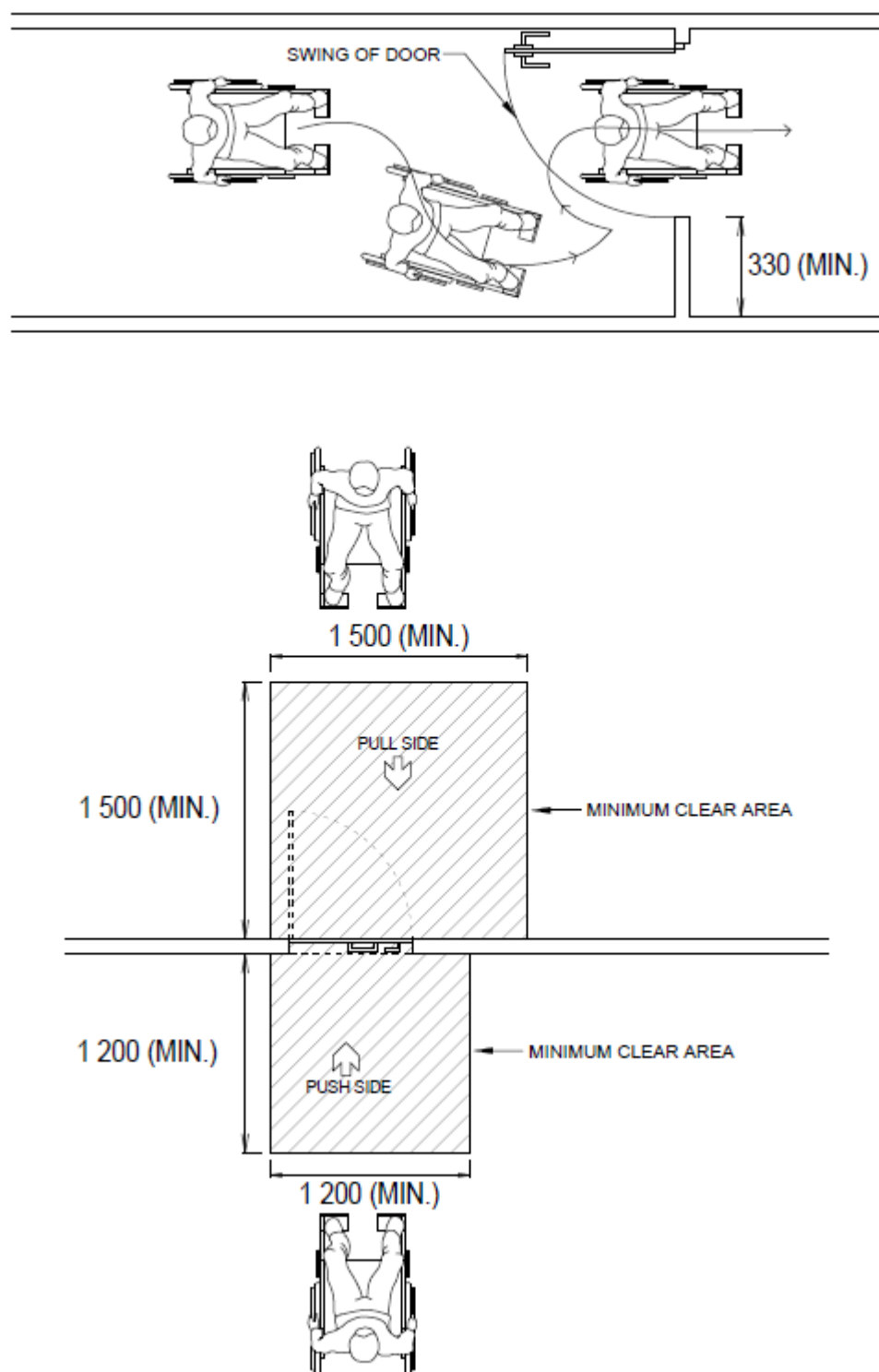
The dimensions of wheelchairs commonly used by PwDs in Hong Kong are provided at **Figure 87** to **Figure 93**.



Type of Wheelchair	Width	Length	Height	Weight
Manual Wheelchair	510-725 mm	665-1100 mm	850-1140 mm	10-27 kg
Electric Wheelchair	520-700 mm	1060-1200 mm	1010-1400 mm	36-100 kg

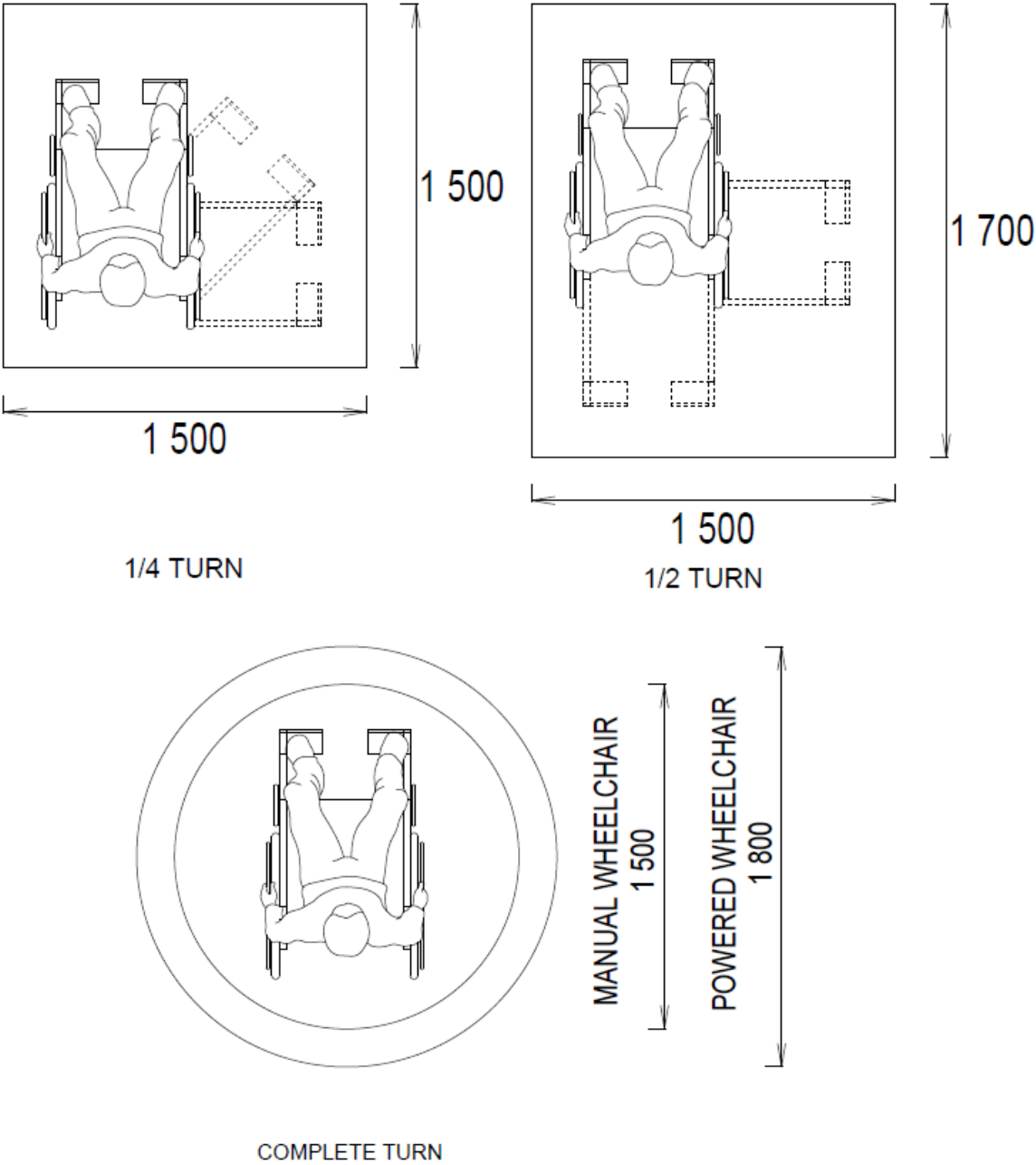
*ALL DIMENSIONS ARE IN mm

Figure 87 - Dimensions of Wheelchair



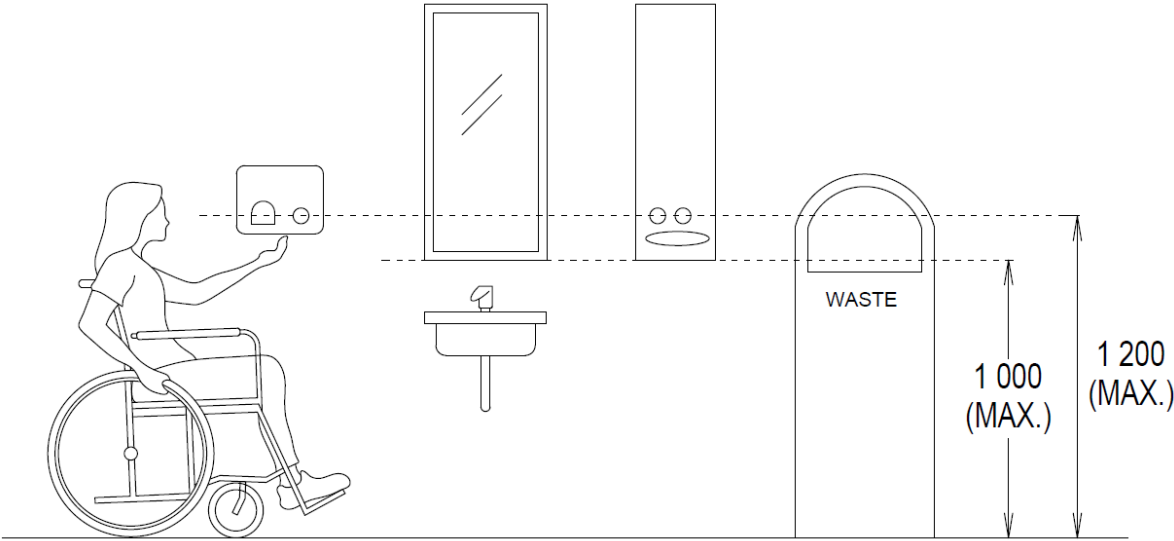
*ALL DIMENSIONS ARE IN mm

Figure 88 - Manoeuvring Spaces for Doorway



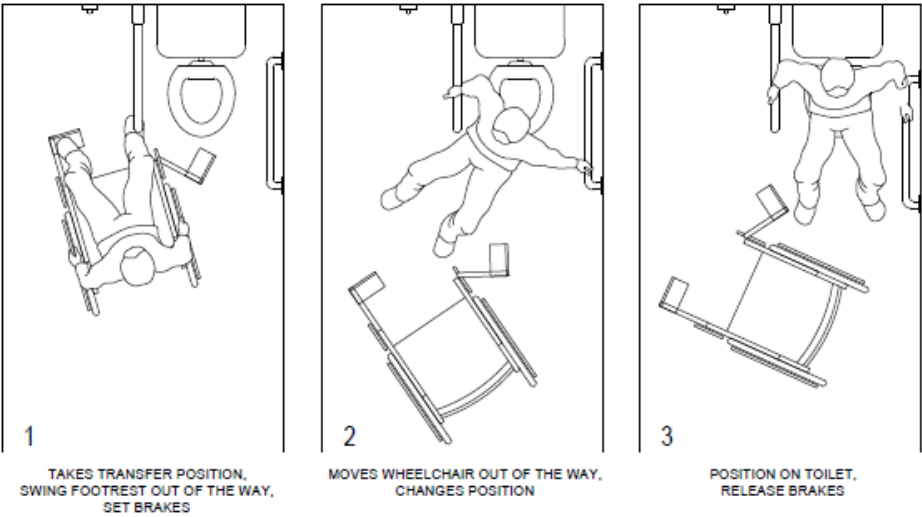
*ALL DIMENSIONS ARE IN mm

Figure 89 - Wheelchair Manoeuvring Space

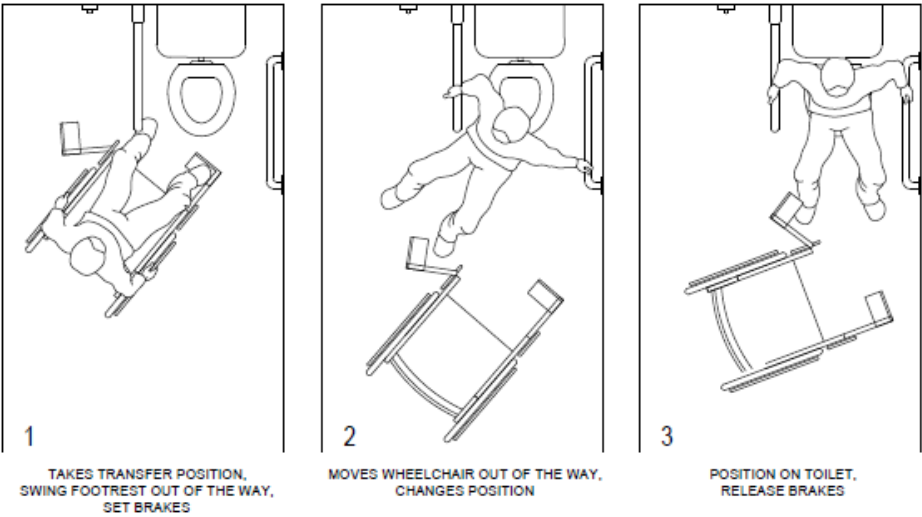


*ALL DIMENSIONS ARE IN mm

Figure 90 - Design Heights for Various Features



ALTERNATIVE APPROACH BY PILOTING THE WHEELCHAIR 45° TO THE TOILET



ALTERNATIVE APPROACH BY PILOTING THE WHEELCHAIR 90° TO THE TOILET

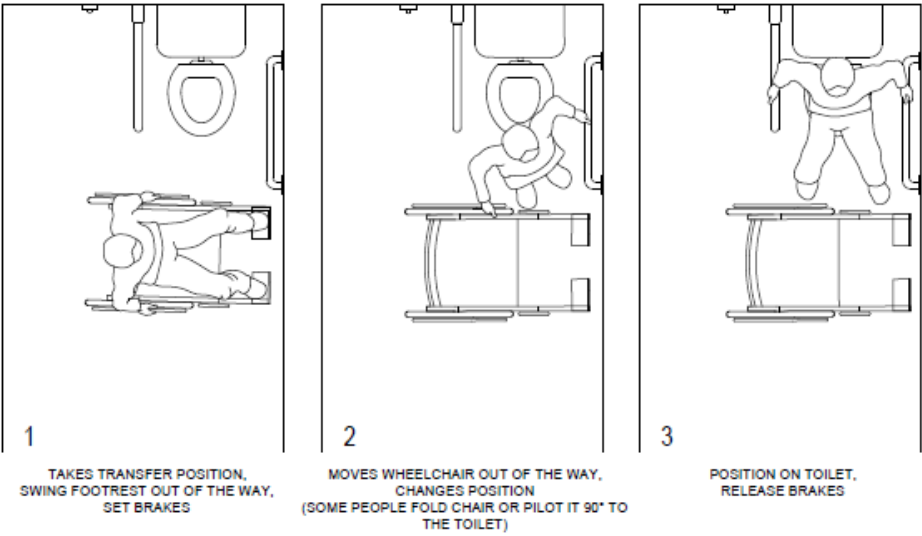
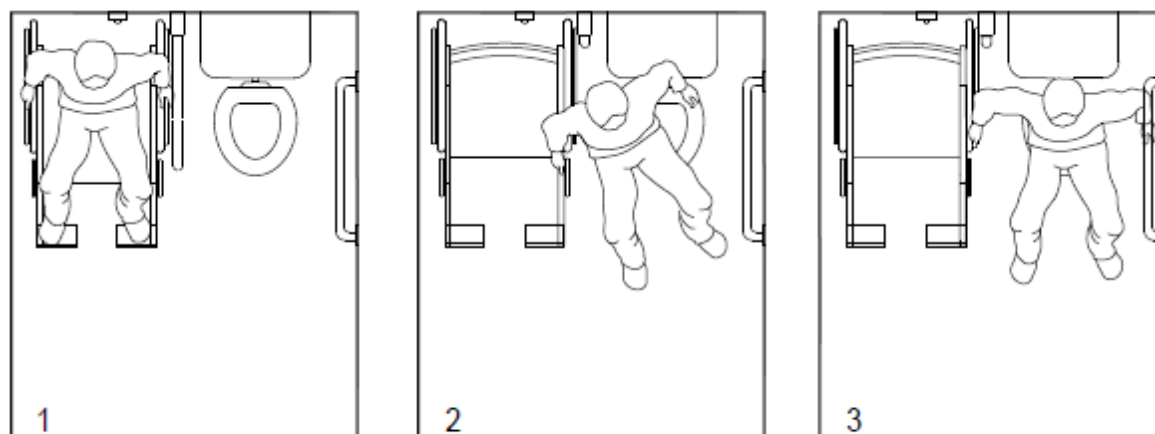


Figure 91 - Diagonal Approach for Transferring from a wheelchair to a W.C.

TRANSFER WITH WHEELCHAIR PARALLEL TO W.C.

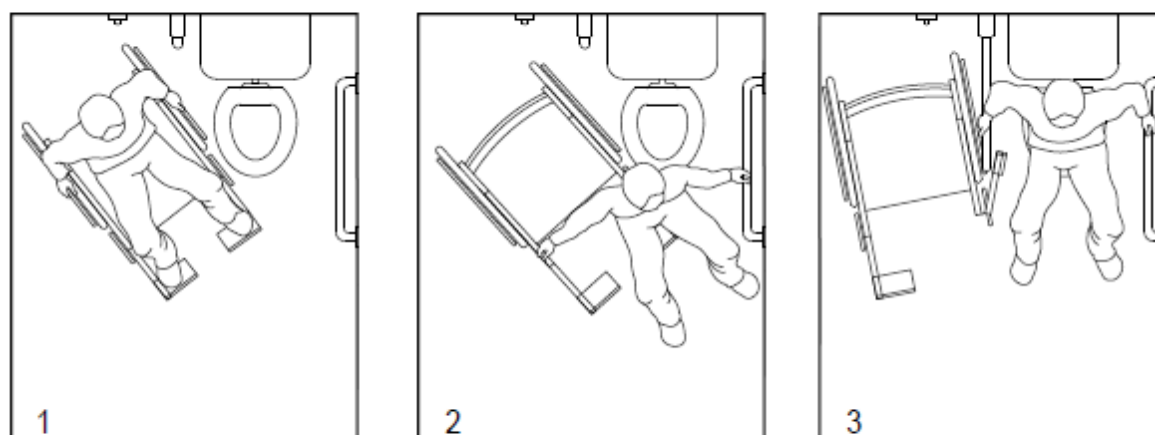


1
TAKES TRANSFER POSITION,
REMOVE ARMREST,
SETS BRAKES

2
TRANSFERS

3
POSITION ON TOILET

ALTERNATIVE TRANSFER POSITION TO W.C.



1
TAKES TRANSFER POSITION,
REMOVE ARMREST,
SETS BRAKES

2
TRANSFER

3
POSITION ON TOILET

Figure 92 - Side Approach for Transferring from a Wheelchair to a W.C.

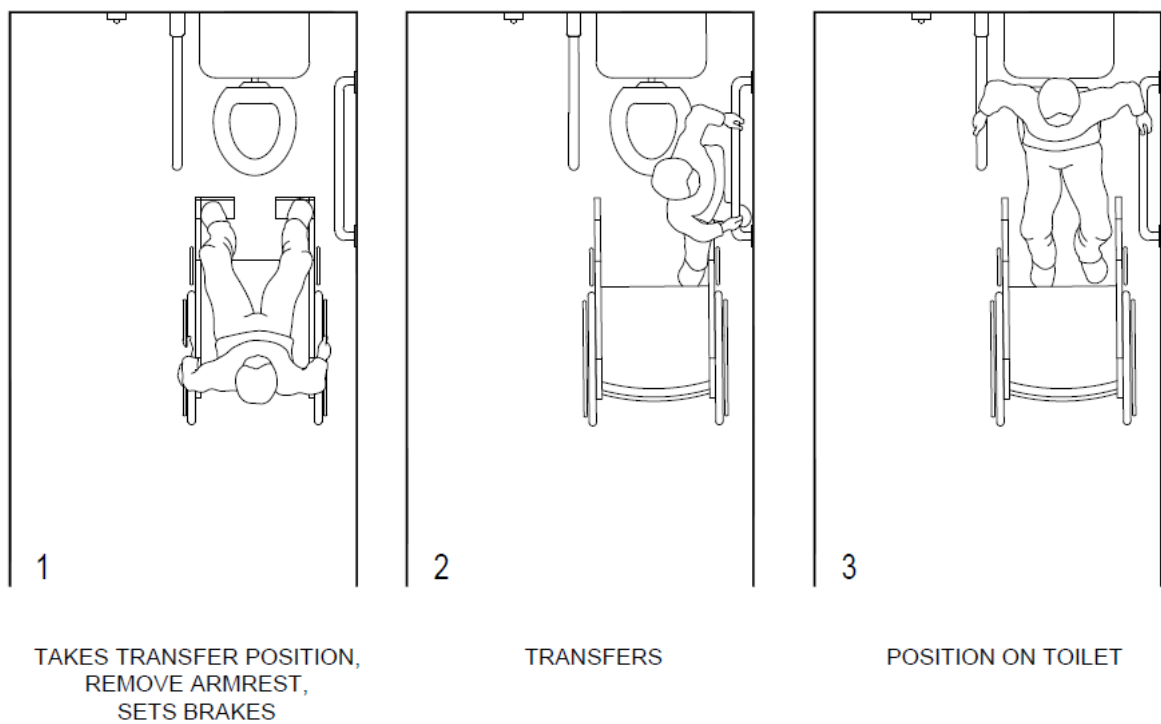


Figure 93 - Frontal Approach for Transferring from a Wheelchair to a W.C.

APPENDIX C SLIP-RESISTANCE OF FLOORING MATERIALS

C.1 Objective

The main goals of slip-resistant flooring materials and tactile guide paths are to improve safety, accessibility, and usability. These materials lower the risk of slips and falls, particularly in wet areas, by providing better traction.

C.2 Considerations

Friction between shoes and the flooring materials is crucial for walking and balance, with insufficient friction leading to slips. The main factors affecting footwear traction are floor surface characteristics, contaminants, footwear type, and gait dynamics. Since controlling footwear and gait is impractical, managing floor surface and cleanliness is more effective for improving slip-resistance.

C.3 Testing of Materials

Floor surfaces required to be slip-resistant shall be tested and classified in accordance with the following standards, or other equivalent standards:

- (a) Australian Standard AS4586:2013, *Slip resistance classification of new pedestrian surface materials*; or
- (b) German Standard DIN 51130:2014-02, *Testing of floor coverings – Determination of the anti-slip property – Workrooms and fields of activities with slip danger – Walking method – Ramp test*.

C.4 Requirements of Slip-resistant Floor Surfaces

When tested in accordance with the standards specified in [paragraph C.3](#), floor surfaces required to be slip-resistant shall attain the respective classification for the wet pendulum test or the oil-wet inclining platform test specified in the tables below. The classifications generally refer to the minimum standards required for slip-resistant floor surfaces in different areas. Higher standards may be adopted depending on the actual site conditions and operational needs, provided that the risk of stumbling or tripping resulting from such higher standards has been taken into account.

Indoor Areas	Classification according to AS 4586 for Wet Pendulum Test	Classification according to AS 4586 or DIN 51130 for Oil-Wet Inclining Platform Test
Accessible route	P3	R10
Ramp (gradient not steeper than 1:14)	P3	R10
Ramp (gradient steeper than 1:14)	P4	R11
Stair tread, nosing and landing	P3	R10
Carpark	P3	R10
Shop / shopping centre / supermarket where it is likely to be wet	P3	R10
Market / food court / fast food / food outlet / canteen dining area	P3	R10
Commercial kitchen	P5	R12
Wet areas inside residential units	P3	R10
Hospitals and aged care facilities – dry areas	P2	R9

Indoor Areas	Classification according to AS 4586 for Wet Pendulum Test	Classification according to AS 4586 or DIN 51130 for Oil-Wet Inclining Platform Test
Hospitals and aged care facilities – ensuites	P3	R10
Communal changing room	P3	A ^(Note 5)
Swimming pool surrounds and communal shower room	P4	B ^(Note 5)
Swimming pool ramp and stair	P5	C ^(Note 5)
Toilet / nursing room	P3	R10
Games court & gymnasium	P3	R10
Refuse room	P3	R10
Plant room ^(Note 6)	P3	R10

Outdoor Areas	Classification according to AS 4586 for Wet Pendulum Test	Classification according to AS 4586 or DIN 51130 for Oil-Wet Inclining Platform Test
Covered accessible route [exposed to weather on side(s)]	P3	R10
Uncovered accessible route	P4	R11
Stair tread, nosing and landing	P4	R11
Ramp (gradient not steeper than 1:14)	P4	R11
Ramp (gradient steeper than 1:14)	P5	R12
Bus stop, taxi stand, passenger pick up	P4	R11
Market / food court / fast food / food outlet / canteen dining area (external)	P4	R11
Swimming pool surrounds and communal shower rooms	P4	B ^(Note 5)
Swimming pool ramps and stairs leading into water	P5	C ^(Note 5)

NOTE 1:

Pursuant to paragraph 10.3(1) of this Manual, the minimum required classification of slip resistance of the tactile guide path shall be not less than that required of the areas it situates and in no case be less than classification P3 or R10 for indoor areas and classification P4 or R11 for outdoor areas.

NOTE 2:

To maintain optimal slip resistance and enhance safety, it is crucial to implement effective drainage systems and other preventive measures. These efforts ensure that dry areas remain free from standing or casual water, minimizing slip hazards and promoting a secure environment.

NOTE 3:

The results from the wet pendulum test and the oil-wet inclining platform test, as detailed in the classification tables, are distinct and not interchangeable. For instance, achieving the classification of P1 in the wet pendulum test does not correspond to the classification of R9 in the oil-wet inclining platform test. Compliance with the classification under either one of the tests for the relevant area is sufficient.

NOTE 4:

Under special circumstances, selecting a lower slip resistance classification may be appropriate to better meet functional requirements or sports-specific demands.

NOTE 5:

Classifications A, B and C, as specified in Table 4 of AS 4586, are derived from the wet-barefoot inclining platform test, which is different from the oil-wet inclining platform test. These classifications offer guidance on slip resistance for areas with frequent barefoot access.

NOTE 6:

Plant rooms, which are regularly accessed by maintenance staff, often present varied floor conditions, including water, oil, or grime. Specifiers should recommend suitable slip-resistant values that account for these dynamic environments. In electrical rooms equipped with raised floor systems, lower slip resistance values may be deemed acceptable due to restricted access and specific operational constraints.

APPENDIX D LUMINANCE AND COLOUR CONTRAST

D.1 Types of contrast

Contrast refers to perceptible differences between different regions of an image or scene. There are two fundamental types of contrast:

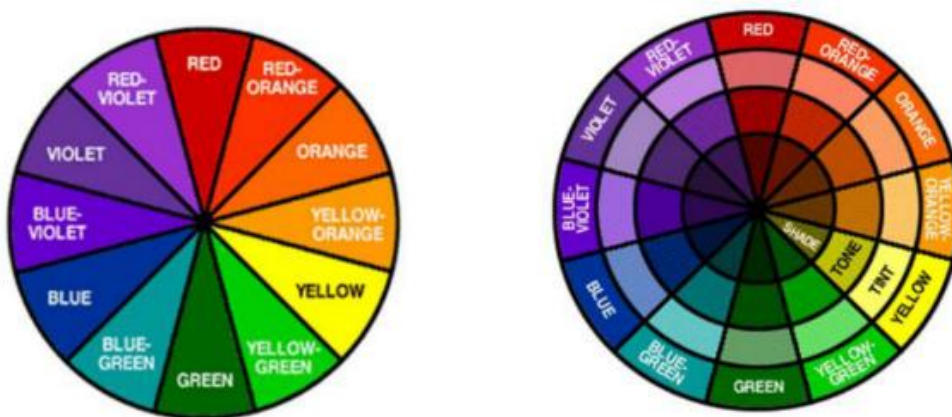
- (1) Colour contrast, and
- (2) Luminance contrast.

D.2 Colour contrast

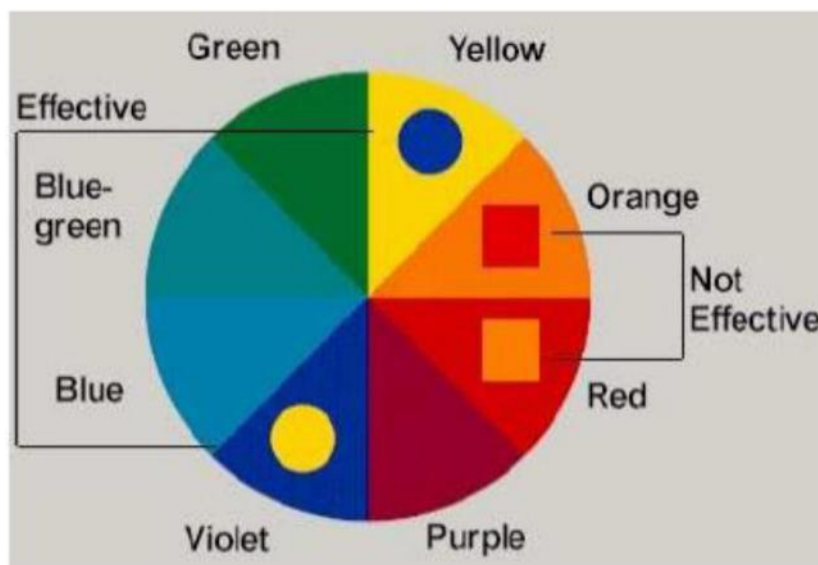
Colour contrast is the difference between colours positioned on a colour wheel, it plays a crucial role in enhancing text readability, boosting the visual impact of graphics, and capturing users' attention. For example, violet is the contrast to yellow, red is contrasted to green. Colours can contrast on the basis of their lightness, saturation or hue.

- (1) lightness measures the relative amount of light reflected from a colour;
- (2) saturation relates to its intensity; and
- (3) hue refers to elementary colour names, that is red, green, blue etc.

The perception of colour, however, can change with the type of light. Some people with visual impairment are unable to see colours as ageing and sight loss diminish their sensitivity of colour perception. Therefore, colour contrast on its own may not be sufficient to provide a visual contrast.



- Tint** : color + white, resulting in a lighter value
- Tone** : color + grey, resulting in a darker value
- Shade** : color + black, delivering the darkest versions of color

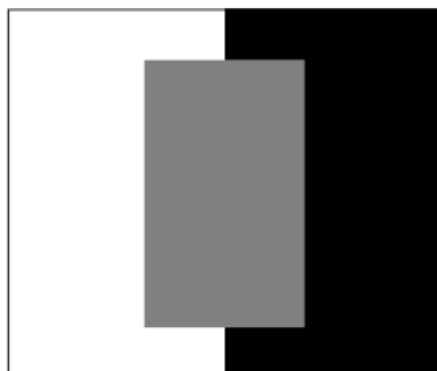


D.3 Luminance Contrast

Luminance contrast refers to the difference in amount of light reflected from the surface of an object and the amount of light reflected from the surface of its surrounding background. It significantly influences the perception of form, depth, and boundaries, proving more reliable than colour contrast for many users with visual impairments.

A minimum luminous contrast of 30% is generally required, as this level is often the threshold discernible by persons with partial sight. For reference:

- Black and white have a 100% Luminance contrast.
- Grey and black or grey and white have a 50% Luminance contrast.



D.4 Measurement of Luminance Contrast using Luminance Factor Value

Luminance contrast is expressed in percent and can be determined by:

$$[(B1-B2) / B1] \times 100$$

where

B1 = light reflectance value (LRV) of the lighter area and

B2 = light reflectance value (LRV) of the darker area.

The light reflectance value (LRV) of a surface can be measured using a luminance meter (photometer) equipped with a 1-degree measurement field. For the method of on-site measurement of luminance contrast using LRV, Appendix B to AS1428.1-2009, or other equivalent standards, should be referred to.

D.5 Lighting

Contrast will only be of assistance to people with visual impairment if there is an appropriate quantity and quality of illumination with which to view the contrasting elements. At low light levels, the perception of contrast diminishes. Lighting levels should generally be relatively uniform and about 25% higher for people with low vision. Strong directional lighting casts shadows that can mask contrasting surfaces. Significant fluctuations in illumination level can reduce visibility due to the slower adaptive response of the eye in someone with low vision.

Effective uses of luminance contrast include:

- (1) Contrasting tactile warning strips to indicate the start and finish of a ramp;
- (2) Contrasting doorframes, doors, skirting boards and architraves to assist with locating doors;
- (3) Contrasting paving at doorways to assist with locating the entry; and

- (4) Contrasting edges of steps, a roadway or poles in play areas to highlight potential hazards.



- (5) Safety colours such as yellow, orange, and red are considered effective warning colours, while green is often associated with safety and environmental friendliness. These colours are highly recognised by the human eye and mind. When selecting colours, it's worth considering their psychological effects.
- (6) Overhead lighting above staircases is preferable and avoid over-lighting to prevent glare and eye adaptation issues.
- (7) Provide uniform lighting at regular intervals along the walkway, and to avoid sudden change in lighting level. Daylight condition should also be taken into consideration.
- (8) Provide lower side-wall lighting or backlit lighting for the escalator steps for user to notice the motion and the signs of the escalator.
- (9) Provision of transitions area to reduce the visual discomfort when entering a dark lobby from bright sunlight or vice-versa, such as canopy for entrance and introduction of daylight into the lobby through skylight, glass wall or window etc.
- (10) Provide adequate lighting for nighttime signage and position light sources above and shield them to prevent glare.

- (11) Use non-reflective, darker-colored materials for sports surfaces.
- (12) Avoid floor lighting on main accessible routes.

APPENDIX E SUMMARY OF OBLIGATORY PROVISIONS PROPOSED TO BE INCLUDED IN THE B(P)R

As mentioned in paragraph 5 of the Preface to this Manual, for those obligatory provisions stipulated in this Manual that are not yet covered by regulations 72 and 73, the Third and Fourth Schedules of the B(P)R, the BD will propose inclusion of such provisions in the B(P)R through legislative amendments. Before the enactment of the legislative amendments, such provisions including interpretations and obligatory design requirements in this Manual are advisory in nature.

In this connection, this Appendix aims to provide a summary of the obligatory provisions in this Manual that are currently advisory in nature such that reference should be made to the corresponding provisions currently in force under regulations 72 and 73, the Third and Fourth Schedules of the B(P)R.

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 2.1	Interpretation “Access” “Access route” “Accessible” “Common areas” “Ground floor” “Large scale specified premises”	Section 1, Third Schedule	Interpretation “access” “accessible” “common areas” “tactile guide path” “tactile warning strip”
	“Luminance contrast” “Public services escalator or passenger conveyor” “Slip-resistance” “Specified premises” “Tactile guide path” “Tactile warning strip” “Wet areas”	Section 1, Fourth Schedule	Interpretation “exempted route” “regulated route”

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Table 1	Category of Buildings & Extent of the Application	Part 4, Third Schedule	Building where Limited Facilities for Persons with a Disability are to be Provided
Para. 3.4(2)	Area only used for building services and maintenance (testing, inspection, verification, repair and overhaul)		Building Design Requirements to Facilitate Elderly Persons
Para. 3.4(7)	Areas to which PwDs may not reasonably be expected to have access in view of the relatively high risk operations in the buildings	Regulation 73(3), 73(4) and 73(5)	
Para. 4.2	Provision of Access Route	Section 11, Third Schedule	Provision of Access Route
		Section 2, Fourth Schedule	Provision of Routes from Lot Boundary to Building Entrances
Para. 4.3(1)	Tactile Guide Path	Section 13, Third Schedule	Tactile Guide Path
Paras. 4.4(1) to 4.4(10)	Car Parking Areas	Sections 8, 9 and 10, Third Schedule	Carparks
Paras. 5.2(1) and 5.2(2)	Floor Surface of Access	Section 12(3), Third Schedule	Requirements for Access Route
		Section 3(2), Fourth Schedule	General Requirements for Regulated Routes

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 5.3(1)	Width of Access	Section 12(1), Third Schedule	Requirements for Access Route
		Section 3(3), Fourth Schedule	General Requirements for Regulated Routes
Para. 5.4(1)(a)	Manoeuvring Space of Access	Section 31(1), Third Schedule	Manoeuvring Space
		Section 3, Fourth Schedule	General Requirements for Regulated Routes
Para. 5.4(3)	Controlled Passage	Section 35(3), Third Schedule	Controlled Passage
Para. 5.5(1)	Width of Doors	Section 38, Third Schedule	Width of Doors
		Section 6, Fourth Schedule	Width of Doors
Para. 5.5(2)	Unobstructed Area	Section 39(1), Third Schedule	Unobstructed Area

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 5.5(3)	Door Thresholds	Section 42, Third Schedule	Door Thresholds
		Section 7, Fourth Schedule	Door Thresholds
Para. 5.5(4)	Automatic Main Entrance Doors	Section 45, Third Schedule	Automatic Main Entrance Doors
		Section 10, Fourth Schedule	Automatic Main Entrance Doors
Paras. 5.6(1) to 5.6(3)	Residential Unit	Sections 11, 12 and 13, Fourth Schedule	Design Requirements for Certain Single-household Buildings or Certain Residential Units
Para. 6.2(1)	Width of Ramp	Section 15, Third Schedule	Width of Ramp
		Section 3(3), Fourth Schedule	General Requirements for Regulated Routes
Para. 6.2(5)(c)	Requirements for Ramps	Section 18(1)(b), Third Schedule	Requirements for Ramps

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 6.2(12)	Protection and Surfaces of Ramps	Section 19, Third Schedule	Protection and Surface
		Section 3(2), Fourth Schedule	General Requirements for Regulated Routes
Para. 6.3(3)	Location of Dropped Kerbs	Section 22, Third Schedule	Location of Dropped Kerbs
Para. 6.4(1)	Steps and Staircases	Section 25, Third Schedule	Dimension and Orientation of Staircase
		Section 4, Fourth Schedule	Dimension of Main Circulation Staircases
		Section 5, Fourth Schedule	Dual-level Handrails of Main Circulation Staircases
Paras. 7.2(2) and 7.2(4)	Accessible Lifts	Section 78, Third Schedule	Special Requirements for Accessible Lifts
		Section 9, Fourth Schedule	Accessible Lifts
Para. 7.3(5)	Indication in Accessible Lift Cars	Section 82(3), Third Schedule	Indication in Accessible Lift Cars

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 7.4(3)	Escalator and Passenger Conveyors	Division 20, Third Schedule	Escalator and Passenger Conveyors
Para. 8.2(2)	Wheelchair Spaces	Section 4, Third Schedule	Wheelchair Spaces
Para. 8.2(5)	Access for Wheelchair Users	Section 5, Third Schedule	Access for Wheelchair Users
Paras. 8.2(8) and 8.2(9)	Braille and Tactile Fire Exit Maps and Tactile Warning Strips; Seating Plate	Section 6, Third Schedule	Braille and Tactile Fire Exit Maps and Tactile Warning Strips
Para. 8.3(2)	Minimum Provision for Accessible Guestrooms	Section 7, Third Schedule	Guest rooms
Para. 9.2(11)	Toilet/Cubicle Doors	Section 52, Third Schedule	Door of Toilets or Cubicles
Para. 9.2(18)	Emergency Call Bell	Section 54, Third Schedule	Emergency Call Bell
Para. 9.2(19)	Urinals	Section 55, Third Schedule	Urinals

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 9.3(3)	Bathtubs	Section 56(3), Third Schedule	Bathtubs in Accessible Bathrooms
Para. 9.3(10)	Roll-in Compartments	Section 62, Third Schedule	Thresholds of Accessible Shower Compartments
Para. 10.2(5)	Handrails	Division 8, Third Schedule	Handrails
		Section 5, Fourth Schedule	Dual-level Handrails of Main Circulation Staircases
Paras. 10.3(1), 10.3(2) and 10.3(3)	Tactile Guide Path	Section 69(3), Third Schedule	Special Design Requirements to assist Persons with Visual or Hearing Impairment
Para. 10.5(1)	Illumination Level	Section 72, Third Schedule	General Illumination Level
		Section 3(6), Fourth Schedule	General Requirements of Regulated Routes
		Section 8, Fourth Schedule	Illumination for Lifts

Obligatory Provisions in this Manual (Currently Advisory in Nature)		Corresponding Provisions currently in Force under the B(P)R	
Para. 10.6(4)	Design of Counter	Section 71, Third Schedule	Design of Counters
Para. 10.7	Assistive Listening System	Section 77, Third Schedule	Assistive Listening System
Para. 10.8(2)	Visual Alarm	Not under prevailing B(P)R. See obligatory design requirements in <u>Appendix F</u> .	
Para. 10.9(5) 10.9(6) 10.9(7) 10.9(9) 10.9(10)	Braille and Tactile Signs	Section 68, Third Schedule	Braille and Tactile Signs
Para. 10.10(1)	Braille and Tactile Floor Plan	Section 69(2), Third Schedule	Special Design Requirements to Assist Persons with Visual or Hearing Impairment

APPENDIX F OBLIGATORY DESIGN REQUIREMENTS FOR FIRE ALARM SYSTEMS

For the purpose of complying with Section 16(1)(b) of the Buildings Ordinance (Cap. 123), the Code of Practice for Minimum Fire Service Installations and Equipment and Inspection, Testing and Maintenance of Installations and Equipment requires provisions of visual alarm and audible alarm to the satisfaction of the Director of Fire Services. The obligatory design requirements of these alarm systems originally stipulated under paragraph 5.2 in Chapter 5 of the DMBFA are excerpted below.

5.2 FIRE ALARM SYSTEMS

5.2.1 Persons with hearing impairment depend on visual cues to alert them of emergencies. Persons with hearing impairment may need a visible as well as audible alarm which is only sufficient to deliver warning to a normal person.

MANDATORY SECTION

5.2.2 Performance Objectives

Where fire alarm systems are provided in areas of a building required to be accessible, the alarm shall emit both audible and visible signals.

5.2.3 Obligatory Design Requirements

Visual Alarm

- (a) Subject to paragraph (b), visual alarm signal shall be provided to form part of the fire alarm system in the form of a flashing red light, labelled “Fire Alarm 火警” in both English and Chinese. It shall be indicated on separate plate affixed nearby or engraved on the light cover. The alarm shall be installed at a prominent location which shall be readily noticeable when activated in the categories of buildings as specified in Table 2 in Chapter 2.
- (b) The provision of visual alarm shall not apply to all exit staircases as required under the Code of Practice for Fire Safety in Buildings including the smoke lobbies adjoining the exit staircase, and the following areas:

Domestic use - domestic units

Office	- areas accessible to staff only, e.g. offices, store rooms, plant rooms, staff toilets, etc.
Department store and shopping complex	- areas not accessible to customers, visitors, e.g. offices, store rooms, plant rooms, staff toilets, etc.
Hotel, guesthouse, hostel and bank	- areas not accessible to guests, customers, visitors, e.g. kitchens, plant rooms, staff toilets, offices, back of house facilities, etc.
Place for worship	- areas not accessible to worshippers, visitors, e.g. offices, staff toilets, etc.
Cinema, theatre, concert hall, stadium, museum, theme park and purpose-built family amusement centre	- areas not accessible to the visitors, spectators, audience, e.g. projector rooms, offices, store rooms, plant rooms, staff toilets, etc.
School, college, university	- areas accessible to staff only, e.g. offices, store rooms, plant rooms, staff toilets, staff quarters, etc.
Factory, workshop and place for industrial use	- areas accessible to staff only, e.g. offices, store rooms, plant rooms, staff toilets, etc.
Sports complex and public swimming pool complex	- areas not accessible to visitors, athletes, spectators, e.g. plant rooms, offices, store rooms, staff toilets, etc.
Restaurant and food court	- areas not accessible to customers, visitors, e.g. offices, kitchens, store rooms, plant rooms, staff toilets, etc.
Indoor market and supermarket	- areas not accessible to customers, visitors, e.g. offices, store rooms, plant rooms, staff toilets, etc.
Hospital, purpose-built clinic, residential home for the elderly and welfare centre	- areas not accessible to patients, inmates, visitors, e.g. kitchens, plant rooms, staff toilets, offices, laboratories, back of house facilities, staff quarters, etc.

Club House	- areas accessible to staff only, e.g. offices, kitchens, store rooms, plant rooms, staff toilets, etc.
Transport station, interchange, passenger terminal	- areas accessible to staff only, e.g. plant rooms, station offices, regulator's offices, staff toilets, etc.
Carpark	- areas not accessible to customers, visitors, e.g. shroff offices, store rooms, plant rooms, staff toilets, etc.

Positioning of Call Points

- (c) The manual fire alarm call points or activation controls, e.g. breakglass units, in places which are intended to be accessible to wheelchair users, shall be positioned at prominent and accessible locations and installed in accordance with the Code of Practice for Minimum Fire Service Installations and Equipment and the Code of Practice for Inspection, Testing and Maintenance of Installations and Equipment or Fire Services Department Circular Letter.

Audible Alarm

- (d) Audible alarm shall comply with the requirements of the Code of Practice for Minimum Fire Service Installations and Equipment and the Code of Practice for Inspection, Testing and Maintenance of Installations and Equipment published by the Director of Fire Services.